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816x VXI*plug&play* Instrument Driver Help Contents

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Using the Instrument Driver

The Keysight Technologies 816x VXI*plug&play* Instrument Driver revision 4.5 complies with the following:

WIN 95/98 and WIN NT and WIN XP System Frameworks

VISA revision F.01.02.00 and newer

This version is also tested and used with Windows 7, including 64-bit systems, and with Keysight VISA 80.3.17281.1, installed with IO Library Suites 16.3.17218

8163A Lightwave Multimeter Firmware revision 5.0

8164A Lightwave Measurement System Firmware revision 5.0

8166A Lightwave Multichannel System Firmware revision 5.0

8163B Lightwave Multimeter Firmware revision 5.0

8164B Lightwave Measurement System Firmware revision 5.0

8166B Lightwave Multichannel System Firmware revision 5.0

N773xA Optical Switch Firmware revision 1.2

N7744A and N7745A Optical Power Meter Firmware revision 1.18

N7747A and N7748A Optical Power Meter Firmware revision 1.30

N775xA-series Optical Attenuator and Power Meter Firmware revision 1.00

N776xA-series Optical Attenuator Firmware revision 1.00

The following information is common to all programs that use the Keysight Technologies 816x VXI*plug&play* instrument driver:

VISA, VXI*plug&play*, and the Keysight Technologies 816x VXI*plug&play* instrument driver

Directory Structure

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VISA, VXI*plug&play*, and the Keysight Technologies 816x Instrument Driver

This Keysight Technologies 816x VXI*plug&play* instrument driver conforms to all portions of the VXI*plug&play* driver standard which are applicable to conventional GP-IB and other non-VXI instruments (that is, rack and stack instruments).

The formal VXI*plug&play* standard only covers VXI instruments, and two elements of the standard do not apply to the Keysight Technologies 816x VXI*plug&play* instrument driver since it is not a VXI instrument. Firstly, there is no soft front panel, as the Keysight Technologies 816x VXI*plug&play* instrument driver can be controlled from the instrument's front panel. Secondly, there is no knowledge base file, which is primarily a physical description of a VXI board.

Aside from these exceptions, you'll find the same features in this driver as in Keysight's other VXI*plug&play* drivers.

1. Conformance with the VXI*plug&play* standard. The only exceptions are that it does not have a soft front panel or a knowledge base file.

2. It is built on top of, and uses the services provided by VISA. VISA supports GP-IB and VXI protocols. The driver can be used with any GP-IB card for which the manufacturer has provided a VISA DLL. Instruments on LAN or USB connections with VISA addresses can also be controlled.

3. It includes a "Function Panel" (.fp) file which allows it to be used with visual programming environments such as Keysight VEE, LabWindows, and LabVIEW.

4. It includes a comprehensive on-line help file which complements the instrument manual. The help file presents application programming examples, a cross-reference between instrument commands and drive functions, and detailed documentation of each function, with examples.

5. It includes a Visual Basic include file (.bas) which contains the function calls in Visual Basic syntax, so that driver functions can be called from Visual Basic. If you use Visual Basic with this driver, you should be familiar with C/C++ function declarations. In particular, care must be taken when working with C/C++ pointers.

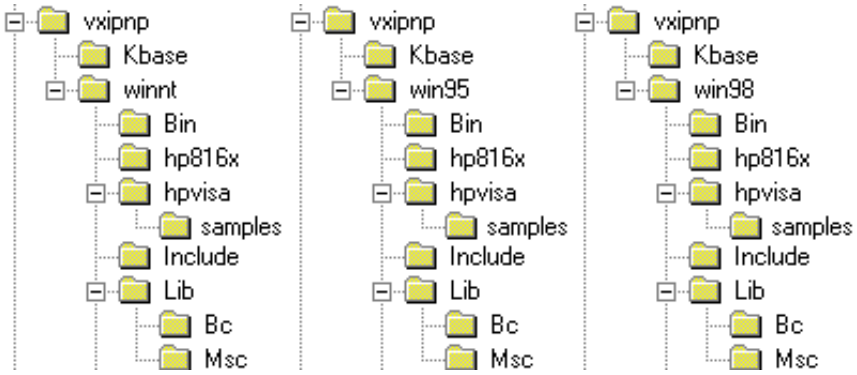
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Directory Structure

The setup program which installs the Keysight Technologies 816x VXI*plug&play* instrument driver creates the VXIPNP directory if it does not already exist. The structures for the Windows NT, Windows 95, and Windows 98 VXIPNP subdirectory tree are:



In the directory example, hp816x is a directory containing the instrument driver. There would be a directory for each instrument driver. In more recent Windows XP installations, the hp816x directory is usually located within the IVI Foundation directory, under VISA.

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Opening an Instrument Session

Introduction

To control an instrument from a program, a communication path between the computer/controller and the instrument must be opened. This path is known as an instrument session and is opened with the hp816x_init function:

ViStatus hp816x_init(ViRsrc *resourceName*, ViBoolean *IDQuery*, ViBoolean *reset*, ViPSession *ihandle*);

Instruments are assigned a handle when the instrument session is opened. The handle, which is a pointer to the instrument, is the first parameter passed in all subsequent calls to driver functions.

The parameters of function hp816x_init include:

ViRsrc *resourceName* - the address of the instrument.

ViBoolean *IDQuery* - a Boolean flag which indicates if in-system verification should be performed. Passing VI_TRUE (1) will perform an in-system verification. Passing VI_FALSE (0) will not. If you set id_query to false it is possible to use the generic functions of the instrument driver with other instruments.

ViBoolean *reset* - a Boolean flag which indicates if the instrument should be reset when it is opened. Passing VI_TRUE (1) will perform a reset when the session is opened. Passing VI_FALSE (0) will not perform a reset.

ViPSession *instrumentHandle* - a pointer to an instrument session. ***instrumentHandle*** is the handle which addresses the instrument and is the first parameter passed in all driver functions.

Successful completion of this function returns VI_SUCCESS.

For more information see:

Opening an Instrument Session - Visual C Example

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Closing an Instrument Session

Sessions (instrumentHandle) opened with the Initialize (hp816x_init) function are closed with the function:

```
hp816x_close (ViSession instrumentHandle);
```

When no further communication with an instrument is required, the session must be explicitly closed using hp816x_close() function. VISA does not remove sessions unless they are explicitly closed. Closing the instrument session frees all data structures and system resources allocated to that session.

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VISA Data Types and Selected Constant Definitions

The driver functions use VISA data types. VISA data types are identified by the 'Vi' prefix in the data type name (e.g. ViInt16, ViUInt16, ViChar). The *visatype.h* file contains a complete listing of the VISA data types, function call casts and some of the common constants. The following is a partial list of the type definitions and constant definitions in *visatype.h*.

Completion and Error Codes

VI_SUCCESS A long equal to zero
VI_ERROR A long equal to hex 0x80000000

Other VISA Definitions

The size of the following three symbols is operating system dependent and is equal to the default integer size.

VI_NULL An integer equal to 0

VI_TRUE An integer equal to 1
VI_FALSE An integer equal to 0

Variable types

ViUInt32 A 32-bit unsigned integer
ViPUInt32 A pointer to a 32-bit unsigned integer
ViAUInt32 An array of 32-bit unsigned integers

ViInt32 A 32-bit signed integer
ViPInt32 A pointer to a 32-bit signed integer
ViAInt32 An array of 32-bit signed integers

ViUInt16 A 16-bit unsigned integer
ViPUInt16 A pointer to a 16-bit unsigned integer
ViAUInt16 An array of 16-bit unsigned integers

ViInt16 A 16-bit signed integer

ViPInt16 A pointer to a 16-bit signed integer

ViAInt16 An array of 16-bit signed integers

ViUInt8 A unsigned byte
ViPUInt8 A pointer to an unsigned byte
ViAUInt8 An arrays of unsigned bytes

ViInt8 A 8-bit signed integer
ViPInt8 A pointer to a 8-bit signed integer
ViAInt8 An array of 8-bit signed

ViChar A character
ViPChar A pointer to a character
ViAChar An array of characters

ViByte A byte
ViPByte A pointer to a byte
ViAByte An array of bytes

ViAddr A Void pointer
ViPAddr A pointer to a void pointer

ViAAddr An array of void pointers

ViReal32 A 32-bit floating point
ViPReal32 A pointer to a 32-bit floating point
ViAReal32 An array of 32-bit floating points

ViReal64 A 64-bit floating point
ViPReal64 A pointer a 64-bit floating point
ViAReal64 A pointer to an array of 64-bit floating points

ViBuf A pointer to an array of bytes
ViPBuf A pointer to an array of bytes
ViABuf An array of byte pointers

ViString An array of characters
ViPString A pointer to an array of characters

ViAString An array of ViStrings

ViRsrc A ViString
ViPRsrc A pointer to a ViString

ViARsrc An array of ViStrings

ViBoolean A 16-bit unsigned boolean value
ViPBoolean A pointer to a 16-bit unsigned boolean
ViABoolean An array of 16-bit unsigned booleans

ViStatus A 32-bit integer status value
ViPStatus A pointer to a 32-bit integer status value
ViAStatus An array of 32-bit integer status values

ViVersion A 32-bit unsigned integer version number
ViPVersion A pointer to a 32-bit unsigned integer version number

ViAVersion An array of 32-bit unsigned integer version numbers

ViObject A 32-bit unsigned integer object
ViPObject A pointer to a 32-bit unsigned integer object
ViAObject An array of 32-bit unsigned objects

ViSession A 32-bit unsigned integer object
ViPSession A pointer to a 32-bit unsigned integer object
ViASession An array of 32-bit unsigned objects

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Error Handling

Events and errors within instrument control program can be detected by polling the instrument. Most of the example programs poll (query) the instrument after each function to determine if an error or other event has occurred. Polling is used in application development environments (ADEs) that do not support asynchronous activities where callbacks can be used. The example programs set up and use polling as shown below.

1. Declare a variable to contain the function completion code.

```
ViStatus errStatus;
```

Every driver function returns the completion code ViStatus. If the function executes with no I/O errors, driver errors, or instrument errors, ViStatus is 0 (VI_SUCCESS). If an error occurs, ViStatus is a negative error code. Warnings are positive error codes, and indicate the operation succeeded but special conditions exist.

2. Enable automatic instrument error checking following each function call.

```
hp816x_errorQueryDetect(instrumentHandle, VI_TRUE);
```

When enabled, the driver queries the instrument for an error condition before returning from the function.

If an error occurred, errStatus (Step 1) will contain a code indicating that an error was detected (hp816x_INSTR_ERROR_DETECTED).

3. Check for an error (or event) after each function.

```
errStatus = hp816x_cmd(instrumentHandle, "MEAS:FREQ");
 check(instrumentHandle, errStatus);
```

After the function executes, errStatus contains the completion code. The completion code and instrument ID are passed to an error checking routine. In the above statement, the routine is called 'check'.

4. Create a routine to respond to the error or event. The following routine is used by the example programs to read errors.

```
void check (ViSession instrumentHandle, ViStatus errStatus)
{
 /* variables for error code and message */
```

```
 ViInt32 inst_err;
 ViChar err_message[256];

 /* VI_SUCCESS is 0 and is defined in VISATYPE.h */
 if(VI_SUCCESS > errStatus)
 {
 /* send a device clear - to ensure communication with the instrument */
  hp816x_dcl(instrumentHandle);

 /* hp816x_INSTR_ERROR_DETECTED defined in hp816x.h */
  if(hp816x_INSTR_ERROR_DETECTED == errStatus)

  {
  /* query the instrument for the error */
  hp816x_error_query(instrumentHandle, &inst_err, err_message);

  /* display the error */
  printf("Instrument Error : %ld, %s\n", inst_err, err_message);
  }
  else /* driver error */
  {
  /* get the driver error message */
  hp816x_error_message(instrumentHandle, errStatus, err_message);

  /* display the error */
  printf("Driver Error : %ld, %s\n", errStatus, err_message);

  }
  /* optionally reset the instrument, close the instrument handle */
  hp816x_reset(instrumentHandle);
  hp816x_close(instrumentHandle);
  exit(1);
 }
 return;
}
```

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Introduction to Programming

To identify the functions you will need to write your instrument control program, select:

Selecting Functions

To install some typical programs, select:

Example Programs

For information on how to create programs and use this driver in a number of common programming environments, select:

Creating, Compiling and Linking 32-bit Programs Using Microsoft Visual C++ Version 4.2

Creating and Running Visual BASIC 4.0 Applications

These sections explain how to begin using the Keysight Technologies 816x VXI*plug&play* instrument driver with these products.

Getting Started with Keysight VEE

Getting Started with LabView

Getting Started with LabWindows

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Selecting Functions

The help file's function reference lists the Keysight Technologies 816x VXIplug&play instrument driver functions alphabetically and hierarchically. There are nine categories of functions based on their intended use. The categories are:

- Utility Functions
- Mainframe Specific Functions
- Attenuator Specific Functions
- Power Sensor Specific Functions
- Fixed Laser Source Specific Functions
- Switch Specific Functions
- Return Loss Module Specific Functions
- Tunable Laser Source Specific Functions
- Applications

For details of which functions are in which category, see the Function Reference (Hierarchical).

The example programs in this helpfile demonstrate most of the functions listed.

Utility Functions

Utility functions perform a variety of standard tasks. For example, Some of the Keysight Technologies 816x VXIplug&play instrument driver's utility functions assist error handling, managing instrument ids, and unit conversion.

Mainframe and Module Specific Functions

Typically, these functions combine multiple SCPI commands into a single, functional operation. They are designed to allow quick and easy access to common instrument command sequences. The functionality they encapsulate should be applicable to a wide variety of functional tasks. These functions are mainframe specific.

Note that this version of the 816x VXIplug&play instrument driver has been extended to support the stand-alone N774xA optical power meter instruments, which do not require a mainframe. These instruments are supported by both relevant mainframe specific functions and power sensor specific functions, as well as the Multi Frame Lambda Scan application function. The relevant mainframe-specific functions and attenuator functions also support the stand-alone N775xA and N776xA series of attenuator instruments. The N775xA-series attenuators also include power meters that are supported by the basic sensor functions of the PnP driver. These latter power meters do not have the functionality for logging or stability measurement and so are also not used by the Multi Frame Lambda Scan function. The relevant mainframe and switch functions also support the stand-alone N773xA series of switch instruments.

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Information for N77 Series Instruments

This version of the 816x VXIplug&play instrument driver has been extended to support the N773xA, N774xA optical power meter and the N775xA and N776xA Optical Attenuator instruments. These are stand-alone instruments that do not require installation in an 816x mainframe. The SCPI command set has been designed for almost complete programming compatibility with the modular instruments of the 816x family. Please be aware of the following details and limitations, when using this driver.

- The N77-series instruments can be controlled via USB, LAN or GPIB interfaces. Especially for transferring large data volumes, the USB and usually LAN interfaces provide faster performance.
- The identification of the individual ports of these multiport instruments in the SCPI and hp816x functions corresponds to the slot number. So for example the ports of the N7745A can be addressed with the slot numbers 1 to 8. The channel number in the commands and functions is not used and can generally be omitted or set to Channel 1 (index 0).
- The N771x-series and 81950A tunable lasers are not supported by the 816x VXIplug&play driver.
- The N773x-series optical switches are supported by the 816x VXIplug&play functions for switches and relevant functions for the mainframes.
- The N774x-series power meters are supported by the 816x VXIplug&play functions for power meters, relevant functions for the mainframes like for standard trigger configuration, and the Multi Frame Lambda Scan application functions.
- The N776x-series optical attenuators are supported by the 816x VXIplug&play functions for attenuators and the relevant functions for the mainframes.
- The N775x-series optical attenuators and power meters are supported by the 816x VXIplug&play functions for attenuators and the relevant functions for the mainframes and the basic functions for the power meters, excluding the advanced functions using data power logging. The choices of averaging time is also more limited for these power meters.
- The averaging time parameter for the power meters is limited in the 816x VXIplug&play functions to a limited set of discrete values, similar to the modular 816x power meters. This set includes: 1, 2, 5, 10, 25, 100, 200, and 500 ìs, 1, 2, 5, 10, 20, 50, 100, 200, 500 ms, and 1, 2, 5, and 10s.
- The maximum number of logging data points that can be acquired is limited by the 816x VXIplug&play data acquisition functions to 100,000 per port.
- The shortest averaging time used by the Multi Frame Lambda Scan application is 25 ìs.
- For the Multi Frame Lambda Scan application, the N774xA instruments can be included simply by registering the instrumet as a mainframe,after the first

mainframe with the tunable laser has been registered.

Further details to available functions are found under the categories below:

Utility Functions

Mainframe Specific Functions

Power Sensor Specific Functions

Applications

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Instrument Addressing

The 8163A/B Lightwave Multimeter, the 8164A/B Lightwave Measurement System and the 8166A/B Lightwave Multichannel System use a GP-IB interface. Later models of the 8163B and 8164B also have a LAN interface. The N773xA Optical Switch, N774xA Optical Power Meter and N775x and N776x Optical Attenuator instruments have USB, LAN and GPIB interfaces. When opening an instrument session using the hp816x_init() function, an instrument address must be specified for the function's *ViRsrc resourceName* parameter.

When addressing the instrument over the GP-IB interface the instrument address is based on the logical address and on the communication interface between the computer/controller and the instrument. The default GP-IB address of your instrument is 20.

For more information see:

Opening an Instrument Session - Visual C Example

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Example Programs

To install example programs when you install the Keysight Technologies 816x VXI*plug&play* Instrument Driver:

1. Choose Custom setup from the Setup Type menu.
2. Select Example Files from the Select Components menu.
3. Select one or more of the following programming environments and complete the installation:
   - Visual Basic
   - VISA
   - VEE
   - LabView
4. If you have chosen the default directory, the examples will be installed in the following directories respectively:
   - <Hard Drive>:\vxipnp\winnt\hp816x\Examples\Vb
   - <Hard Drive>:\vxipnp\winnt\hp816x\Visa\VC
   - <Hard Drive>:\vxipnp\winnt\hp816x\Examples\VEE
   - <Hard Drive>:\vxipnp\winnt\hp816x\Examples\LabView

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Creating, Compiling, and Linking 32-bit Programs Using Microsoft Visual C++ Version 4.2

The example that follows is a step-by-step list of instructions on how to create a 32-bit console application project using Microsoft Visual C++ Version 4.2 or higher. Any project type such as a MFC application or windows application could be created if desired. The name of the program is *SimpleExample*. The example program opens a GP-IB connection to the instrument, querys the instruments ID string, prints the string, closes the GP-IB connection and terminates. It is assumed that the VISA I/O configuration has already been successfully completed.

To create a new project:

Select *File* | *New* | *Project Workspace*
Click *OK*

Select the project type, for this example we will build a simple console applications, but any other project type could be used:

Select Console Application as the Type

Enter a name for the project, for example *SimpleExample*
Change the location the project will be created in if desired.
Click *Create*

Create the program source file:

Select *File* | *New* | *Text File*

Enter the following program into the text file window:

```
#include <stdio.h>
#include "hp816x.h"
main()
{
 ViStatus vistatReturn;
 ViSession instrumentHandle;
```

```
        ViChar viszResponse[256];

        // open session, no verification, no reset
        if ((vistatReturn = hp816x_init("GPIB::20::INSTR",
         VI_FALSE, VI_FALSE, &instrumentHandle)) != VI_SUCCESS)
        {
         // connection failed
         printf("Connection Failed!\n");
         return vistatReturn;
        }
        // query instrument ID string
        vistatReturn = hp816x_cmdString_Q(instrumentHandle,"*IDN?",
                256,viszResponse);
        printf("The instrument is: %s",viszResponse);
        // close session
        vistatReturn = hp816x_close(instrumentHandle);
        printf("Program complete.\n");
        return 0;
       }
```

Save the file:

Select *File* | *Save*
Enter *SimpleExample.cpp* for the file name.
Click *Save*

Insert the source file into the project:

Select *Insert* | *Files into Project...*
Click on *SimpleExample.cpp*
Click *Add*

Set the include file search path:

Select *Tools* | *Options...*
Select the *Directory* tab
Select *Show Directories for Include Files*
Scroll through the *Directory* list looking for the directory that the *hp816x.h* file is in
If the *hp816x.h* path isn't already in the list add it to the bottom of the list
Click *OK*

Insert the hp816x library into the project link libraries:

Select *Build* | *Settings* | *Link* | *General*
Insert *hp816x.lib* in object/libraries modules edit box.
Click *OK*

Set the library search path:

Select *Tools* | *Options...*
Select the *Directory* tab
Select *Show Directories for Library Files*
Scroll through the *Directory* list looking for the directory that the *hp816x.lib* file is in
If the *hp816x.lib* path isn't already in the list add it to the bottom of the list
Click *OK*

Compile the project:

Click *Build* | *Build SimpleExample.exe*

Run the program

Select *Build* | *Debug* | *Go*

You may want to put a break point on the last line of the program to pause the program while the output window is still visible. Alternatively open a DOS command window, switch to the directory that contains the executable file (SimpleExample/Debug) and run the program from the command line.

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Opening an Instrument Session - Visual C Example

```
/* The following example demonstrates how to establish communication using
VXIplug&play driver calls. */
#include <stdio.h>
#include <conio.h>

/*header of the vxipnp driver*/
#include <hp816x.h>

/*GPIB address 20 on local controller 0*/
#define INSTR_ADDRESS "GPIB0::20::INSTR"

/*function prototypes*/
void check_err (ViSession vi, ViStatus errStatus);



int main (void )

{

ViStatus err_status;
ViSession iHandle;
ViChar driver_rev[256];
ViChar instr_rev[256];
ViBoolean do_reset = VI_TRUE;

/* Initialize the instrument. Note that this function will always verify */
/* that the instrument specified is an Keysight Technologies 816x Series instrument and will send a reset*/
/* to the instrument if do_reset=VI_TRUE. */

err_status = hp816x_init(INSTR_ADDRESS , VI_FALSE, do_reset, &iHandle);
if(err_status < VI_SUCCESS )
{
check_err(iHandle, err_status);
return -1;
}

err_status = hp816x_revision_Q(iHandle, driver_rev, instr_rev);
if(err_status < VI_SUCCESS )
```

```
{
check_err(iHandle, err_status);
return -1;
}
else printf ("Driver Revision:%s\nFirmware Revision:%s\n",driver_rev, instr_rev);
/* wait till a key is pressed */
while(!kbhit());
hp816x_close(iHandle);
return 0;
}


/*============================================================================*/
void check_err (ViSession vi, ViStatus errStatus)
/*============================================================================*/

{
ViInt32 inst_err;
ViChar err_message[256];

if(VI_SUCCESS > errStatus)
{
/*check if an instrument error occured*/
if(hp816x_INSTR_ERROR_DETECTED == errStatus)
{
/*get a readable message*/
hp816x_error_query(vi, &inst_err, err_message);
printf("Instrument Error : %lx, %s\n",inst_err, err_message);
}
else /*query the driver to get a readable message*/
{
hp816x_error_message(vi, errStatus, err_message);
printf("Driver Error :%lx, %s\n", errStatus, err_message);
}

}

return;
}
```

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Creating and Running Visual BASIC Applications

Typically Visual Basic (VB) programs are created as event-driven applications. The code representing the main body of the program is stored in a .FRM file and executes when a VB form receives an event (i.e. a button click). The example that follows is a step-by-step list of instructions on how to create a program using Microsoft Visual Basic Version 5.0. The name of the program is Project1. When the user clicks the Run button the program opens a GP-IB connection to the instrument, queries the instruments ID string, prints the string, closes the GP-IB connection and terminates. It is assumed that the VISA I/O configuration has already been successfully completed.

See Also:

Visual BASIC Data Types

Visual BASIC Important Notes for Programmers

To create a new project:

Select *File* | *New Project Workspace*
Click *OK*

Add the hp816x.bas file to the project::

Select *File* | *Add File*
Browse to and select the *hp816x.bas* file
Click *Open*

Change the focus to Form1 and insert a button on the form:

Click on *Form1*
Click on the command button on the toolbox
With the mouse point over Form1 hold down the right mouse button a drag to create a rectangle where the button will be placed.
Release the right mouse button and a command button will be placed on Form1

Change the name on the button:

Click on the command button just created.
Change the focus to *Properties-Form1*, with *Command1 CommandButton* visible in the drop down list box
Edit the name of the caption to be Run
Press <*Enter*> to enter the change

Open the event handler for Command1

Change the focus to the project window
Highlight *Form1*
Click *View Code*
Select *Command1* as the Object

Insert the following code between the Private Sub Command1_Click() and End Sub statements:

```
Dim lReturn As Long
Dim iHandle As Long
Dim szResponse As String *256
Dim lLength As Long

' open session, no verification, no reset
lReturn = hp816x_init("GPIB::20::INSTR", VI_FALSE, VI_FALSE, iHandle)
If (lReturn <> VI_SUCCESS) Then
 ' connection failed
 Print "Connection Failed!"
 Return
End If
' query instrument ID string
lLength = 256
lReturn = hp816x_cmdString_Q(iHandle, "*IDN?", lLength, szResponse)
Print "The instrument is: ", szResponse

' close session
lReturn = hp816x_close(iHandle)
Print "Program complete."
```

Save the project

Select *File* | *Save Project*
Navigate to the desire directory to save the project in.
Click *Save* to save the form.
Click *Save* to save the project.

Run the program:

Select *Run* | *Start*

Execute the command handler:

Click *Run*

The results will be printed on Form1.

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Visual BASIC Data Types

The Visual BASIC data types for VXI*plug&play* functions are contained in the driver's Visual BASIC header (.BAS) file (e.g. hp816x.bas). In a standard this file is located in the directory:

c:\vxipnp\win95\include for Win95

c:\vxipnp\win98\include for Win98

c:\vxipnp\winnt\include for WinNT

c:\Program Files\VXIpnp\WinNT\include for Win2000

c:\Program Files\IVI Foundation\VISA\WinNT\include for WinXP

The VISA types and their Visual BASIC equivalents are shown below. You can also determine a function's data types by selecting the Visual BASIC header file from the Project window and then selecting View Code. The data types by function are located in the 'Function Declarations' section of the header file.

VISA Type Visual BASIC Type Declaration

ViBoolean

ViPBoolean

ViBoolean[ ] Dim As Integer

ViInt16

ViPInt16

ViInt16[ ]

ViInt32

ViPInt32

ViInt32[ ] Dim ... As Long

ViSession

ViPSession

ViStatus

ViReal64

ViPReal64 Dim ... As Double

ViReal64[ ]

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Visual BASIC Important Notes for Programmers

When programming in Visual BASIC note the following:

6. When passing strings to Keysight Technologies 816x VXI*plug&play* instrument driver functions the strings must be allocated with a fixed length. For example to allocate a 256 character string use the statement:

   Dim szMyString As String *256

7. When passing arrays to Keysight Technologies 816x VXI*plug&play* instrument driver functions pass the array by reference by specifying the first element in the array. For example:

   lMyArray(1)

8. Visual BASIC defaults to passing parameters by reference. The ByVal keyword is used with numeric types to specify that the parameter is to be passed by value. The ByVal keyword is also used with Visual BASIC String parameters to indicate that the string should be converted to a C string that is then passed by reference.

. Visual BASIC uses signed integers only.

10. For boolean operations that affect program control, use the VISA VI_TRUE and VI_FALSE conditions rather than the True/False states of the Visual BASIC boolean data type. VI_TRUE and VI_FALSE are available with the VISATYPE.BAS header file.

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Getting Started with Keysight VEE

Keysight Visual Engineering Environment (Keysight VEE) is a visual programming language optimized for instrument control applications. To develop programs in Keysight VEE, you connect graphical "objects" instead of writing lines of code. These programs resemble easy-to-understand block diagrams with lines.

Keysight VEE allows you to leverage your investment in textual languages by integrating with languages such as C, C++, Visual Basic, FORTRAN, Pascal and BASIC. Keysight VEE is used in a wide variety of applications such as test and measurement, data acquisition and monitoring and control. Keysight VEE controls GP-IB, VXI, USB, Serial, GPIO, PC Plug-in and LAN instruments directly over the interfaces or by using instrument drivers.

VEE version 3.2 through VEE 6.0 support drivers that comply with the Windows 95/98, or Windows NT VXI*plug&play* specification version 3.0 or later. VEE version 7.0 does not support HP-UX or Windows 95. VEE version 8.0 supports Windows XP or Windows 2000, but not Windows 98 or Windows NT. VEE version 8.5 supports Windows Vista, Windows 2000 and Windows XP. In addition, VEE versions 3.2 and greater support the graphical Function Panel interface in VXI*plug&play* drivers, which provides a function tree of the hierarchy of the driver. In addition VEE automatically calls the "initialize" and "close" functions and performs automatic error checking. Keysight VEE gives you an easy-to-use high-level interface to VXI*plug&play* drivers while also allowing you low-level control. Our customers say Keysight VEE is the easiest way to call VXI*plug&play* drivers.

The 32-bit Keysight Technologies 816x VXI*plug&play* instrument driver can be used with VEE 3.2 and above.

To access the functions of the Keysight Technologies 816x VXI*plug&play* instrument driver from within VEE, consult the manual Using VXI*plug&play* Drivers with VEE, which comes with VEE 3.2 or the VEE on-line help.

VEE also supports RS-232 port for interfacing with the instrument. The following needs to be performed before one can use the RS-232 port from the VEE 4.0

In IO menu you have Instrument Manager and within that you have Add Button to help you add a new instrument.Double clicking the add button presents you with a Device Configuration screen wherein you are requested the following

Name of instrument - you can key in any name which you like,

Interface - Select GP-IB even when you wish to use the serial port,

Address of the instrument - you can key in any number as you will only be using one of the serial ports,

Gateway - should read This host.

The *Advanced I/O Config* button in this screen will allow you to select the Keysight Technologies 816x VXI*plug&play* instrument driver from a drop down list provided the driver is installed properly. You have to specify the address of the instrument as ASRLx if you are planning to use the COMx port in the machine.You can also select whether *Reset* and *Instrument Name Check* should be performed whenever VEE opens the instrument for interaction, from this screen.

You need to save this configuration, after getting back to the Instrument Manager Screen by clicking *OK* buttons, for VEE to remember the configuration.

For more specific and up-to-date instructions using newer versions of VEE, please consult the VEE on-line help.

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Getting Started with LabView

The 32-bit Keysight Technologies 816x VXI*plug&play* instrument driver can be used with LabVIEW 4.0 and above. LabVIEW 4.0 is a 32-bit version of LabVIEW which runs on Windows 95/98 and Windows NT.

To access the functions of the Keysight Technologies 816x VXI*plug&play* instrument driver from within LabVIEW 4.0, select File from the main menu, then select the *Convert CVI FP File* submenu item. In the file selection dialog box which appears, select *hp816x.fp* and click on the OK button. LabVIEW will create a series of VI's, one per driver function. It will create a file called *hp816x.llb* which contains these VI's. This library of VI's can then be accessed like any other VI library in LabVIEW.

With newer versions of LabView, the wrapper for the driver can be generated with the **Instrument Driver Import Wizard** available from National Instruments.

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Getting Started with LabWindows

You must use the 32-bit version of the Keysight Technologies 816x VXI*plug&play* instrument driver with LabVIEW 4.0.

The 32-bit Keysight Technologies 816x VXI*plug&play* instrument driver can be used with LabWindows 4.0 and above. LabWindows 4.0 is a 32-bit version of LabWindows which runs on Windows 95/98 and Windows NT.

To access the functions of the Keysight Technologies 816x VXI*plug&play* instrument driver from within LabWindows, select *Instrument* from the main menu, and then select the *LOAD...* submenu item. In the file selection dialog box which appears, select *hp816x.fp* and click on the OK button. LabWindows will load the function panel and instrument driver. The driver now appears as a selection on the Instrument menu, and can be treated like any LabWindows driver.

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Application Functions

These functions combine multiple SCPI commands into a single, functional operation. They are designed to allow quick and easy access to common instrument command sequences for coordinating two or more instruments, especially a tunable laser with one or more power meters for spectral measurements.

These application functions allow you to perform one of the following applications:

A Lambda Scan - a spectrum analysis operation where a single 8164A/B, 8163A/B or 8166A/B mainframe, with a back-loadable or 819x0-series compact Tunable Laser module and one or more Power Meters modules installed, performs a wavelength sweep and the Tunable Laser module and Power Meters are coordinated with each other. If more than one tunable laser is installed in the mainframe, only the laser with the lowest slot number (e.g. Slot 0 for the back-loadable laser) can be used.

A Multi Frame Lambda Scan - a spectrum analysis operation where an back-loadable tunable laser or 819x0-series compact tunable laser module in one 8164A/B, 8163A/B or 8166A/B mainframe performs a wavelength sweep and is coordinated with Power Meters that are installed in the same or other mainframes. The stand-alone N7744A and N7745A Optical Power Meter instruments are also supported in this version. These additional mainframes or instruments must be connected to the GPIB bus or other interface and have their Input Trigger Connector connected to the Output Trigger Connector of the mainframe with the tunable laser. If more than one tunable laser is installed in the mainframe, only the laser with the lowest slot number (e.g. Slot 0 for the back-loadable laser) can be used. If two lasers are present in separate mainframes, the mainframe to be used for the laser can be selected by the program.

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Equally Spaced Datapoints

A linear interpolation is performed on all wavelength and power data for the Lambda Scan Application and is optional for the Multi Frame Lambda Scan Application.

The advantage of spacing all measurements equally is that presenting results through use of a spreadsheet is greatly simplified. The operation returns one wavelength array and a power array for each power meter channel.



The disadvantage of using equally spaced datapoints is that the linear interpolation is analogous to the use of a low pass filter. The figure above shows two curves: the original curve, in red, as measured directly by a Power Meter and the interpolated curve, in green, which is the result of the linear interpolation. Interpolation will always tend to produce a smoother curve by rounding off any peaks in the curve.

Equally Spaced Datapoints is enabled by default for the Multi Frame Lambda Scan Application but you may choose not to use it, see the Equally Spaced Datapoints (hp816x_returnEquidistantData) function.

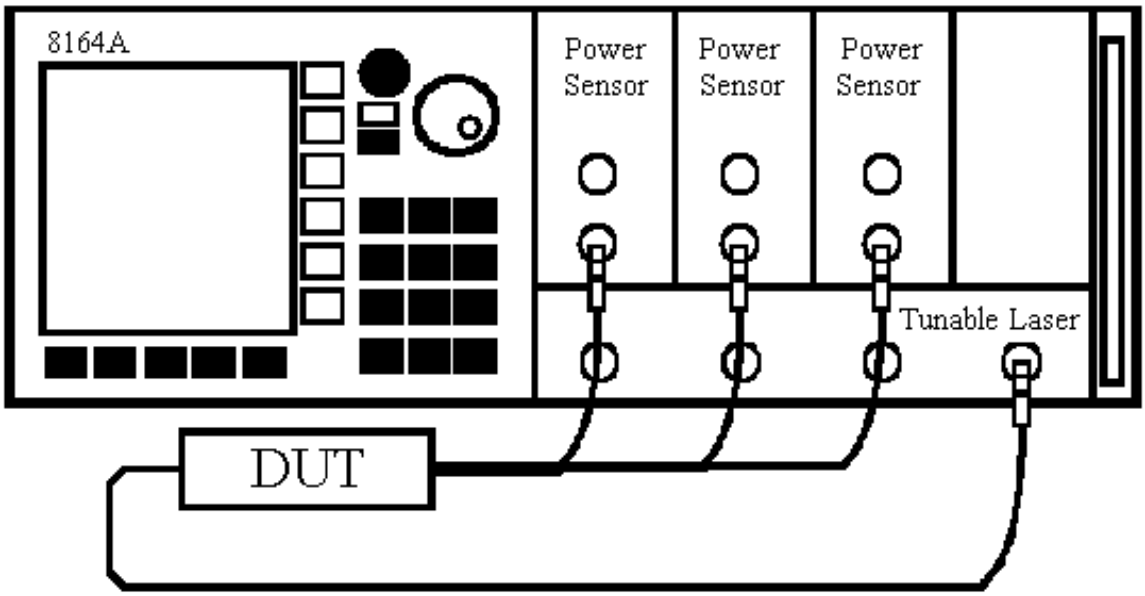
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Lambda Scan



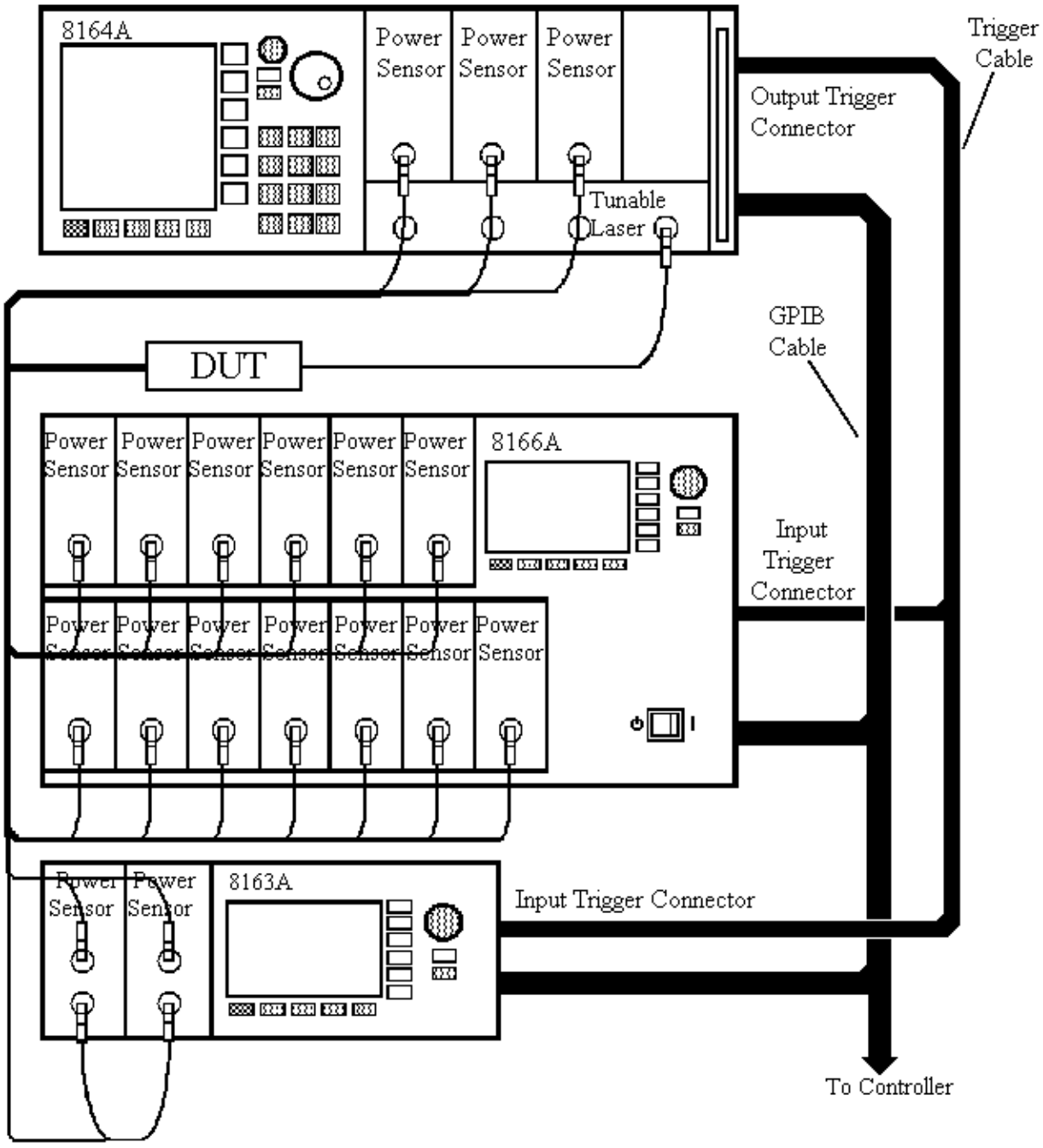
A Lambda Scan Operation performs a wavelength sweep where the Tunable Laser module and Power Sensors installed in the 8164A/B Lightwave Measurement System are coordinated with each other. When an 819x0-series compact tunable laser is used, the modules may also be installed in an 8163A/B or 8166A/B mainframe.

Equally Spaced Datapoints is enabled as part of the Lambda Scan operation.

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Multi Frame Lambda Scan



A Multi Frame Lambda Scan Operation coordinates a Lambda Scan for multiple mainframe instruments (up to 150 mainframes) including the stand-alone N774xA optical power meter instruments. All mainframes must be connected to the GPIB, LAN, or USB interface and have their Input Trigger Connector connected to the Output Trigger Connector of the

8164A/B Lightwave Measurement System mainframe containing the tunable laser. If an 819x0-series tunable laser is used, then it may also be installed in an 8163A/B or 8166A/B mainframe.

Equally Spaced Datapoints is enabled by default for the Multi Frame Lambda Scan Application but you may choose not to use it, see the Equally Spaced Datapoints (hp816x_returnEquidistantData) function.

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Example Programs

To install example programs when you install the Keysight Technologies 816x VXI*plug&play* Instrument Driver:

1. Choose Custom setup from the Setup Type menu.
2. Select Example Files from the Select Components menu.
3. Select one or more of the following programming environments and complete the installation:
   - Visual Basic
   - VISA
   - VEE
   - LabView
4. If you have chosen the default directory, the examples will be installed in the following directories respectively:
   - <Hard Drive>:\vxipnp\winnt\hp816x\Examples\Vb
   - <Hard Drive>:\vxipnp\winnt\hp816x\Visa\VC
   - <Hard Drive>:\vxipnp\winnt\hp816x\Examples\VEE
   - <Hard Drive>:\vxipnp\winnt\hp816x\Examples\LabView

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hp816x_init

Syntax:

ViStatus _VI_FUNC hp816x_init(ViRsrc *resourceName*, ViBoolean *IDQuery*, ViBoolean *reset*, ViPSession *ihandle*);

The Initialize function (hp816x_init) connects the instrument to the VISA driver and initializes VISA driver resources.

The function identifies the instrument and sets up an internal parameter structure that reads data from the instrument, for example, the current configuration.

Changing the instrument configuration without calling the Initialize function (hp816x_init) again will lead to unpredictable results. For example, inserting a different module changes the instrument configuration, so the driver may not recognise the new module if you do not call the Initialize function again.

Parameter	Description
resourceName	The Resource Name represents the connection type and address (ASRL or GPIB). For example,"GPIB0::20::INSTR", where GPIB0 identifies the GPIB board 0 (installed in your computer), 20 is the mainframe's GPIB address, and INSTR is a suffix needed by the VISA driver. If you use serial interface, make sure the parameters of the instrument match the COM ports parameter. Use the control panel (icon PORTS) to adjust the serial interface parameters. **Data Type:** ViRsrc **Input/Output:** IN
IDQuery	Use the ID Query control, a boolean input, to perform an identification check, by choosing On (VI_TRUE). This instrument series ignores this parameter. The identification check is always executed. **Data Type:** ViBoolean **Input/Output:** IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

reset

Use the Reset control, a boolean input, to reset the instrument, by choosing On (VI_TRUE).

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

ihandle

This output, *ihandle*, returns the session handle. A new session handle is created every time the mainframe is initialized.

Data Type: ViPSession

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_getHandle

Syntax:

ViStatus _Status hp816x_getHandle(ViSession *IHandle*, ViSession * *Instrument_Handle*);

The Get Session Handle function (hp816x_getHandle) returns the handle for the hp816x driver session.

This function allows, for example, Keysight VEE users to pass the session handle to other functions, such as those provided by the Keysight Photonic Foundation Library (pfl).

Parameter	Description
IHandle	IHandle is the session handle returned from the Initialize Mainframe function (hp816x_init). A new session handle is created every time the mainframe is initialized. **For LabView Programmers:** This input terminal must be connected to the session handle, *ihandle*, that is returned from the Initialize Mainframe function (hp816x_init). **Data Type:** ViSession **Input/Output:** IN
Instrument_Handle	Returns the session handle. Use this handle as an input parameter in subsequent calls. Passed by reference **Data Type:** ViSession * **Input/Output:** OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine

"hp816x_error_message"

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hp816x_revision_Q

Syntax:

ViStatus _VI_FUNC hp816x_revision_Q(ViSession *ihandle*, ViChar*driverRevision*[ ], ViChar*firmwareRevision*[ ]);

The Revision Query function (hp816x_revision_Q) returns the revision number of the instrument driver and of the instrument firmware.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
driverRevision	Driver Revision returns the revision number of the instrument driver. **Data Type:** ViChar[] **Input/Output:** OUT
firmwareRevision	Firmware Revision returns the revision number of the instrument firmware. **Data Type:** ViChar[] **Input/Output:** OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_self_test

Syntax:

ViStatus _VI_FUNC hp816x_self_test(ViSession *ihandle*, ViPInt16 *self-testResult*, ViChar*self-testMessage*[ ]);

The Self Test function (hp816x_self_test) makes the instrument perform a self-test on the mainframe and all slots. The final result is returned as Self-Test Message.

If the self-test fails, the instrument returns an error code as the Self-Test Result. Look up the *TST? command in your instrument's Programming Guide for an explanation of the Self-Test Result.

No further commands are allowed while the test is running. After the self-test the instrument is returned to the setting that was active at the time the self-test query was processed.

Remark: This function is equivalent to the Mainframe Self-Test function (hp816x_mainframeSelftest).

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
self-testResult	Self-Test Result returns the error code, this is an integer. A value of zero indicates no errors. Look up the *TST? command in your instrument's Programming Guide for an explanation of the Self-Test Result. **Data Type:** ViPInt16 **Input/Output:** OUT

self-testMessage

Self-Test Message returns the self-test status message string. The maximum message length is 256 characters.

Data Type: ViChar[]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_close

Syntax:

ViStatus _VI_FUNC hp816x_close(ViSession *ihandle*);

The Close (hp816x_close) function terminates the software connection to the instrument and deallocates system resources.

It is advisable to use this function when the program has finished using the instrument.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_timeOut

Syntax:

ViStatus _VI_FUNC hp816x_timeOut(ViSession *ihandle*, ViInt32 *timeout*);

The Timeout function (hp816x_timeOut) sets the driver timeout value in milliseconds (ms). This value specifies the time a VXiPnP driver function will wait till a timeout error occurs.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
timeout	This input specifies the time in milliseconds a function will wait before a timeout error is flagged. **Data Type:** ViInt32 **Input/Output:** IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_timeOut_Q

Syntax:

ViStatus _VI_FUNC hp816x_timeOut_Q(ViSession *ihandle*, ViPInt32 *timeout*);

The Query Timeout function (hp816x_timeOut_Q) returns the timeout value for driver I/O transactions in milliseconds. Use the Timeout function to set the timeout value. The default value is 20 seconds (20000 milliseconds).

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
timeout	Timeout returns the minimum timeout period that the driver is set to in milliseconds (ms). **Data Type:** ViPInt32 **Input/Output:** OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_getInstrumentId_Q

Syntax:

ViStatus _VI_FUNC hp816x_getInstrumentId_Q(ViString *busAddress*, ViChar*IDNString*[ ]);

The Get Instrument ID function (hp816x_getInstrumentId_Q) returns the IDN string of an instrument.

For example, the IDN string might include the instrument's manufacturer, model, serial number, and firmware revision.

Parameter	Description
busAddress	Use Bus Address to enter the bus address of the instrument. **Data Type:** ViString **Input/Output:** IN
IDNString	IDN String returns the IDN string of the intrument. **Data Type:** ViChar[] **Input/Output:** OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_listVisa_Q

Syntax:

ViStatus _VI_FUNC hp816x_listVisa_Q(ViInt32 *interface*, ViBoolean *select*, ViPInt32 *numberOfDevices*, ViChar*listofAddresses*[ ]);

The List Visa Addresses (hp816x_listVisa_Q) function returns a comma separated list of VISA interface addresses.

You can use these addresses to initialize the hp816x VXIplug&play Instrument Driver.

Parameter	Description
interface	Use the Interface control to select one of following: GPIB (hp816x_INTF_GPIB), the function returns a list of the GPIB addresses of all instruments that are connected to the controller. These addresses are returned in the form "GBIBxx::YY::INSTR", where xx is the board number and YY is the GPIB address. Serial (hp816x_INTF_ASRL), the function returns a list of the Serial addresses. These addresses are returned in the form "ASRLxx::INSTR", where xx is the serial port number. All (hp816x_INTF_ALL), the function returns a list of all interface types available for visa, that is, VXI, GPIB and Serial addresses. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_INTF_GPIB	0	GPIB
hp816x_INTF_ASRL	1	Serial
hp816x_INTF_ALL	2	All

select

Use the Select control to choose to return the following addresses:

816x (hp816x_SEL_816X), only 816x series instruments or

All (hp816x_SEL_ALL), all instruments connected to the controller.

Remark:

If you select 816x, an instrument ID query (*IDN) is sent to each instrument. This ensures that the instrument is attached and powered on.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
hp816x_SEL_816X	1	816x
hp816x_SEL_ALL	0	All

numberOfDevices

Number of Devices returns the number of VISA addresses found.

Data Type: ViPInt32

Input/Output: OUT

listofAddresses

List of Addresses returns a list of VISA addresses separated by commas.

Data Type: ViChar[]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine

"hp816x_error_message"

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hp816x_setBaudrate

Syntax:

ViStatus _VI_FUNC hp816x_setBaudrate(ViRsrc *interfaceIdentifier*, ViUInt32 *baudrate*);

The Set Baudrate function (hp816x_setBaudrate) sets the baudrate of the instrument serial port.

Remark: This function does not needs an instrument handle because getting a handle implies an init call, which requires communication to be running. This function cannot be used with VEE.

Parameter	Description
interfaceIdentifier	Use the Interface Identifier control to specify the communication port for which the baudrate should be set. The string you pass must be of the form ASRLx::INSTR where x is the port number (1,..). **Data Type:** ViRsrc **Input/Output:** IN
baudrate	Use the Baudrate control to choose a baudrate. **Data Type:** ViUInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_BAUD_9600	0	9600
hp816x_BAUD_19200	1	19200
hp816x_BAUD_38400	2	38400

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_dcl

Syntax:

ViStatus _VI_FUNC hp816x_dcl(ViSession *ihandle*);

The Device Clear function (hp816x_dcl) clears the following:

1. The Instrument Error Queue

2. The Standard Event Status Register

3. The Status Byte

See the your instrument's Programming Guide for further information. This function performs the same purpose as the *CLS GPIB command.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_error_query

Syntax:

ViStatus _VI_FUNC hp816x_error_query(ViSession *ihandle*, ViPInt32 *instrumentErrorCode*, ViChar*errorMessage*[ ]);

The Instrument Error Query function (hp816x_error_query) returns the error number and corresponding error message from the error queue of the instrument. This type of error is caused by the instrument or one of the installed modules.

See your instrument's Programming Guide for a listing of the instrument error numbers and messages.

Instrument errors can occur because you have placed the instrument in a bad state. For example, if you send an invalid sequence of coupled commands.

Automatic error detection can be accomplished by using the Automatic Error Detection function (**hp816x_errorQueryDetect**).

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
instrumentErrorCode	Error Code returns the instrument error code. See your instrument's Programming Guide for a listing of the instrument error numbers and messages. **Data Type:** ViPInt32 **Input/Output:** OUT

errorMessage

Error Message returns an error message string from the instrument.

Data Type: ViChar[]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_error_message

Syntax:

ViStatus _VI_FUNC hp816x_error_message(ViSession *ihandle*, ViStatus *errorCode*, ViString *errorMessage*);

The Error Message function (hp816x_error_message) transforms the error code returned from any instrument driver to a user-readable string.

Connect the Status output of any function to the Error Code input of this function to get information about any instrument or driver errors.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
errorCode	Use Error Code to input an error code. If Status for any function returns an error code other than 0 (VI_SUCCESS), input this value as the Error Code. **Data Type:** ViStatus **Input/Output:** IN
errorMessage	Error Message returns the error message string.

Data Type: ViString

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_errorQueryDetect

Syntax:

ViStatus _VI_FUNC hp816x_errorQueryDetect(ViSession *ihandle*, ViBoolean *automaticErrorDetection*);

The Automatic Error Detection function (hp816x_errorQueryDetect) checks for instrument errors.

After each command the Error Status Register is read and checked for errors.

If you want to increase execution speed, it is recommended to turn Automatic Error Dectection OFF. After an application is tested, instrument errors are less likely.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
automaticErrorDetection	Switch Automatic Error Detection to On (VI_TRUE) to automatically check for instrument error. Switch Automatic Error Detection to Off (VI_FALSE) to disable this function. **Data Type:** ViBoolean **Input/Output:** IN **Values:** Name Value Description

VI_TRUE	(1)	On
VI_FALSE	(0)	Off

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_dbmToWatt

Syntax:

ViStatus _VI_FUNC hp816x_dbmToWatt(ViSession *ihandle*, ViReal64 *dbm*, ViPReal64 *watt*);

The dBm to Watt function (hp816x_dBmToWatt) converts a power value in dBm into Watts.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
dbm	Use dBm to input a power value in dBm. **Data Type:** ViReal64 **Input/Output:** IN
watt	Watt returns a power value in Watts (W) that has been converted from dBm. **Data Type:** ViPReal64 **Input/Output:** OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_wattToDBm

Syntax:

ViStatus _VI_FUNC hp816x_WattToDBm(ViSession *ihandle*, ViPReal64 *watt,* ViReal64 *dbm*);

The Watt to dBm function (hp816x_WattToDBm) converts a power value in Watt into dBm.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
watt	Use Watt to input a power value in Watts (W) **Data Type:** ViPReal64 **Input/Output:** IN
dbm	dBm returns a power value in dBm that has been converted from Watts. **Data Type:** ViReal64 **Input/Output:** OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_setDate

Syntax:

ViStatus _VI_FUNC hp816x_setDate(ViSession *ihandle*, ViInt32 *year*, ViInt32 *month*, ViInt32 *day*);

The Set Date function (hp816x_setDate) sets the current date of the instrument.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
year	Use Year to set the current year of the instrument. **Data Type:** ViInt32 **Input/Output:** IN
month	Use Month to set the current month of the instrument. **Data Type:** ViInt32 **Input/Output:** IN
day	

Use Day to set the current day of the instrument.

Data Type: ViInt32
Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_getDate

Syntax:

ViStatus _VI_FUNC hp816x_getDate(ViSession *ihandle*, ViPInt32 *year*, ViPInt32 *month*, ViPInt32 *day*);

The Get Date function (hp816x_getDate) returns the current date of the instrument.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
year	Year returns the current year from the instrument. **Data Type:** ViPInt32 **Input/Output:** OUT
month	Month returns the current month from the instrument. **Data Type:** ViPInt32 **Input/Output:** OUT
day	

Day returns the current day from the instrument.

Data Type: ViPInt32
Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_setTime

Syntax:

ViStatus _VI_FUNC hp816x_setTime(ViSession *ihandle*, ViInt32 *hour*, ViInt32 *minute*, ViInt32 *second*);

The Set Time function (hp816x_setTime) sets the current time of the instrument.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
hour	Use Hour to set the current hour of the instrument. **Data Type:** ViInt32 **Input/Output:** IN
minute	Use Minute to set the current minute of the instrument. **Data Type:** ViInt32 **Input/Output:** IN
second	

Use Second to set the current second of the instrument.

Data Type: ViInt32

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_getTime

Syntax:

ViStatus _VI_FUNC hp816x_getTime(ViSession *ihandle*, ViPInt32 *hour*, ViPInt32 *minute*, ViPInt32 *second*);

The Get Time function (hp816x_getTime) returns the current time of the instrument.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
hour	Hour returns the current hour of the instrument. **Data Type:** ViPInt32 **Input/Output:** OUT
minute	Minute returns the current minute of the instrument. **Data Type:** ViPInt32 **Input/Output:** OUT
second	

Second returns the current second of the instrument.

Data Type: ViPInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_cls

Syntax:

ViStatus _VI_FUNC hp816x_cls(ViSession *ihandle*);

The Clear Error Queue function (hp816x_cls) clears the instrument's error queue. This function performs the same function as the GPIB command, "*CLS".

See your instrument's Programming Guide for more information.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_forceTransaction

Syntax:

ViStatus _VI_FUNC hp816x_forceTransaction(ViSession *ihandle*, ViBoolean *forceTransaction*);

The Force Transaction function (hp816x_forceTransaction) enables you to increase execution speed.

If Force Transaction is turned Off, a command is only transmitted to the instrument, if a difference between an actual parameter and a previous setting is noticed.

If Force Transaction is turned On, each parameter setting is transmitted to the instrument, independent of previous settings.

Turning the Force Transaction function Off causes the driver to run faster.

REMARKS The default value is ON. If Force Transactions is turned Off, manual adjustments are not noticed by the driver and can falsify results.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
forceTransaction	If Force Transaction is turned Off (VI_FALSE), a command is only transmitted to the instrument, if a difference between an actual parameter and a previous setting is noticed. If Force Transaction is turned On (VI_TRUE), each setting of parameters is transmitted to the instrument, independent of previous setting.

REMARKS The default value is On. If Force Transaction is turned Off, manual adjustments are not noticed by the driver and can falsify results.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_driverLogg

Syntax:

ViStatus _VI_FUNC hp816x_driverLogg(ViSession *ihandle*, ViString *filename*, ViBoolean *logging*, ViBoolean *includeReplies*);

The Enable Driver Logging function (hp816x_driverLogg) logs all commands sent to the instrument to a file in text format, with the exception of data aquisition commands.

If the file exists, the output will be appended.

This log file can become very large, especially if the instrument is polled for some reason.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
filename	Use Filename to enter the name of the file to which all of the commands will be logged. **Data Type:** ViString **Input/Output:** IN
logging	Switch Logging to Enable (VI_TRUE) to enable driver logging. Switch Logging to Disable (VI_FALSE) to disable

driver logging.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Enable
VI_FALSE	(0)	Disable

includeReplies

Use the Include Replies control to decide if the response from the instrument should be also logged.

Switch the Include Replies control to Yes (VI_TRUE) if you want to log the response to commands.

Switch the Include Replies control to No (VI_FALSE) if you do not want to log the response to commands.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Yes
VI_FALSE	(0)	No

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_cmd

Syntax:

ViStatus _VI_FUNC hp816x_cmd(ViSession *ihandle*, ViChar*commandString*[ ]);

The Send String Command function (hp816x_cmd) sends a GPIB command (conforming to the SCPI standard) to the instrument. It does not look for a response.

See your instrument's Programming Guide for information on GPIB commands.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
commandString	Use Command String to enter a string containing a GPIB command. See your instrument's Programming Guide for information on GPIB commands. **Data Type:** ViChar[] **Input/Output:** IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_cmdString_Q

Syntax:

ViStatus _VI_FUNC hp816x_cmdString_Q(ViSession *ihandle*, ViChar*inputQuery*[ ], Integer *stringSize*, ViChar*result*[ ]);

The Send String Query function (hp816x_cmdString_Q) sends a GPIB query string to the instrument and waits for a response which must be a string (character data).

A minimum of one byte will be read from the instrument and the string will be NULL terminated (so its length can be found with strlen, the String Length C function).

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
inputQuery	Use Input Query to enter a query string containing a GPIB query. This GPIB query should return a character string. All GPIB queries have a question mark, ?, as their last character. See your instrument's Programming Guide for information on GPIB commands. **Data Type:** ViChar[] **Input/Output:** IN
stringSize	String Size sets the size of the character array

that the function uses to store the Result.

Data Type: Integer
Input/Output: IN

Values:

String Size_MIN	2
String Size_MAX	32767

result

Result returns a string from the instrument, that is, the reply to your Input Query.

See your instrument's Programming Guide for information on GPIB commands.

Data Type: ViChar[]
Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_cmdInt

Syntax:

ViStatus _VI_FUNC hp816x_cmdInt(ViSession *ihandle*, ViChar*integerCommand*[ ], ViInt32 *integerValue*);

The Send Integer Command function (hp816x_cmdInt) sends an instrument command which requires one integer parameter.

See your instrument's Programming Guide for information on GPIB commands.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
integerCommand	Use Integer Command to enter a command string containing a GPIB command. The command string should require a single integer parameter as the Integer Value. For example, the GPIB command to set the display brightness to 65 is: disp:brig 65 In this case, enter disp:brig as Integer Command. See your instrument's Programming Guide for information on GPIB commands. **Data Type:** ViChar[]

Input/Output: IN

integerValue

Use Integer Value to enter the integer value you wish to send to the instrument.

For example, the GPIB command to set the display brightness to 65 is: disp:brig 65

In this case, enter 65 as Integer Value.

See your instrument's Programming Guide for information on GPIB commands.

Data Type: ViInt32

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_cmdInt32_Q

Syntax:

ViStatus _VI_FUNC hp816x_cmdInt32_Q(ViSession *ihandle*, ViChar*integerQuery*[ ], ViPInt32 *integerResult*);

The Send Integer Query function (hp816x_cmdInt32_Q) sends a GPIB query string to the instrument and waits for a response that must be representable as a ViInt32. A non-numeric instrument response returns zero as the result.

See your instrument's Programming Guide for information on GPIB commands.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
integerQuery	Use Integer Query to enter a query string containing a GPIB query. This GPIB query should return a ViInt32. All GPIB queries have a question mark, ?, as their last character. See your instrument's Programming Guide for information on GPIB commands. **Data Type:** ViChar[] **Input/Output:** IN
integerResult	Integer Result returns a ViInt32 value from the

instrument, that is, the reply to your Integer Query.

A non-numeric instrument response returns zero as the result.

See your instrument's Programming Guide for information on GPIB commands.

Data Type: ViPInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_cmdReal

Syntax:

ViStatus _VI_FUNC hp816x_cmdReal(ViSession *ihandle*, ViString *realCommand*, ViReal64 *realValue*);

The Send Real Command function (hp816x_cmdReal) sends an instrument command which requires one real parameter.

See your instrument's Programming Guide for information on GPIB commands.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
realCommand	Use Real Command to a enter a string containing a GPIB command. The command string should require a single real parameter. For example, the GPIB command to set the averaging time is: sens1:pow:atime 1.234s In this case, enter sens1:pow:atime as Real Command. This command sets the averaging time of the Power Sensor in slot 1. See your instrument's Programming Guide for information on GPIB commands. **Data Type:** ViString **Input/Output:** IN

realValue

Use Real Value to enter the real value you want to send to the instrument.

For example, the GPIB command to set the averaging time is: sens1:pow:atime 1.234s

In this case, enter 1.234 as Real Value. This sets the averaging time of the Power Sensor in slot 1 to 1.234 seconds.

You cannot use any units, default units are presumed by the function.

See your instrument's Programming Guide for information on GPIB commands.

Data Type: ViReal64

Input/Output: IN

Values:

Real Value_MIN	-1.000000E+300
Real Value_MAX	1.000000E+300

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_cmdReal64_Q

Syntax:

ViStatus _VI_FUNC hp816x_cmdReal64_Q(ViSession *ihandle*, ViString *realQuery*, ViPReal64 *realResult*);

The Send Real Query function (hp816x_cmdReal64_Q) sends a GPIB query string to the instrument and waits for a response that must be representable as a ViReal64.

See your instrument's Programming Guide for information on GPIB commands.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
realQuery	Use Real Query to enter a query string containing a GPIB query. This GPIB query should return a ViReal64 value. The command string cannot exceed 256 bytes in length. All GPIB queries have a question mark, ?, as their last character. See your instrument's Programming Guide for information on GPIB commands. **Data Type:** ViString **Input/Output:** IN
realResult	Real Result returns a ViReal64 value from the

instrument, that is, the reply to your Real Query.

A non-numeric instrument response returns zero as the result.

See your instrument's Programming Guide for information on GPIB commands.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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Information for N77 Series Instruments

This version of the 816x VXIplug&play instrument driver has been extended to support the N773xA, N774xA optical power meter and the N775xA and N776xA Optical Attenuator instruments. These are stand-alone instruments that do not require installation in an 816x mainframe. The SCPI command set has been designed for almost complete programming compatibility with the modular instruments of the 816x family. Please be aware of the following details and limitations, when using this driver.

- The N77-series instruments can be controlled via USB, LAN or GPIB interfaces. Especially for transferring large data volumes, the USB and usually LAN interfaces provide faster performance.
- The identification of the individual ports of these multiport instruments in the SCPI and hp816x functions corresponds to the slot number. So for example the ports of the N7745A can be addressed with the slot numbers 1 to 8. The channel number in the commands and functions is not used and can generally be omitted or set to Channel 1 (index 0).
- The N771x-series and 81950A tunable lasers are not supported by the 816x VXIplug&play driver.
- The N773x-series optical switches are supported by the 816x VXIplug&play functions for switches and relevant functions for the mainframes.
- The N774x-series power meters are supported by the 816x VXIplug&play functions for power meters, relevant functions for the mainframes like for standard trigger configuration, and the Multi Frame Lambda Scan application functions.
- The N776x-series optical attenuators are supported by the 816x VXIplug&play functions for attenuators and the relevant functions for the mainframes.
- The N775x-series optical attenuators and power meters are supported by the 816x VXIplug&play functions for attenuators and the relevant functions for the mainframes and the basic functions for the power meters, excluding the advanced functions using data power logging. The choices of averaging time is also more limited for these power meters.
- The averaging time parameter for the power meters is limited in the 816x VXIplug&play functions to a limited set of discrete values, similar to the modular 816x power meters. This set includes: 1, 2, 5, 10, 25, 100, 200, and 500 ìs, 1, 2, 5, 10, 20, 50, 100, 200, 500 ms, and 1, 2, 5, and 10s.
- The maximum number of logging data points that can be acquired is limited by the 816x VXIplug&play data acquisition functions to 100,000 per port.
- The shortest averaging time used by the Multi Frame Lambda Scan application is 25 ìs.
- For the Multi Frame Lambda Scan application, the N774xA instruments can be included simply by registering the instrumet as a mainframe,after the first

mainframe with the tunable laser has been registered.

Further details to available functions are found under the categories below:

Utility Functions

Mainframe Specific Functions

Power Sensor Specific Functions

Applications

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hp816x_reset

Syntax:

ViStatus _VI_FUNC hp816x_reset(ViSession *ihandle*);

The Reset function (hp816x_reset) resets the instrument and the GPIB bus to their initial state. The parameters of the instrument and its modules are set to their default values.

This function is identical to the "*RST" GPIB command. See your instrument's Programming Guide for information on GPIB commands.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_preset

Syntax:

ViStatus _VI_FUNC hp816x_preset(ViSession *ihandle*);

The Preset function (hp816x_preset) resets the instrument to its initial state. The parameters of the instrument and its modules are set to their default values.

This function is the same as pressing the preset hardkey on the instrument's front panel.

This function is also identical to the ":syst:preset" GPIB command. See your instrument's Programming Guide for information on GPIB commands.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_mainframeSelftest

Syntax:

ViStatus _VI_FUNC hp816x_mainframeSelftest(ViSession *ihandle*, ViPInt16 *self-testResult*, ViChar*self-testMessage*[ ]);

The Mainframe Self-Test function (hp816x_mainframeSelftest) makes the instrument perform a self-test on the mainframe and all slots. The final result is returned as Self-Test Message.

If the self-test fails, the instrument returns an error code as the Self-Test Result. Look up the *TST? command in your instrument's Programming Guide for an explanation of the Self-Test Result.

No further commands are allowed while the test is running. After the self-test the instrument is returned to the setting that was active at the time the self-test query was processed.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
self-testResult	Self-Test Result returns the error code, this is an integer. A value of zero indicates no errors. Look up the *TST? command in your instrument's Programming Guide for an explanation of the Self-Test Result. **Data Type:** ViPInt16 **Input/Output:** OUT

self-testMessage

Self-Test Message returns the self-test status message string. The maximum message length is 256 characters.

Data Type: ViChar[]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_moduleSelftest

Syntax:

ViStatus _VI_FUNC hp816x_moduleSelftest(ViSession *ihandle*, ViInt32 *slottoTest*, ViPInt16 *result*, ViChar*self-testMessage*[ ]);

The Module Self-Test (hp816x_moduleSelftest) returns the latest selftest results for a module inserted in a given slot.

NOTE: This command does not perform a selftest. Use the Mainframe Self-Test function (**hp816x_mainframeSelftest**) to make the instrument perform a self-test on the mainframe and all slots.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
slottoTest	Use the Slot to Test control to choose an installed module. Data Type: ViInt32 Input/Output: IN **Values:**

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3

hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

result

Result returns the error code, this is an integer. A value of zero indicates no errors.

See your instrument's Programming Guide for a listing of the instrument error numbers and messages.

Data Type: ViPInt16

Input/Output: OUT

self-testMessage

Self-Test Message returns the self-test status message string. The maximum message length is 256 characters.

Data Type: ViChar[]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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E0615, Jun 2015



hp816x_revision_query

Syntax:

ViStatus _VI_FUNC hp816x_revision_query(ViSession *ihandle*, ViChar*driverRevision*[ ], ViChar*firmwareRevision*[ ]);

The revision query function (hp816x_revision_query) returns the revision number of the instrument driver and of the instrument firmware.

Remark: This function is equivalent to the Revision Query function (hp816x_revision_Q).

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
driverRevision	Driver Revision returns the revision number of the instrument driver. **Data Type:** ViChar[] **Input/Output:** OUT
firmwareRevision	Firmware Revision returns the revision number of the instrument firmware. **Data Type:** ViChar[] **Input/Output:** OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_opc_Q

Syntax:

ViStatus _VI_FUNC hp816x_opc_Q(ViSession *ihandle*, ViPBoolean *operationComplete*);

The Operation Complete function (hp816x_opc_Q) sends the "*OPC?" GPIB command to the instrument and returns 1 when all pending operations are complete.

See your instrument's Programming Guide for more information.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
operationComplete	Operation Complete returns VIStatus when all pending operations are complete. **Data Type:** ViPBoolean **Input/Output:** OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_lockUnlockInstument

Syntax:

ViStatus _VI_FUNC hp816x_lockUnlockInstument(ViSession *ihandle*, ViBoolean *softlock*, ViString *password*);

The Lock-Unlock Instrument function (hp816x_lockUnlockInstrument) locks or unlocks the software lock for high power source modules inserted in the mainframe.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
softlock	If you choose Lock (VI_TRUE), high power source modules will be locked. If you choose Unlock (VI_FALSE), high power source modules will be unlocked. **Data Type:** ViBoolean **Input/Output:** IN **Values:** Name / Value / Description **VI_TRUE** (1) Lock **VI_FALSE** (0) Unlock
password	

Use Password to enter the instrument password which is used to lock or unlock high power source modules.

1234 is your instrument's default password.

Data Type: ViString

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_getLockState

Syntax:

ViStatus _VI_FUNC hp816x_getLockState(ViSession *Ihandle*, ViBoolean * *Soft_Lock*, ViBoolean * *Remote_Interlock*);

The Get Lock State function (hp816x_getLockState) returns the status of the mainframe's softlock and its remote interlock. If either the softlock or the remote interlock is locked, or both are locked, high-power laser sources cannot be turned on.

Parameter	Description
Ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
Soft_Lock	Returns the status of the softlock. If 0 (VI_FALSE) is returned, the softlock is unlocked. If 1 (VI_TRUE) is returned, the softlock is locked. Set the status of the softlock using the Lock-Unlock Instrument function (**hp816x_lockUnlockInstrument**). **Data Type:** ViBoolean **Input/Output:** OUT

Remote_Interock

Returns the status of the remote interlock.

If 0 (VI_FALSE) is returned, the remote interlock is unlocked.

If 1 (VI_TRUE) is returned, the remote interlock is locked.

Data Type: ViBoolean

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_enableDisableDisplay

Syntax:

ViStatus _VI_FUNC hp816x_enableDisableDisplay(ViSession *ihandle*, ViBoolean *display*);

The Enable-Disable Display function (hp816x_enableDisableDisplay) turns the display of the instrument On or Off.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
display	If you select On (VI_TRUE), the Enable Disable Display function turns the display of the instrument On. If you select Off (VI_FALSE), the Enable Disable Display function turns the display of the instrument Off. **Data Type:** ViBoolean **Input/Output:** IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_getSlotInformation_Q

Syntax:

ViStatus _VI_FUNC hp816x_getSlotInformation_Q(ViSession *ihandle*, ViInt32 *arraySize*, ViInt32*slotInformation*[ ]);

The Get Slot Information function (hp816x_getSlotInformation_Q) returns information about installed modules. The Slot Information can be used to write applications that are independent of the physical location of the inserted modules.

REMARKS You must assure that the Array Size is properly allocated.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
arraySize	Use Array Size to set the size of the memory buffer needed for the function. Depending on your mainframe, you should set Array Size to the following integer value: 8163A/B: 3 8164A/B: 5 8166A/B: 18 **Data Type:** ViInt32 **Input/Output:** IN

slotInformation

Slot Information is both an input and a output.

As an input, allocate Slot Information as a one-dimensional integer array of the following dimensions:

8163A/B: 1 x 3

8164A/B: 1 x 5

8166A/B: 1 x 18

As a return, Slot Information returns a one dimensional array. Each component consists of the array component number and the module type number. The array component number corresponds to the slot number, starting at 0 and ending at the highest slot numbe r. The following numbers define the module type number:

0 The slot is empty hp816x_UNDEF

1 A single-channel Power Sensor hp816x_SINGLE_SENSOR

2 A dual-channel Power Sensor hp816x_DUAL_SENSOR

3 A single Laser Source hp816x_FIXED_SINGLE_SOURCE

4 A dual-wavelength Laser Source hp816x_FIXED_DUAL_SOURCE

5 A Tunable Laser module hp816x_TUNABLE_SOURCE

6 A Return Loss Module hp816x_RETURN_LOSS

7 A Return Loss Combo Module hp816x_RETURN_LOSS_COMBO

For example, if the result below is returned for an 8163A/B, it means that a single-channel Power Sensor Module is installed in slot 1 and a single-

channel Laser Source Module is installed in slot 2
(slot 0 does not exist for the 8163A/B):

0: 0

1: 1

2: 3

Data Type: ViInt32[]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_getModuleStatus_Q

Syntax:

ViStatus _VI_FUNC hp816x_getModuleStatus_Q(ViSession *ihandle*, ViPBoolean *statusSummary*, ViInt32*moduleStatusArray*[ ], ViPInt32 *maxMessageLength*);

The Get Module Status function (hp816x_getModuleStatus_Q) checks the questionable status condition register of all modules inserted into the instrument. The questionable status condition is the current status of all modules. See your instrument's Programming Guide for information on the instrument's Status System.

Questionable status information is information about the following module errors:

If excessive power is set by the user, if zeroing failed, if temperature is out of range, if laser protection is switched on, if the module has not settled, if the module is out of specifications, if ARA is recommended, or if the duty cycle is out of range.

REMARKS You must assure that the Array Size is properly allocated.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
statusSummary	If Status Summary equals 0, there are no current questionable events for all inserted modules. If Status Summary equals 1, there is at least one current questionable event for one inserted module. **Data Type:** ViPBoolean

Input/Output: OUT

moduleStatusArray

Module Status Array is both an input and an output.

As an input, Module Status Array accepts a one-dimensional integer array that determines the size of the Module Status Array output. Allocate a one-dimensional integer array of the following dimensions for your mainframe:

8163A/B: 1 x 3

8164A/B: 1 x 5

8166A/B: 1 x 18

As an output, Module Status Array returns a one dimensional array. Each array component consists of the questionable status condition value for the corresponding slot. The array includes a result for Slot 0 even if Slot 0 does not exist, that is, for the 8163A/B and 8166A/B.

The questionable status condition value is the current questionable status of the slot. The value is continuously updated. It is not changed by being read.

A questionable status of 0 means that are no current questionable events for the slot.

For any other value, set the Questionable Status Input of the Convert Questionable Status function (**hp816x_convertQuestionableStatus_Q**) to the displayed value to get a message about the questionable status.

Data Type: ViInt32[]

Input/Output: OUT

maxMessageLength

Max Message Length returns the maximum number of bytes needed to translate the status information; use this value or a bigger one to

allocate a message buffer for the Message Status function (hp816x_messageStatus_Q).

Data Type: ViPInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_convertQuestionableStatus_Q

Syntax:

ViStatus _VI_FUNC hp816x_convertQuestionableStatus_Q(ViSession *ihandle*, ViInt32 *questionableStatusInput*, ViChar*message*[ ]);

The Convert Questionable Status function (hp816x_convertQuestionableStatus_Q) converts a questionable Status word to a message string describing the questionable event.

The questionable status word for all slots is retuned by the Module Status Array output of the Get Module Status function (**hp816x_getModuleStatus_Q**).

You must allocate a message buffer to assure that the array Message is connected to the Max Message Length output returned by the Get Module Status function (**hp816x_getModuleStatus_Q**). This allocates a size to the message buffer for the Message output.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
questionableStatusInput	The Questionable Status Input should be connected to the questionable status value for a single slot. The questionable status condition values for all slots are returned by the Module Status Array output of the Get Module Status function (**hp816x_getModuleStatus_Q**). Questionable Status Input only reads integer values.

Data Type: ViInt32

Input/Output: IN

message

Status Message is both an input and an output.

The Status Message input is a message buffer. Allocate a message buffer with the same size as the Max Message Length output of the Get Module Status function (**hp816x_getModuleStatus_Q**).

As an output, Status Message returns the translated status message of the questionable Status Input value. Because each bit set in the status word marks a condition, the explanations for each condition are separated with a comma.

Data Type: ViChar[]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_standardTriggerConfiguration

Syntax:

ViStatus _VI_FUNC hp816x_standardTriggerConfiguration(ViSession *ihandle*, ViInt32 *triggerConfiguration*, ViUInt32 *nodeAInputConfig*, ViUInt32 *nodeBInputConfig*, ViUInt32 *outputMatrixConfiguration*);

The Standard Trigger Configuration (hp816x_standardTriggerConfiguration) function configures the mainframe trigger system.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
triggerConfiguration	The Trigger Configuration control allows you to choose the trigger configuration. The following describe the triggering modes: Disabled (hp816x_TRIG_DISABLED), which you should choose if you do not want to use triggering. Default (hp816x_TRIG_DEFAULT), which you should choose if you want to enable the trigger connectors. Passthrough (hp816x_TRIG_PASSTHROUGH), which you should choose if you want an input trigger to automatically generate an output trigger. This allows you to trigger another instrument almost simultaneously.

Loopback (hp816x_TRIG_LOOPBACK), which you should choose if you want an output trigger to automatically generate an input trigger. For example, using this mode, you could trigger each step of a wavelength sweep with just one externally generated input tr igger.

Custom (hp816x_TRIG_CUSTOM), which you should choose if you want to configure the trigger function yourself.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_TRIG_DISABLED	0	Disabled
hp816x_TRIG_DEFAULT	1	Default
hp816x_TRIG_PASSTHROUGH	2	Passthrough
hp816x_TRIG_LOOPBACK	3	Loopback
hp816x_TRIG_CUSTOM	4	Custom

nodeAInputConfig

If you choose Custom as the Trigger Configuration, Node A Input Config must contain a 32-bit value that configures Node A of the trigger system.

Use your instrument's Programming Guide to obtain information about the bits to set.

See the Trigger Input Configuration function (**hp816x_nodeInputConfiguration**) for information on setting this value.

Data Type: ViUInt32

Input/Output: IN

nodeBInputConfig

If you choose Custom as the Trigger Configuration, Node B Input Config must contain a 32-bit value that configures Node B of the trigger system.

Use your instrument's Programming Guide to

obtain information about the bits to set.

See the Trigger Input Configuration function (**hp816x_nodeInputConfiguration**) for information on setting this value.

Data Type: ViUInt32

Input/Output: IN

outputMatrixConfiguration

If you choose Custom as the Trigger Configuration, Output Matrix Configuration must contain a 32-bit value that sets the node interconnection.

Use your instrument's Programming Guide to obtain information about the bits to set.

See the Trigger Output Configuration function (**hp816x_trigOutConfiguration**) for information on setting this value.

Data Type: ViUInt32

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_standardTriggerConfiguration_Q

Syntax:

ViStatus _VI_FUNC hp816x_standardTriggerConfiguration_Q(ViSession *ihandle*, ViInt32 *triggerConfiguration*, ViUInt32 *nodeAInputConfig*, ViUInt32 *nodeBInputConfig*, ViUInt32 *outputMatrixConfiguration*);

The Standard Trigger Configuration Query (hp816x_standardTriggerConfiguration_Q) function queries the mainframe trigger system configuration.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
triggerConfiguration	The Trigger Configuration returns the trigger configuration. The following describe the triggering modes: Disabled (hp816x_TRIG_DISABLED), if the trigger input and output are disabled. Default (hp816x_TRIG_DEFAULT), when input triggers are accepted and output triggers are sent over the trigger ports. Passthrough (hp816x_TRIG_PASSTHROUGH), when an input trigger automatically generates an output trigger. This allows you to trigger another instrument almost simultaneously. Loopback (hp816x_TRIG_LOOPBACK), when an output trigger automatically generates an input

trigger. For example, using this mode, you could trigger each step of a wavelength sweep with just one externally generated input tr igger. Loopback is not supported on the N774xA, N775xA and N776xA and acts the same as Passthrough.

Custom (hp816x_TRIG_CUSTOM), when a custom configuration of the trigger function is set.

Data Type: ViInt32

Input/Output: OUT

Values:

Name	Value	Description
hp816x_TRIG_DISABLED	0	Disabled
hp816x_TRIG_DEFAULT	1	Default
hp816x_TRIG_PASSTHROUGH	2	Passthrough
hp816x_TRIG_LOOPBACK	3	Loopback
hp816x_TRIG_CUSTOM	4	Custom

nodeAInputConfig

If the Trigger Configuration is Custom, the Node A Input Config returns a 32-bit value that configures Node A of the trigger system.

Use your instrument's Programming Guide to obtain information about the bits to set.

See the Trigger Input Configuration function (**hp816x_nodeInputConfiguration**) for information on setting this value.

Data Type: ViUInt32

Input/Output: OUT

nodeBInputConfig

If the Trigger Configuration is Custom, the Node B Input Config returns a 32-bit value that configures Node B of the trigger system.

Use your instrument's Programming Guide to obtain information about the bits to set.

See the Trigger Input Configuration function

(**hp816x_nodeInputConfiguration**) for information on setting this value.

Data Type: ViUInt32

Input/Output: OUT

outputMatrixConfiguration

If the Trigger Configuration is Custom, the Output Matrix Configuration returns a 32-bit value that sets the node interconnection.

Use your instrument's Programming Guide to obtain information about the bits to set.

See the Trigger Output Configuration function (**hp816x_trigOutConfiguration**) for information on setting this value.

Data Type: ViUInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_generateTrigger

Syntax:

ViStatus _VI_FUNC hp816x_generateTrigger(ViSession *ihandle*, ViBoolean *triggerAt*);

The Generate Trigger function (hp816x_generateTrigger) triggers the selected trigger node.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
triggerAt	Use the Trigger At control to trigger Node A (hp816x_NODE_A) or Node B (hp816x_NODE_B). **Data Type:** ViBoolean **Input/Output:** IN

Values:

Name	Value	Description
hp816x_NODE_A	0	Node A
hp816x_NODE_B	1	Node B

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_nodeInputConfiguration

Syntax:

ViStatus _VI_FUNC hp816x_nodeInputConfiguration(ViBoolean *connectionFunctionNodeA*, **ViBoolean** *BNCTriggerConnector*, **ViBoolean** *nodeBTriggerOutput*, **ViBoolean** *slot0*, **ViBoolean** *slot1*, **ViBoolean** *slot2*, **ViBoolean** *slot3*, **ViBoolean** *slot4*, **ViBoolean** *connectionFunctionNodeB*, **ViBoolean** *BNCTriggerConnector*, **ViBoolean** *nodeATriggerOutput*, **ViBoolean** *slot0*, **ViBoolean** *slot1*, **ViBoolean** *slot2*, **ViBoolean** *slot3*, **ViBoolean** *slot4*, **ViUInt32** *nodeAInputConfiguration*, **ViUInt32** *nodeBInputConfiguration*);

The Trigger Input Configuration function (hp816x_nodeInputConfiguration) helps you use the "Custom" Trigger Configuration of the Standard Trigger Configuration function (**hp816x_standardTriggerConfiguration**).

The "Custom" configuration requires you to enter three 32-bit values as input parameters. This function calculates two of these 32-values, Node A Input Config and Node B Input Config.

You should use the Outputs of this function as Inputs for the Standard Trigger Configuration function (**hp816x_standardTriggerConfiguration**).

Parameter	Description
connectionFunctionNodeA	Connection Function Node A determines if the trigger inputs to Node A use OR logic or AND logic. If you choose OR logic, any active trigger input can trigger Node A. If you choose AND logic, all active trigger inputs must be fulfilled before Node A is triggered. **Data Type:** ViBoolean **Input/Output:** IN **Values:** Name / Value / Description **VI_TRUE** / (1) / AND **VI_FALSE** / (0) / OR
BNCTriggerConnector	

BNC Trigger Connector determines if the Input Trigger Connector can trigger Node A.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Active
VI_FALSE	(0)	Inactive

nodeBTriggerOutput

Node B Trigger Output determines if a trigger at Node B can trigger Node A.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Active
VI_FALSE	(0)	Inactive

slot0

Slot 0 determines if a trigger event at slot 0 can trigger Node A.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Active
VI_FALSE	(0)	Inactive

slot1

Slot 1 determines if a trigger event at slot 1 can trigger Node A.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Active
VI_FALSE	(0)	Inactive

slot2

Slot 2 determines if a trigger event at slot 2 can trigger Node A.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Active
VI_FALSE	(0)	Inactive

slot3

Slot 3 determines if a trigger event at slot 3 can trigger Node A.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Active
VI_FALSE	(0)	Inactive

slot4

Slot 4 determines if a trigger event at slot 4 can trigger Node A.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Active
VI_FALSE	(0)	Inactive

connectionFunctionNodeB

Connection Function Node B determines if the trigger inputs to Node B use OR logic or AND logic.

If you choose OR logic, any active trigger input can trigger Node B.

If you choose AND logic, all active trigger inputs must be fulfilled before Node B is triggered.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	AND
VI_FALSE	(0)	OR

BNCTriggerConnector

BNC Trigger Connector determines if the Input Trigger Connector can trigger Node B.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Active
VI_FALSE	(0)	Inactive

nodeATriggerOutput

Node A Trigger Output determines if a trigger at Node B can trigger Node B.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Active
VI_FALSE	(0)	Inactive

slot0

Slot 0 determines if a trigger event at slot 0 can trigger Node B.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Active
VI_FALSE	(0)	Inactive

slot1

Slot 1 determines if a trigger event at slot 1 can trigger Node B.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Active
VI_FALSE	(0)	Inactive

slot2

Slot 2 determines if a trigger event at slot 2 can trigger Node B.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Active

VI_FALSE	(0)	Inactive

slot3

Slot 3 determines if a trigger event at slot 3 can trigger Node B.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Active
VI_FALSE	(0)	Inactive

slot4

Slot 4 determines if a trigger event at slot 4 can trigger Node B.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Active
VI_FALSE	(0)	Inactive

nodeAInputConfiguration

Node A Input Configuration returns the trigger configuration of Node A as a 32-bit value.

Pass this value as a parameter to the function Standard Trigger Configuration function (**hp816x_standardTriggerConfiguration**).

Data Type: ViUInt32

Input/Output: OUT

nodeBInputConfiguration

Node B Input Configuration returns the trigger configuration of Node B as a 32-bit value.

Pass this value as a parameter to the function Standard Trigger Configuration function (**hp816x_standardTriggerConfiguration**).

Data Type: ViUInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_trigOutConfiguration

Syntax:

ViStatus _VI_FUNC hp816x_trigOutConfiguration(ViBoolean *nodeswitchedtoBNCOutput*, ViBoolean *slot0*, ViBoolean *slot1*, ViBoolean *slot2*, ViBoolean *slot3*, ViBoolean *slot4*, ViUInt32 *outputMatrixConfiguration*);

The Trigger Output Configuration function (hp816x_nodeOutputConfiguration) helps you use the "Custom" Trigger Configuration of the Standard Trigger Configuration function (**hp816x_standardTriggerConfiguration**).

The "Custom" configuration requires you to enter three 32-bit values as input parameters. This function calculates one of these 32-values, Output Matrix Configuration.

You should use the Output of this function as an Input for the Standard Trigger Configuration function (**hp816x_standardTriggerConfiguration**).

Parameter | **Description**

nodeswitchedtoBNCOutput

Node switched to BNC Output determines whether Node A or Node B can generate a trigger at the Output Trigger Connector.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Node B
VI_FALSE	(0)	Node A

slot0

Slot 0 determines which node can trigger slot 0.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Node B
VI_FALSE	(0)	Node A

slot1

Slot 1 determines which node can trigger slot 1.

Data Type: ViBoolean
Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Node B
VI_FALSE	(0)	Node A

slot2

Slot 2 determines which node can trigger slot 2.

Data Type: ViBoolean
Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Node B
VI_FALSE	(0)	Node A

slot3

Slot 3 determines which node can trigger slot 3.

Data Type: ViBoolean
Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Node B
VI_FALSE	(0)	Node A

slot4

Slot 4 determines which node can trigger slot 4.

Data Type: ViBoolean
Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Node B
VI_FALSE	(0)	Node A

outputMatrixConfiguration

Output Matrix Configuration returns the output matrix configuration as a 32-bit value.

Pass this value as a parameter to the function Standard Trigger Configuration function (**hp816x_standardTriggerConfiguration**).

Data Type: ViUInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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Information for N77 Series Instruments

This version of the 816x VXIplug&play instrument driver has been extended to support the N773xA, N774xA optical power meter and the N775xA and N776xA Optical Attenuator instruments. These are stand-alone instruments that do not require installation in an 816x mainframe. The SCPI command set has been designed for almost complete programming compatibility with the modular instruments of the 816x family. Please be aware of the following details and limitations, when using this driver.

- The N77-series instruments can be controlled via USB, LAN or GPIB interfaces. Especially for transferring large data volumes, the USB and usually LAN interfaces provide faster performance.
- The identification of the individual ports of these multiport instruments in the SCPI and hp816x functions corresponds to the slot number. So for example the ports of the N7745A can be addressed with the slot numbers 1 to 8. The channel number in the commands and functions is not used and can generally be omitted or set to Channel 1 (index 0).
- The N771x-series and 81950A tunable lasers are not supported by the 816x VXIplug&play driver.
- The N773x-series optical switches are supported by the 816x VXIplug&play functions for switches and relevant functions for the mainframes.
- The N774x-series power meters are supported by the 816x VXIplug&play functions for power meters, relevant functions for the mainframes like for standard trigger configuration, and the Multi Frame Lambda Scan application functions.
- The N776x-series optical attenuators are supported by the 816x VXIplug&play functions for attenuators and the relevant functions for the mainframes.
- The N775x-series optical attenuators and power meters are supported by the 816x VXIplug&play functions for attenuators and the relevant functions for the mainframes and the basic functions for the power meters, excluding the advanced functions using data power logging. The choices of averaging time is also more limited for these power meters.
- The averaging time parameter for the power meters is limited in the 816x VXIplug&play functions to a limited set of discrete values, similar to the modular 816x power meters. This set includes: 1, 2, 5, 10, 25, 100, 200, and 500 ìs, 1, 2, 5, 10, 20, 50, 100, 200, 500 ms, and 1, 2, 5, and 10s.
- The maximum number of logging data points that can be acquired is limited by the 816x VXIplug&play data acquisition functions to 100,000 per port.
- The shortest averaging time used by the Multi Frame Lambda Scan application is 25 ìs.
- For the Multi Frame Lambda Scan application, the N774xA instruments can be included simply by registering the instrumet as a mainframe,after the first

mainframe with the tunable laser has been registered.

Further details to available functions are found under the categories below:

Utility Functions

Mainframe Specific Functions

Power Sensor Specific Functions

Applications

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hp816x_set_ATT_attenuation

Syntax:

ViStatus _VI_FUNC hp816x_set_ATT_attenuation(ViSession *ihandle*, ViInt32 *ATTSlot*, ViInt32 *selection*, ViReal64 *attenuation*, ViBoolean *waitforCompletion*);

Use the Set Attenuation function (hp816x_set_ATT_attenuation) to set the attenuation factor.

Adjusting attenuation also adjusts the filter attenuation:

a Filter(new) = a(new) -- a Offset

and power CALCULATED using:

P[dBm] = Pref[dBm] -- a Filter

While the filter is moving to its new position, the status line displays "Busy" (8164) or "B" (8163, 8166).

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN **Values:** Name — Value — Description

hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

selection

Sets attenuation factor to device limits or device default. If Manual Entry is activated the entry of Attenuation is used.

Data Type: ViInt32
Input/Output: IN

Values:

Name	Value	Description
hp816x_SELECT_MIN	0	Minimum
hp816x_SELECT_MAX	1	Maximum
hp816x_SELECT_DEF	2	Default
hp816x_SELECT_MANUAL	3	Manual Entry

attenuation

The attenuation factor in dB

Data Type: ViReal64
Input/Output: IN

waitforCompletion

Use this control to turn "Wait for Completion"

On(Off). If "Wait for Completion" is set to Yes(1) the function will not return until the attenuation is set. If "Wait for Completion" is set to "No"(0) the function returns immediately. In this case it is not guaranteed that the attenuation adjustment has been completed. You may use **hp816x_opc_Q** to ask for the command completion.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Yes
VI_FALSE	(0)	No

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_get_ATT_attenuation_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_attenuation_Q(ViSession *ihandle*, ViInt32 *ATTSlot*, ViPReal64 *minimum*, ViPReal64 *maximum*, ViPReal64 *default*, ViPReal64 *current*);

Use the Get Attenuation function (hp816x_get_ATT_attenuation_Q) to query, for a specified Attenuator module, the current user-set attenuation factor and its limits.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

minimum

Returns the minimum attenuation.

Data Type: ViPReal64
Input/Output: OUT

maximum

Returns the maximum attenuation.

Data Type: ViPReal64
Input/Output: OUT

default

Returns the default attenuation.

Data Type: ViPReal64
Input/Output: OUT

current

Returns the current user-set attenuation factor.

Data Type: ViPReal64
Input/Output: OUT

Return Value:

VI_SUCCESS: No error
Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine

"hp816xerror_message"

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hp816x_set_ATT_attenuationOffset

Syntax:

ViStatus _VI_FUNC hp816x_set_ATT_attenuationOffset(ViSession *ihandle*, ViInt32 *selection*, ViReal64 *offset*, ViInt32 *ATTSlot*);

Use the Set Attenuation Offset function (hp816x_set_ATT_attenuationOffset) to set the attenuation offset factor.

Remark: This function changes the range of the attenuator and recalculates its limits.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
selection	Sets the attenuation offset factor to device limits or device default. If Manual Entry is activated the Offset entry is used. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SELECT_MIN	0	Minimum
hp816x_SELECT_MAX	1	Maximum
hp816x_SELECT_DEF	2	Default
hp816x_SELECT_MANUAL	3	Manual Entry

offset	The offset factor in dB.

Data Type: ViReal64

Input/Output: IN

ATTSlot

Use the ATT Slot control to specify the slot used to host the Attenuator module.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_get_ATT_attenuationOffset_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_attenuationOffset_Q(ViSession *ihandle*, ViInt32 *ATTSlot*, ViPReal64 *minimum*, ViPReal64 *maximum*, ViPReal64 *default*, ViPReal64 *current*);

Use the Get Attenuation Offset function (hp816x_get_ATT_attenuationOffset_Q) to query, for a specified Attenuator module, the current user-set attenuation offset factor and its limits.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

minimum

Returns the minimum offset.

Data Type: ViPReal64
Input/Output: OUT

maximum

Returns the maximum offset.

Data Type: ViPReal64
Input/Output: OUT

default

Returns the default offset.

Data Type: ViPReal64
Input/Output: OUT

current

Returns the current offset.

Data Type: ViPReal64
Input/Output: OUT

Return Value:

VI_SUCCESS: No error
Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine

"hp816xerror_message"

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hp816x_set_ATT_attenuatorSpeed

Syntax:

ViStatus _VI_FUNC hp816x_set_ATT_attenuatorSpeed(ViSession *ihandle*, ViInt32 *ATTSlot*, ViInt32 *selection*, ViReal64 *speed*);

Use the Set Attenuation Speed function (hp816x_set_ATT_attenuatorSpeed) to set the filter transition speed. That is, the speed at which the filter moves from one attenuation setting to another, in dB/s

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8

hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

selection

Set filter speed to device limits or use the manual Speed entry.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_SELECT_MIN	0	Minimum
hp816x_SELECT_MAX	1	Maximum
hp816x_SELECT_DEF	2	Default
hp816x_SELECT_MANUAL	3	Manual Entry

speed

New transition speed.

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_get_ATT_attenuatorSpeed_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_attenuatorSpeed_Q(ViSession *ihandle*, ViInt32 *ATTSlot*, ViPReal64 *minimum*, ViPReal64 *maximum*, ViPReal64 *default*, ViPReal64 *current*);

Use the Get Attenuator Speed function (hp816x_get_ATT_attenuatorSpeed_Q) to query the attenuation filter's transition speed and its limits.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

minimum

Minimum transition speed.

Data Type: ViPReal64
Input/Output: OUT

maximum

Maximum transition speed.

Data Type: ViPReal64
Input/Output: OUT

default

Default transition speed.

Data Type: ViPReal64
Input/Output: OUT

current

Current transition speed.

Data Type: ViPReal64
Input/Output: OUT

Return Value:

VI_SUCCESS: No error
Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine

"hp816xerror_message"

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hp816x_ATT_displayToOffset

Syntax:

ViStatus _VI_FUNC hp816x_ATT_displayToOffset(ViSession *ihandle*, ViInt32 *ATTSlot*);

The Display To OffSet function (hp816x_ATT_displayToOffset) reads the currently displayed attenuation then sets this, as a negative value, as the attenuation offset for the specified Attenuator.

Remark: The range limits of the Attenuator are displaced by this offset value.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_set_ATT_wavelength

Syntax:

ViStatus _VI_FUNC hp816x_set_ATT_wavelength(ViSession *ihandle*, ViInt32 *ATTSlot*, ViInt32 *selection*, ViReal64 *wavelength*);

Use the Set Wavelength function (hp816x_set_ATT_wavelength) to set the wavelength used by the Attenuator module. The Attenuator module is factory calibrated for wavelength dependencies.

If a powermeter is integrated into the Attenuator module, this wavelength setting applies to both the Attenuator filter calibration and the powermeter.

If the wavelength:offset table is activated, this wavelenght setting also applies the corresonding offset.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN **Values:** Name / Value / Description **hp816x_SLOT_1** / 1 / Slot 1 **hp816x_SLOT_2** / 2 / Slot 2 **hp816x_SLOT_3** / 3 / Slot 3 **hp816x_SLOT_4** / 4 / Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

selection

Set the wavelength to device limits or uses the Wavelength entry.

Data Type: ViInt32
Input/Output: IN

Values:

Name	Value	Description
hp816x_SELECT_MIN	0	Minimum
hp816x_SELECT_MAX	1	Maximum
hp816x_SELECT_DEF	2	Default
hp816x_SELECT_MANUAL	3	Manual Entry

wavelength

Wavelength value

Data Type: ViReal64
Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_ATT_wavelength_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_wavelength_Q(ViSession *ihandle*, ViInt32 *ATTSlot*, ViPReal64 *minimum*, ViPReal64 *maximum*, ViPReal64 *default*, ViPReal64 *current*);

Use the Get Wavelength function (hp816x_get_ATT_wavelength_Q) to query the wavelength calibration for the specified Attenuator module, including its device limits.

If a powermeter is integrated into the Attenuator module, this wavelength setting applies to both the Attenuator filter calibration and the powermeter.

If the wavelength:offset table is activated, this wavelenght setting also applies the corresonding offset.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN **Values:** Name / Value / Description **hp816x_SLOT_1** / 1 / Slot 1 **hp816x_SLOT_2** / 2 / Slot 2 **hp816x_SLOT_3** / 3 / Slot 3

hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

minimum

Returns the minimum wavelength calibration.

Data Type: ViPReal64
Input/Output: OUT

maximum

Returns the maximum wavelength calibration.

Data Type: ViPReal64
Input/Output: OUT

default

Returns the default wavelength calibration.

Data Type: ViPReal64
Input/Output: OUT

current

Returns the current wavelength calibration.

Data Type: ViPReal64
Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_set_ATT_powerUnit

Syntax:

ViStatus _VI_FUNC hp816x_set_ATT_powerUnit(ViSession *ihandle*, ViInt32 *ATTSlot*, ViBoolean *unit*);

Use the Set Power Unit function (hp816x_set_ATT_powerUnit) to set the power unit used by the Attenuator module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8

hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

unit

Use this control to seletc either Watt (1) or dBm (0) as power unit.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Watt
VI_FALSE	(0)	Dbm

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_get_ATT_powerUnit_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_powerUnit_Q(ViSession *ihandle*, ViPBoolean *unit*, ViInt32 *ATTSlot*);

Use the Get Power Unit function (hp816x_get_ATT_powerUnit_Q) to query the power unit set for the specified Attenuator module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
unit	This controls returns the current power unit set for the Attenuator module hosted by *ATTSLot* **Data Type:** ViPBoolean **Input/Output:** OUT
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1

hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_set_ATT_absPowerMode

Syntax:

ViStatus _VI_FUNC hp816x_set_ATT_absPowerMode(ViSession *ihandle*, ViInt32 *ATTSlot*, ViBoolean *absolutePowerMode*);

Use the Set Absolute Power Mode function (hp816x_set_ATT_absPowerMode) to set the operation mode of the specified Attenuator module. This command exists for compatibility with the 8156A Attenuator.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8

hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

absolutePowerMode

OFF: The Attenuator module attempts to restore attenuation to the last set attenuation after bootup.

ON: The Attenuator module attempts to restore power to the last set power after bootup.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_get_ATT_absPowerMode_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_absPowerMode_Q(ViSession *ihandle*, ViInt32 *ATTSlot*, ViPBoolean *absolutePowerMode*);

Use the Get Absolute Power Mode function (hp816x_get_ATT_absPowerMode_Q) to query the current operational mode of a specified Attenuator module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8

hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

absolutePowerMode

Returns the current operational mode of the specified Attenuator module.

If 0 (VI_FALSE) is returned, the Attenuator module attempts to restore attenuation to the last set attenuation after bootup.

If 1 (VI_TRUE) is returned, the Attenuator module attempts to restore power to the last set power after bootup.

Data Type: ViPBoolean

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_set_ATT_power

Syntax:

ViStatus _VI_FUNC hp816x_set_ATT_power(ViSession *ihandle*, ViInt32 *ATTSlot*, ViBoolean *powerControl*, ViBoolean *powerUnit*, ViInt32 *selection*, ViReal64 *power*);

Use the Set Power (hp816x_set_ATT_power) to set a new power.

If you have an Attenuator module with power control, you can activate the power control mode so that the Attenuator module automatically corrects for changes to input power, so maintaining the desired output power.

If you have an Attenuator module without power control, a reference calibration must be set using hp816x_set_ATT_powerReference.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN **Values:** Name / Value / Description **hp816x_SLOT_1** / 1 / Slot 1 **hp816x_SLOT_2** / 2 / Slot 2 **hp816x_SLOT_3** / 3 / Slot 3

hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

powerControl

OFF: The filter position is not automatically corrected.

ON : Internal control loop activated. The filter position is automatically corrected to compensate for changes to input power, so maintaining the desired output power.

Remark: This control is ignored by Attenuator modules with no integrated powermeter.

Data Type: ViBoolean
Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

powerUnit

Sets the power unit.

Data Type: ViBoolean
Input/Output: IN

Values:

Name	Value	Description

VI_TRUE	(1)	Watt
VI_FALSE	(0)	Dbm

selection

Sets the through power to device limits or, if Manual Entry is selected, the value entered in the power control is used.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_SELECT_MIN	0	Minimum
hp816x_SELECT_MAX	1	Maximum
hp816x_SELECT_DEF	2	Default
hp816x_SELECT_MANUAL	3	Manual Entry

power

If selection equals hp816x_SELECT_MANUAL, sets the the desired output power.

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_get_ATT_power_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_power_Q(ViSession *ihandle*, ViInt32 *ATTSlot*, ViPBoolean *powerControl*, ViPBoolean *powerUnit*, ViPReal64 *minimum*, ViPReal64 *maximum*, ViPReal64 *default*, ViPReal64 *current*);

Use the Get Power function (hp816x_get_ATT_power_Q) to query the power values and device limits.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

powerControl

Returns the power control mode

Data Type: ViPBoolean
Input/Output: OUT

powerUnit

Returns the power unit.

If VI_FALSE (0) the power unit is dBm,
If VI_TRUE (1) the power unit is Watt.

Data Type: ViPBoolean
Input/Output: OUT

minimum

Returns the minimum power.

Data Type: ViPReal64
Input/Output: OUT

maximum

Returns the maximum power.

Data Type: ViPReal64
Input/Output: OUT

default

Returns the default power.

Data Type: ViPReal64
Input/Output: OUT

current

Returns the current power.

Data Type: ViPReal64
Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_cp_ATT_refFromExtPM

Syntax:

ViStatus _VI_FUNC hp816x_cp_ATT_refFromExtPM(ViSession *ihandle*, ViInt32 *ATTSlot*, ViInt32 *PWMSlot*, ViInt32 *channel*);

The Copy Ref. from Powermeter function (hp816x_cp_ATT_refFromExtPM) reads the power value from the reference powermeter and copies this value to the reference power of the specified Attenuator.

Remark: This function is only applies to Attenuator modules without power control.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. The Attenuator module specified by ATTSlot must be hosted by the same mainframe as the Power Meter specified by PWM Slot. **Data Type:** ViInt32 **Input/Output:** IN **Values:** Name / Value / Description **hp816x_SLOT_1** / 1 / Slot 1 **hp816x_SLOT_2** / 2 / Slot 2 **hp816x_SLOT_3** / 3 / Slot 3

hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

PWMSlot

Use the PWM Slot control to specify the slot hosting the Power Meter (a Power Sensor or Optical Head Interface Module) used to provide the external reference measurement for the Attenuator module.

The Power Meter specified by PWM Slot must be hosted by the same mainframe as the Attenuator module specified by ATTSlot.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11

hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channel

If your reference Power Meter is a dual-channel Power Sensor module, or a dual-channel Optical Head Interface module, use the Channel control to select either Channel 1 or Channel 2.

If your reference Power Meter is a single-channel Power Sensor module, or a Dual Optical Head Interface module, select Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_set_ATT_powerReference

Syntax:

ViStatus _VI_FUNC hp816x_set_ATT_powerReference(ViSession *ihandle*, ViInt32 *ATTSlot*, ViInt32 *selection*, ViReal64 *reference*);

Use the Set Power Reference function (hp816x_set_ATT_powerReference) to set the reference power value.

Remark: This function only applies to Attenuator modules without power control.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

selection

Sets the power reference to device limits or if Manual Entry selected, the value entered in the reference control is used..

Data Type: ViInt32
Input/Output: IN

Values:

Name	Value	Description
hp816x_SELECT_MIN	0	Minimum
hp816x_SELECT_MAX	1	Maximum
hp816x_SELECT_DEF	2	Default
hp816x_SELECT_MANUAL	3	Manual Entry

reference

Reference power value in dBm or Watt.

Data Type: ViReal64
Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_get_ATT_powerReference_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_powerReference_Q(ViSession *ihandle*, ViInt32 *ATTSlot*, ViPReal64 *minimum*, ViPReal64 *maximum*, ViPReal64 *default*, ViPReal64 *current*);

Use the Get Power Reference function (hp816x_get_ATT_powerReference_Q) to query the reference power and the device limits.

Remark: This function does not apply to Attenuator modules with power control.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

minimum

Minimum value for the reference power.

Data Type: ViPReal64
Input/Output: OUT

maximum

Maximum value for the reference power.

Data Type: ViPReal64
Input/Output: OUT

default

Default value for the reference power.

Data Type: ViPReal64
Input/Output: OUT

current

Current value for the reference power.

Data Type: ViPReal64
Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_set_ATT_shutterState

Syntax:

ViStatus _VI_FUNC hp816x_set_ATT_shutterState(ViSession *ihandle*, ViInt32 *ATTSlot*, ViBoolean *shutterState*);

Use the Set Shutter State function (hp816x_set_ATT_shutterState) to set the output path state.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN **Values:** Name / Value / Description **hp816x_SLOT_1** / 1 / Slot 1 **hp816x_SLOT_2** / 2 / Slot 2 **hp816x_SLOT_3** / 3 / Slot 3 **hp816x_SLOT_4** / 4 / Slot 4 **hp816x_SLOT_5** / 5 / Slot 5 **hp816x_SLOT_6** / 6 / Slot 6 **hp816x_SLOT_7** / 7 / Slot 7 **hp816x_SLOT_8** / 8 / Slot 8

hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

shutterState

Open: Enables the output; the shutter is opened
Close: Disables the output; the shutter is closed

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Open
VI_FALSE	(0)	Close

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_get_ATT_shutterState_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_shutterState_Q(ViSession *ihandle*, ViInt32 *ATTSlot*, ViPBoolean *shutterState*);

Use the Get shutter State function (hp816x_get_ATT_shutterState_Q) to query the state of output path for the specied Attenuator module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8

hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

shutterState

Returns the Attenuator module's shutter state.

If VI_TRUE (1) the shutter is open.
If VI_FALSE (0) the shutter is closed.

Data Type: ViPBoolean
Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_set_ATT_shutterAtPowerOn

Syntax:

ViStatus _VI_FUNC hp816x_set_ATT_shutterAtPowerOn(ViSession *ihandle*, ViInt32 *ATTSlot*, ViBoolean *shutterState*);

Use the Set Shutter State at Power On function (hp816x_set_ATT_shutterAtPowerOn) to set the output path state at power on.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8

hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

shutterState

Open: Enables the output at power on; the shutter is open
Closed: Disables the output at power on; the shutter is closed

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Open
VI_FALSE	(0)	Closed

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_get_ATT_shutterStateAtPowerOn_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_shutterStateAtPowerOn_Q(ViSession *ihandle*, ViInt32 *ATTSlot*, ViPBoolean *shutterState*);

Use the Get Shutter State at Power On function (hp816x_get_ATT_shutterStateAtPowerOn_Q) to query the state of the specified Attenuator module's output path at power on.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8

hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

shutterState

Returns the shutter state at power-on for the specified Attenuator module.

If VI_TRUE (1) the shutter is open at power on.
If VI_FALSE (0) the shutter is closed at power on.

Data Type: ViPBoolean
Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_get_ATT_operStatus

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_operStatus(ViSession *ihandle*, ViInt32 *ATTSlot*, ViUInt32 *value*, ViChar*conditionalInfo*[ ]);

Use the Get Operational Status function (hp816x_get_ATT_operStatus) to query the operational status register and return attenuator-specific status information. To be certain that the Attenuator module is not in an inoperable state, you should also call the Get Module Status function (hp816x_getModule Status_Q).

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

value

Returns the contents of the operational status register. Only bits 3 to 7 of the first byte contain attenuator-specific information:

Bit 3 set - zeroing

Bit 4 set - shutter opened

Bit 5 set - lambda offset active

And, if Bit5 is set (lambda offset active):

Bit6 Bit7

0 0 exact value

1 0 extrapolation below

0 1 extrapolation above

1 1 interpolation

Data Type: ViUInt32

Input/Output: OUT

conditionalInfo

Returns a comma-separated string listing the contents of the operational status register.

Remark:Make sure you have allocated sufficient space for this string.

Data Type: ViChar[]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_set_ATT_wlOffsRefPowermeter

Syntax:

ViStatus _VI_FUNC hp816x_set_ATT_wlOffsRefPowermeter(ViSession *ihandle*, ViInt32 *ATTSlot*, ViInt32 *PWMSlot*, ViInt32 *channel*);

Use the Set Wavel. Offs. Ref. PM function (hp816x_set_ATT_wlOffsRefPowermeter) to specify the external powermeter (hosted by the same mainframe as the Attenuator module) used for determine offsets for particular wavelengths. This funtion works in conjunction with the Set Wavelength Offset function when the TOREF control is active. (**hp816x_set_ATT_wavelengthOffset**)

Remark: This function only applies to Attenuator modules with power control.

Parameter

ihandle

IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized.

For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**).

Data Type: ViSession

Input/Output: IN

ATTSlot

Use the ATT Slot control to specify the slot used to host the Attenuator module.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

PWMSlot

Use the PWM Slot control to select a reference powermeter (a Power Sensor or Optical Head Interface Module).

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channel

Use the Channel control to select the channel that you want to reference.

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel control to choose between Channel 1 and Channel 2 of your module.

If you choose a single--channel Power Meter, use the Channel control to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_get_ATT_wlOffsRefPowermeter_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_wlOffsRefPowermeter_Q(ViSession *ihandle*, ViInt32 *ATTSlot*, ViPInt32 *slot*, ViPInt32 *channel*);

Use the Get Power Reference function (hp816x_get_ATT_wlOffsRefPowermeter_Q)to query the slot and channel used to host the reference powermeter.

Remark: This function only applies to Attenuator modules with power control.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN **Values:** Name / Value / Description **hp816x_SLOT_1** / 1 / Slot 1 **hp816x_SLOT_2** / 2 / Slot 2 **hp816x_SLOT_3** / 3 / Slot 3 **hp816x_SLOT_4** / 4 / Slot 4 **hp816x_SLOT_5** / 5 / Slot 5 **hp816x_SLOT_6** / 6 / Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

slot

Returns the slot number of the powermeter used as reference powermeter.

Data Type: ViPInt32

Input/Output: OUT

channel

Returns the channel of the powermeter used as the reference powermeter.

Data Type: ViPInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_set_ATT_wavelengthOffsetState

Syntax:

ViStatus _VI_FUNC hp816x_set_ATT_wavelengthOffsetState(ViSession *ihandle*, ViInt32 *ATTSlot*, ViBoolean *offsetDependant*);

Use the Set Wavelength Offset State function (hp816x_set_ATT_wavelengthOffsetState) to turn the Attenuator module's wavelength offset table on(off).

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8

hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

offsetDependant

Use this control to turn on (off) the Attenuator module's wavelength-offset table.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_get_ATT_wavelengthOffsetState_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_wavelengthOffsetState_Q(ViSession *ihandle*, ViInt32 *ATTSlot*, ViPBoolean *offsetDependancy*);

Use the Get Wavelength Offset State function (hp816x_get_ATT_wavelengthOffsetState_Q) to query whether a wavelength-offset table is being used by the specified Attenuator module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8

hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

offsetDependancy

Returns whether a wavelength-offset table is used by the specified Attenuator module.

If VI_TRUE (1) a wavelength-offset table is used.

If VI_FALSE (0) a wavelength-offset table is not used.

Data Type: ViPBoolean

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_set_ATT_wavelengthOffset

Syntax:

ViStatus _VI_FUNC hp816x_set_ATT_wavelengthOffset(ViSession *ihandle*, ViInt32 *ATTSlot*, ViReal64 *wavelength*, ViBoolean *offsetSource*, ViReal64 *offsetValue*);

Use the Set Wavelength Offset function (hp816x_set_ATT_wavelengthOffset) to set for an offset for a particular wavelength. This function generates a wavelength-offset pair entry in the Attenuator module's Wavelength Offset table.

Remark: If a wavelength-offset pair entry already exists in the Attenuator module's Wavelength Offset table, the offset for this wavelength entry is overwritten.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN **Values:** Name / Value / Description **hp816x_SLOT_1** / 1 / Slot 1 **hp816x_SLOT_2** / 2 / Slot 2 **hp816x_SLOT_3** / 3 / Slot 3 **hp816x_SLOT_4** / 4 / Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

wavelength

Use this control to enter the wavelength.

Data Type: ViReal64

Input/Output: IN

offsetSource

This control selects whether the value entered in the Offset Value input control (Offset Value), or the power reading from an external reference powermeter (TOREF), is stored.

Remark: If TOREF is selected the following conditions must be met:

1. The Attenuator module must include an integrated powermeter.

2. A reference powermeter must have been previously seleected using hp816x_set_ATT_wlOffsRefPM.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	TOREF
VI_FALSE	(0)	Offset Value

offsetValue

Use this control to enter an offset for the specified wavelength.

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_get_ATT_wavelengthOffsetIndex_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_wavelengthOffsetIndex_Q(ViSession *ihandle*, ViInt32 *ATTSlot*, ViUInt32 *tableIndex*, ViPReal64 *wavelength*, ViPReal64 *offset*);

Use the Get WL/Offset from Index function (hp816x_get_ATT_wavelengthOffsetIndex_Q) to query the wavelength offset applied for a particular wavelength offset table index for a specified Attenuator module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

tableIndex

Use this control to specify a particular index for a wavelength offset table entry. Index entries start from 1.

Data Type: ViUInt32

Input/Output: IN

wavelength

Returns the wavelength for the *tableIndex*.

Data Type: ViPReal64

Input/Output: OUT

offset

This control returns the offset for the *tableIndex*.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_get_ATT_offsetFromWavelength_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_offsetFromWavelength_Q(ViSession *ihandle*, ViInt32 *ATTSlot*, ViReal64 *wavelength*, ViPReal64 *offset*);

Use the Get Offset from Wavelength function (hp816x_get_ATT_offsetFromWavelength_Q) to query the attenuation offset applied to a particular wavelength.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**) **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8

hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

wavelength

The wavelength for which you wish to query the offset applied.

Data Type: ViReal64

Input/Output: IN

offset

The offset applied to *wavelength*

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_delete_ATT_offsetTblEntries

Syntax:

ViStatus _VI_FUNC hp816x_delete_ATT_offsetTblEntries(ViSession *ihandle*, ViInt32 *ATTSlot*, ViBoolean *noOfEntries*, ViReal64 *wavelengthOrIndex*);

Use the Delete Table Entries function (hp816x_delete_ATT_offsetTblEntries) to delete all entries, or a particular entry, from an attenuator module's wavelength-offset table.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8

hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

noOfEntries

Use the noOfEntries control to specify whether you want all wavelength-offset entries deleted (VI_TRUE) deleted, or a single wavelength-offset entry deleted (VI_FALSE).

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	All
VI_FALSE	(0)	Single

wavelengthOrIndex

Use the wavelengthOrIndex control to specify the wavelength or index entry in the wavelength-offset table that you want to delete.

If the input is less than 1, the entry corresponding to the specified wavelength is deleted.

If the input is greater or equal 1, the entry corresponding to the specified index is deleted.

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_get_ATT_offsetTable_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_offsetTable_Q(ViSession *ihandle*, ViInt32 *ATTSlot*, ViReal64*wavelengthArray*[ ], ViReal64*offsetArray*[ ]);

Use the Get Offset Table Entries function (hp816x_get_ATT_offsetTable_Q) to return the wavelength offset table of the specified Attenuator module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8

hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

wavelengthArray

Returns the wavelengths saved in the wavelength-offset table as a ViReal64 array.

Remark: Make sure that the array is large enough to hold all the entries.

Use **hp816x_get_ATT_offsetTblSize_Q** to query the number of entries that will be returned.

Data Type: ViReal64[]

Input/Output: OUT

offsetArray

Returns the offsets saved in the wavelength-offset table as a ViReal64 array.

Remark: Make sure that the array is large enough to hold all the entries.

Use **hp816x_get_ATT_offsetTblSize_Q** to query the number of entries that will be returned.

Data Type: ViReal64[]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_get_ATT_offsetTblSize_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_offsetTblSize_Q(ViSession *ihandle*, ViInt32 *ATTSlot*, ViUInt32 *currentSize*, ViUInt32 *maximumSize*);

Use the Get Table Size function (hp816x_get_ATT_offsetTblSize_Q) to query the dimensions of the wavelength-offset table for the specified Attenuator module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8

hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

currentSize

This control returns the number of entries currently saved in the wavelength-offset table.

Data Type: ViUInt32

Input/Output: OUT

maximumSize

This parameter returns the maximum capacity of the wavelength-offset table .

Data Type: ViUInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_read_ATT_outputPower

Syntax:

ViStatus _VI_FUNC hp816x_read_ATT_outputPower(ViSession *ihandle*,ViPInt32 *ATT_Slot*, Real64 * *Output_Power*[ ]);

The Read ATT Output Power function (hp816x_read_ATT_outputPower) forces the specified attenuator module's builtin powermeter to begin a new measurement cycle. The evaluated output power value is not returned until the measurement averaging time has past.

Remark: Applies to Attenuator modules with Power Control only.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATT_Slot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Output_Power

Returns the current power in W or dBm.

Data Type: ViPReal64
Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_fetch_ATT_outputPower

Syntax:

ViStatus _VI_FUNC hp816x_fetch_ATT_outputPower(ViSession *ihandle*,ViPInt32 *ATT_Slot*, Real64 * *Output_Power*[ ]);

The Fetch ATT Output Power function (hp816x_fetch_ATT_outputPower) forces the specified attenuator module's builtin powermeter to return a power value without averaging power measurements over the averaging time period.

If this function is called more than once within an averaging cycle, the returned values will be identical.

Remark: Applies to Attenuator modules with Power Control only.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATT_Slot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN **Values:** Name / Value / Description **hp816x_SLOT_1** / 1 / Slot 1 **hp816x_SLOT_2** / 2 / Slot 2 **hp816x_SLOT_3** / 3 / Slot 3

hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Output_Power

Returns the current power in W or dBm.

Data Type: ViPReal64
Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_ATT_powerOffset

Syntax:

ViStatus _VI_FUNC hp816x_set_ATT_powerOffset(ViSession *ihandle*, ViInt32 *ATTSlot*, ViInt32 *selection*, ViReal64 *offset*);

Use the Set Power Offset function (hp816x_set_ATT_powerOffset) to set the offset factor applied to the power measured at the attenuator module. This factor does not affect the Attenuator module's filter, or its power output.

Pset = Pout -Poffset

Remark: This function only applies to Attenuator modules with power control.

Parameter | **Description**

ihandle

IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized.

For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**).

Data Type: ViSession

Input/Output: IN

ATTSlot

Use the ATT Slot control to specify the slot used to host the Attenuator module.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

selection

Sets the offset reference to device limits or if Manual Entry selected, the value entered in the offset control is used.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_SELECT_MIN	0	Minimum
hp816x_SELECT_MAX	1	Maximum
hp816x_SELECT_DEF	2	Default
hp816x_SELECT_MANUAL	3	Manual Entry

offset

Use this control to specify the power offset applied.

Remark: Applying a power offset changes the apparent output power limits.

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_get_ATT_powerOffset_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_powerOffset_Q(ViSession *ihandle*, ViInt32 *ATTSlot*, ViPReal64 *minimum*, ViPReal64 *maximum*, ViPReal64 *default*, ViPReal64 *current*);

Use the Get Power Offset function (hp816x_get_ATT_powerOffset_Q) to query the power offset limits and the current power offset setting for the specified Attenuator module.

Pset = Pout -Poffset

Remark: This function only applies to Attenuator modules with power control.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

minimum

Minimum value of power offset.

Remark: This value is changed whenever the power offset is changed.

Data Type: ViPReal64
Input/Output: OUT

maximum

Maximum value of power offset.

Remark: This value is changed whenever the power offset is changed.

Data Type: ViPReal64
Input/Output: OUT

default

Default value of power offset.

Data Type: ViPReal64
Input/Output: OUT

current

Current value of power offset.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_set_ATT_pwrOffsRefPM

Syntax:

ViStatus _VI_FUNC hp816x_set_ATT_pwrOffsRefPM(ViSession *ihandle*, ViInt32 *ATTSlot*, ViInt32 *PWMSlot*, ViInt32 *PWMChannel*);

Use the Set Power Offset Ref. PM function (hp816x_set_ATT_pwrOffsRefPM) to calculate a power offset factor using the builtin Powermeter and an external reference powermeter. The difference P offset = P out -- P ext is copied to the Power offset parameter.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

PWMSlot

Use the PWM Slot control to specify the slot used to host a reference powermeter (a Power Sensor or Optical Head Interface Module).

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

PWMChannel

Use the Channel control to specify the channel that you want to reference.

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel control to choose between Channel 1 and Channel 2 of your module.

If you choose a single--channel Power Meter, specify Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_set_ATT_offsByRefPM

Syntax:

ViStatus _VI_FUNC hp816x_set_ATT_offsByRefPM(ViSession *ihandle*, ViInt32 *ATTSlot*, ViInt32 *PWMSlot*, ViInt32 *PWMChannel*);

Use the Set Att. Off. by EXT PM Loop function (hp816x_set_ATT_offsByRefPM) to calculate an attenuation offset factor by subtracting the power value measured by an external reference powermeter from the power value measured by the Attenuator module's integrated powermeter. The result is stored as the attenuation offset factor.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

PWMSlot

Use the PWM Slot control to specify the slot used to host the external reference powermeter (a Power Sensor or Optical Head Interface Module).

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

PWMChannel

Use the Channel control to specify the channel that you want to reference.

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel control to choose between Channel 1 and Channel 2 of your module.

If you choose a single--channel Power Meter, specify Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_set_ATT_controlLoopState

Syntax:

ViStatus _VI_FUNC hp816x_set_ATT_controlLoopState(ViSession *ihandle*, ViInt32 *ATTSlot*, ViBoolean *controlLoop*);

Use the Set Control Loop State function (hp816x_set_ATT_controlLoopState) to enable or disable an Attenuator module's power control loop.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8

hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

controlLoop

Use this control to enable (VI_TRUE) or disable (VI_FALSE) the power control loop.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Enable
VI_FALSE	(0)	Disable

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_get_ATT_controlLoopState_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_controlLoopState_Q(ViSession *ihandle*, ViInt32 *ATTSlot*, ViPBoolean *controlLoopState*);

Use the Get Control Loop State function (hp816x_get_ATT_controlLoopState_Q) to query the power control loop state.

Remark: Applies to Attenuator modules with Power Control only.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN **Values:** Name / Value / Description **hp816x_SLOT_1** / 1 / Slot 1 **hp816x_SLOT_2** / 2 / Slot 2 **hp816x_SLOT_3** / 3 / Slot 3 **hp816x_SLOT_4** / 4 / Slot 4 **hp816x_SLOT_5** / 5 / Slot 5 **hp816x_SLOT_6** / 6 / Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

controlLoopState

This control return the state of the power control loop.

If VI_TRUE is returned the power control loop is enabled,

If VI_FALSE is returned the power control loop is disabled.

Data Type: ViPBoolean

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_set_ATT_avTime

Syntax:

ViStatus _VI_FUNC hp816x_set_ATT_avTime(ViSession *ihandle*, ViInt32 *ATTSlot*, ViReal64 *averagingTime*);

Use the Set Averaging Time function (hp816x_set_ATT_avTime) to set the averaging time of an Attenuator module's integrated powermeter.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8

hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

averagingTime

Use this control to set the averaging time of the integrated powermeter.

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_get_ATT_avtime_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_avtime_Q(ViSession *ihandle*, ViInt32 *ATTSlot*, ViPReal64 *averagingTime*);

Use the Get Averaging Time function (hp816x_get_ATT_avtime_Q) to query the averaging time set for the Attenuator module's built-in powermeter.

Remark: Applies to Attenuator modules with Power Control only.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

averagingTime

This control returns the averaging time set for the Attenuator module's built-in powermeter.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_set_ATT_TriggerConfig

Syntax:

ViStatus _VI_FUNC hp816x_set_ATT_triggerCongfig(ViSession *ihandle*, ViInt32 *ATT_Slot*, ViInt32 *Trigger_In*);

The Set ATT Trigger Configuration (hp816x_set_ATT_triggerConfig) function configures input trigger handling for a builtin powermeter of an attenuator.

Remark: Applies to Attenuator modules with Power Control only.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATT_Slot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN **Values:**

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Trigger_In

Use the Trigger In control to select one of the following input trigger configurations:

Ignore (hp816x_PWM_TRIGIN_IGN), input triggers are ignored,

Single Measurement (hp816x_PWM_TRIGIN_SME), an input trigger triggers a single power measurement, or

Complete Measurement (hp816x_PWM_TRIGIN_CME), as (hp816x_PWM_TRIGIN_SME), an input trigger triggers a single complete power measurement.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_PWM_TRIGIN_IGN	0	Ignore
hp816x_PWM_TRIGIN_SME	1	Single Measurement
hp816x_PWM_TRIGIN_CME	2	Complete Measurement

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_ATT_triggerConfig_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_triggerConfig_Q(ViSession *ihandle*, ViInt32 *ATT_Slot*, ViPInt32 * *Trigger_In*);

The Get ATT Trigger Configuration (hp816x_get_ATT_triggerConfig_Q) returns the input trigger configuration of a builtin powermeter of an attenuator.

Remark: Applies to Attenuator modules with Power Control only.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATT_Slot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Trigger_In

Trigger In returns one of the following input trigger configurations:

0 (hp816x_PWM_TRIGIN_IGN), input triggers are ignored,

1 (hp816x_PWM_TRIGIN_SME), an input trigger triggers a single power measurement, or

2 (hp816x_PWM_TRIGIN_CME), as (hp816x_PWM_TRIGIN_SME), an input trigger triggers a single complete power measurement.

Data Type: ViPInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_zero_ATT_powermeter

Syntax:

ViStatus _VI_FUNC hp816x_zero_ATT_powermeter(ViSession *ihandle*, ViInt32 *ATTSlot*);

Use the Zero Builtin Powermeter function (hp816x_zero_ATT_powermeter) to zero the powermeter built-in to the specified Attenuator modue

Remark: Applies to Attenuator modules with Power Control only.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_get_ATT_zeroResult_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_ATT_zeroResult_Q(ViSession *ihandle*, ViInt32 *ATTSlot*, ViPInt32 *lastZeroResult*);

Use the Get Last Zero Result function (hp816x_get_ATT_zeroResult_Q) to query the result of the last zeroing operation for the powermeter built-in to the specified Attenuator module.

Remark: Applies to Attenuator modules with Power Control only.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATTSlot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN **Values:** Name / Value / Description **hp816x_SLOT_1** / 1 / Slot 1 **hp816x_SLOT_2** / 2 / Slot 2 **hp816x_SLOT_3** / 3 / Slot 3 **hp816x_SLOT_4** / 4 / Slot 4 **hp816x_SLOT_5** / 5 / Slot 5 **hp816x_SLOT_6** / 6 / Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

lastZeroResult

Returns the result of the last zeroing operation.

If the operation was successful 0 is returned.

Data Type: ViPInt32
Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_zero_ATT_all

Syntax:

ViStatus _VI_FUNC hp816x_zero_ATT_all(ViSession *ihandle*);

Use the Zero All function (hp816x_zero_ATT_all) to zero all powermeters (including the powermeter built-in to each Attenuator module with Power Control, and returnloss modules).

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_spectralCalibration

Syntax:

ViStatus _VI_FUNC hp816x_spectralCalibration(ViSession *ihandle,* ViInt32 *Slot*, ViUInt32 *Size_of_Spectrum*, ViReal64 *Wavelength*[ ], ViReal64 *Power*[ ], ViReal64 * *Wavelength_Result*, ViChar *Error_Diagnose*);

Use the Spectral Calibration function (hp816x_spectralCalibration) to calibrate an attenuator module's built-in power meter or a power meter to an effective wavelength for a multiwavelength or broadband signal. For accurate power measurement of a multiwavelength or broadband signal, you can set the power meter to its effective wavelength (corresponding to the required weighted-averageresponsivity factor) for the signal.

Remark: This function only applies to power meters or Attenuator modules with power control, specifically the 81576A and 81577A. These modules use a filter technology with minimum wavelength dependence, but incorporate a power meter whose responsivity varies with wavelength.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
Slot	Use the Slot control to specify the slot used to host the Attenuator module (or powermeter). **Data Type:** ViInt32 **Input/Output:** IN **Values:** Name: **hp816x_SLOT_1** — Value: 1 — Description: Slot 1

hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Size_of_Spectrum

Use the Size of Spectrum control to specifies the number of wavelength:power pairs in the multiwavelength DWDM signal.

Data Type: ViUInt32

Input/Output: IN

Wavelength

Input the array of wavelength values

Data Type: ViReal64[ ]

Input/Output: IN

Power

Input the array of power values in W

Data Type: ViReal64[ ]

Input/Output: IN

Wavelength_Result

Returns the effective wavelength corresponding to the required powermeter responsivity factor for the DWDM signal.

If 0.0 is returned, the spectral calibration failed.

Data Type: ViReal64

Input/Output: OUT

Error_Diagnose

Returns a diagnostic message, which may help you understand why a spectral calibration failed.

Data Type: ViChar

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_getWlRespTblSize

Syntax:

ViStatus _VI_FUNC hp816x_getWlRespTblSize (ViSession *ihandle*, ViInt32 *ATT_Slot*, ViUInt32 * *Size*, ViUInt32 * *CSV_Size*);

Use the Get Wavelength Resp. Table Size function (hp816x_getWlRespTblSize) to query the size of the wavelength response table used to calibrate a power meter or an Attenuator module's built-in power meter.

Remark: This function only applies to power meters or Attenuator modules with power control, specifically the 81576A and 81577A. These modules use a filter technology with minimum wavelength dependence, but incorporate a power meter whose responsivity varies with wavelength.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATT_Slot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Size

Returns the size of the powermeter's wavelength response calibration table.

Remark: Use this value to allocate memory prior to using the Read Wavelength Response Table function (**hp816x_readWlRespTable**).

Data Type: ViUInt32

Input/Output: OUT

CSV_Size

Returns the number of bytes required by the powermeter's wavelength response calibration table in CSV format.

Remark: Use this value to allocate memory prior to using the Read Wavelength Response Table function (**hp816x_readWlRepTblCSV**).

Data Type: ViUInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_readWlRespTable

Syntax:

ViStatus _VI_FUNC hp816x_getWlRespTable (ViSession *ihandle*, ViInt32 *ATT_Slot*, ViReal64 *Wavelength* [ ], ViReal64 *Response_Factor* [ ]);

Use the Read Wavelength Resp. Table function (hp816x_readWlRespTable) to read the wavelength response table used to calibrate a power meter or an Attenuator module's built-in power meter.

Use the Get Wavelength Resp. Table Size function (hp816x_getWvlRespTblSize) to query the size of this table.

Remark: This function only applies to power meters or Attenuator modules with power control, specifically the 81576A and 81577A. These modules use a filter technology with minimum wavelength dependence, but incorporate a power meter whose responsivity varies with wavelength.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATT_Slot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN **Values:** Name / Value / Description **hp816x_SLOT_1** / 1 / Slot 1 **hp816x_SLOT_2** / 2 / Slot 2

hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Wavelength

Wavelength is both an input and an output.

As an input, use a Real64 array with size given by the Size result from the hp816x_getWRespTblSize query.

Returns the wavelength entries from the powermeter's wavelength response calibration table.

Data Type: ViReal64[ ]

Input/Output: OUT

Response_Factor

Response_Factor is both an input and an output.

As an input, use a Real64 array with size given by the Size result from the hp816x_getWRespTblSize query.

Returns the response factor entries from the powermeter's wavelength response calibration table.

Data Type: ViReal64[ ]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_readWlRepTblCSV

Syntax:

ViStatus _VI_FUNC hp816x_getWlRepTblCSV (ViSession *ihandle*, ViInt32 *ATT_Slot*, ViChar *CSV_List* [ ]);

Use the Read Wl Rep. Table CSV function (hp816x_readWlRepTblCSV) to read the wavelength response table used to calibrate a power meter or an Attenuator module's built-in power meter in CSV format.

Use the Get Wavelength Resp. Table Size function (hp816x_getWvlRespTblSize) to query the size of this table.

Remark: This function only applies to power meters or Attenuator modules with power control, specifically the 81576A and 81577A. These modules use a filter technology with minimum wavelength dependence, but incorporate a power meter whose responsivity varies with wavelength.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
ATT_Slot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2

hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

CSV_List

CSV_List is both an input and an output.

As an input, use a text string with length given by the CSV_Size result from the hp816x_getWRespTblSize query.

Returns the wavelength, response_factor entries from the powermeter's wavelength response calibration table in CSV format.

Data Type: ViChar[ ]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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Information for N77 Series Instruments

This version of the 816x VXIplug&play instrument driver has been extended to support the N773xA, N774xA optical power meter and the N775xA and N776xA Optical Attenuator instruments. These are stand-alone instruments that do not require installation in an 816x mainframe. The SCPI command set has been designed for almost complete programming compatibility with the modular instruments of the 816x family. Please be aware of the following details and limitations, when using this driver.

- The N77-series instruments can be controlled via USB, LAN or GPIB interfaces. Especially for transferring large data volumes, the USB and usually LAN interfaces provide faster performance.
- The identification of the individual ports of these multiport instruments in the SCPI and hp816x functions corresponds to the slot number. So for example the ports of the N7745A can be addressed with the slot numbers 1 to 8. The channel number in the commands and functions is not used and can generally be omitted or set to Channel 1 (index 0).
- The N771x-series and 81950A tunable lasers are not supported by the 816x VXIplug&play driver.
- The N773x-series optical switches are supported by the 816x VXIplug&play functions for switches and relevant functions for the mainframes.
- The N774x-series power meters are supported by the 816x VXIplug&play functions for power meters, relevant functions for the mainframes like for standard trigger configuration, and the Multi Frame Lambda Scan application functions.
- The N776x-series optical attenuators are supported by the 816x VXIplug&play functions for attenuators and the relevant functions for the mainframes.
- The N775x-series optical attenuators and power meters are supported by the 816x VXIplug&play functions for attenuators and the relevant functions for the mainframes and the basic functions for the power meters, excluding the advanced functions using data power logging. The choices of averaging time is also more limited for these power meters.
- The averaging time parameter for the power meters is limited in the 816x VXIplug&play functions to a limited set of discrete values, similar to the modular 816x power meters. This set includes: 1, 2, 5, 10, 25, 100, 200, and 500 ìs, 1, 2, 5, 10, 20, 50, 100, 200, 500 ms, and 1, 2, 5, and 10s.
- The maximum number of logging data points that can be acquired is limited by the 816x VXIplug&play data acquisition functions to 100,000 per port.
- The shortest averaging time used by the Multi Frame Lambda Scan application is 25 ìs.
- For the Multi Frame Lambda Scan application, the N774xA instruments can be included simply by registering the instrumet as a mainframe,after the first

mainframe with the tunable laser has been registered.

Further details to available functions are found under the categories below:

Utility Functions

Mainframe Specific Functions

Power Sensor Specific Functions

Applications

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hp816x_PWM_slaveChannelCheck

Syntax:

ViStatus _VI_FUNC hp816x_PWM_slaveChannelCheck(ViSession *ihandle*, ViInt32 *PWMSlot*, ViBoolean *slaveChannelCheck*);

The Slave Channel Check function (hp816x_PWM_slaveChannelCheck) allows you to automatically correct parameters that are passed mistakenly to the slave channel of a dual Power Meter.

Some parameters for dual Power Meters can only be passed to the master channel, channel 1. The parameter affects the slave channel, channel 2. See your mainframe's programming guide for a table of GPIB commands that can only be configured for the master channel.

If Slave Channel Check is On, the driver generates a run-time error, if a command is passed mistakenly to the slave channel of a dual Power Meter.

If Slave Channel Check is Off, the driver corrects parameters that are passed mistakenly to the slave channel of a dual Power Meter. Off is the default.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

slaveChannelCheck

If Slave Channel Check is On, the driver generates a runtime error, if a command is passed mistakenly to the slave channel of a dual Power Meter.

If Slave Channel Check is Off, the driver corrects parameters that are passed mistakenly to the slave channel of a dual Power Meter. Off is the default.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine

"hp816x_error_message"

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hp816x_set_PWM_parameters

Syntax:

ViStatus _VI_FUNC hp816x_set_PWM_parameters(ViSession *ihandle*, **ViInt32** *PWMSlot*, **ViInt32** *channelNumber*, **ViBoolean** *rangeMode*, **ViBoolean** *powerUnit*, **ViBoolean** *internalTrigger*, **ViReal64** *wavelength*, **ViReal64** *averagingTime*, **ViReal64** *powerRange*);

The Set PWM Parameters function (hp816x_set_PWM_parameters) sets the most important parameters for a sensor, preparing the instrument to start a measurement.

If a value is not yet applied to the instrument and the Force Transaction function (hp816x_forceTransaction) is switched off, the new values will be set.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10

hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

rangeMode

Use the MinMax control to choose one of the following range modes:

Auto (hp816x_PWM_RANGE_AUTO), the auto-ranging mode, ensures that the result has a displayed value between 9% and 100% of full scale. The default state is for automatic ranging to be enabled.

Manual (hp816x_PWM_RANGE_MANUAL), which allows you to set a user-defined range.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
hp816x_PWM_RANGE_AUTO	1	Auto
hp816x_PWM_RANGE_MANUAL	0	manual

powerUnit

Use the Power Unit control to choose between power units of Watts (hp816x_PU_WATT) and dBm (hp816x_PU_DBM)

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
hp816x_PU_WATT	1	Watt
hp816x_PU_DBM	0	dBm

internalTrigger

Use the Internal Trigger control to choose between Continuous triggering and Immediate triggering.

If you select Immediate triggering (hp816x_PWM_IMMEDIATE) one measurement cycle is performed. When the averaging time is over, the result is available. No further measurement cycles are started. If the result is read with the Fetch PWM Value function, the value will always be the same.

If you select Continous triggering (hp816x_PWM_CONTINUOUS) a new measuremet cycle is immediately triggered after each measurement cycle is completed.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
hp816x_PWM_CONTINUOUS	1	Continuous
hp816x_PWM_IMMEDIATE	0	Immediate

wavelength

Use Wavelength to set the measurement wavelength of the Power Meter in meters (m).

This is the Wavelength value. The responsivity of the Power Meter varies with wavelength. For accurate power measurement, you need to input the wavelength of the optical input

Data Type: ViReal64

Input/Output: IN

averagingTime

Use Averaging Time to set the length of time over which a signal is averaged.

Longer averaging times increase the accuracy and improve the noise rejection of the measurement. Longer averaging times also decrease sensitivity.

If you use a 8153A Series Power Meter and you set Trigger In to Single Measurement using the Set PWM Trigger Configuration (hp816x_set_PWM_triggerConfiguration) function, it is possible set the averaging time to 5 ms.

Data Type: ViReal64

Input/Output: IN

powerRange

If you choose Manual (hp816x_PWM_RANGE_MANUAL) using the Range Mode control, you must set a Power Range.

The Power Range is the highest power level that can be measured. If the measured power is less than the Power Range, measured power is displayed.

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_PWM_parameters_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_PWM_parameters_Q(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*, ViPBoolean *rangeMode*, ViPBoolean *powerUnit*, ViPInt32 *internalTrigger*, ViPReal64 *wavelength*, ViPReal64 *averagingTime*, ViPReal64 *powerRange*);

The Get PWM parameters function (hp816x_get_PWM_parameters_Q) returns the most important parameters for a sensor.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

rangeMode

Range Mode returns one of the following range modes:

1, the auto-ranging mode (hp816x_PWM_RANGE_AUTO), ensures that the result has a displayed value between 9% and 100% of full scale.

0, manual-ranging mode (hp816x_PWM_RANGE_MANUAL), you set the range.

Data Type: ViPBoolean

Input/Output: OUT

powerUnit

Power Unit returns the following:

0 (hp816x_PU_DBM) units of dBm or

1 (hp816x_PU_WATT) units of Watts.

Data Type: ViPBoolean

Input/Output: OUT

internalTrigger

Internal Trigger returns one of the following internal trigger configurations:

0 (hp816x_TRIG_IMMEDIATE), where only one power measurement is triggered, or

1 (hp816x_TRIG_CONTINUOUS), where power measurements are continuously triggered.

Data Type: ViPInt32

Input/Output: OUT

wavelength

Wavelength returns the measurement wavelength of the Power Meter in meters (m).

This is the Wavelength value. The responsivity of the Power Meter varies with wavelength. For accurate power measurement, you need to input the wavelength of the optical input.

Data Type: ViPReal64

Input/Output: OUT

averagingTime

Averaging Time returns the length of time in seconds (s) over which a signal is averaged.

Longer averaging times increase the accuracy and improve the noise rejection of the measurement. Longer averaging times also decrease sensitivity.

Data Type: ViPReal64

Input/Output: OUT

powerRange

Power Range returns the power range in dBm.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_PWM_averagingTime

Syntax:

ViStatus _VI_FUNC hp816x_set_PWM_averagingTime(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*, ViReal64 *averagingTime*);

The Set PWM Averaging Time function (hp816x_set_PWM_averagingTime) sets the averaging time of a Power Meter.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

averagingTime

Use Averaging Time to set the length of time over which a signal is averaged.

Longer averaging times increase the accuracy and improve the noise rejection of the measurement. Longer averaging times also decrease sensitivity.

If you use a 8153A Series Power Meter and you set Trigger In to Single Measurement using the Set PWM Trigger Configuration (**hp816x_set_PWM_triggerConfiguration**) function, it is possible set the averaging time to 5 ms.

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_PWM_averagingTime_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_PWM_averagingTime_Q(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*, ViPReal64 *averagingTime*);

The Get PWM Averaging Time (hp816x_get_PWM_averagingTime_Q) function returns the averaging time of a Power Meter.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

averagingTime

Averaging Time returns the length of time over which a signal is averaged.

Longer averaging times increase the accuracy and improve the noise rejection of the measurement. Longer averaging times also decrease sensitivity.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_PWM_wavelength

Syntax:

ViStatus _VI_FUNC hp816x_set_PWM_wavelength(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*, ViReal64 *wavelength*);

The Set PWM Wavelength function (hp816x_set_PWM_wavelength) sets the wavelength of a sensor.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

wavelength

Use Wavelength to set the measurement wavelength of the Power Meter in meters (m).

The responsivity of the Power Meter varies with wavelength. For accurate power measurement, you need to input the wavelength of the optical input.

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_PWM_wavelength_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_PWM_wavelength_Q(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*, ViPReal64 *minWavelength*, ViPReal64 *maxWavelength*, ViPReal64 *currentWavelength*);

The Get PWM Wavelength function (hp816x_get_PWM_wavelength_Q) returns the minimum, maximum, and current wavelength of a Power Meter.

The responsivity of the Power Meter varies with wavelength. For accurate power measurement, you need to input the wavelength of the optical input.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3

hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

minWavelength

Min Wavelength returns the lowest wavelength that can be set for the Power Meter in meters (m).

Data Type: ViPReal64

Input/Output: OUT

maxWavelength

Max Wavelength returns the highest wavelength that can be set for the Power Meter in meters (m).

Data Type: ViPReal64

Input/Output: OUT

currentWavelength

Current Wavelength returns the wavelength in meters (m) that the Power Meter is set to.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_PWM_calibration

Syntax:

ViStatus _VI_FUNC hp816x_set_PWM_calibration(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*, ViReal64 *calibration*);

The Set PWM Calibration function (hp816x_set_PWM_calibration) sets the calibration factor of a Power Meter.

This is a calibration offset that you can enter to compensate for external optical circuitry, for example, the 81022FF Integrating Sphere or a 3 dB coupler. This value is automatically subtracted from the input signal.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

calibration

Use the Calibration input to set the calibration factor of the sensor in decibels (dB).

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_PWM_calibration_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_PWM_calibration_Q(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*, ViPReal64 *calibration*);

The Get PWM Calibration function (hp816x_get_PWM_calibration_Q) returns the calibration factor of your Power Meter in dBm.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

calibration

Calibration returns the calibration factor of the Power Meter in decibels (dB).

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_PWM_powerRange

Syntax:

ViStatus _VI_FUNC hp816x_set_PWM_powerRange(ViSession *ihandle*, **ViInt32** *PWMSlot*, **ViInt32** *channelNumber*, **ViBoolean** *rangeMode*, **ViReal64** *powerRange*);

The Set PWM Range function (hp816x_set_PWM_powerRange) sets the power range of a Power Meter.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15

hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

rangeMode

Use the Range Mode control to choose one of the following range modes:

Auto, the auto-ranging mode, ensures that the result has a displayed value between 9% and 100% of full scale. The default state is for automatic ranging to be enabled.

Manual, which allows you to set a user-defined range.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
hp816x_PWM_RANGE_AUTO	1	Auto
hp816x_PWM_RANGE_MANUAL	0	Manual

powerRange

If you choose Manual using the Range Mode control, you must set a Power Range in dBm.

The Power Range determines the range of power levels that can be measured.

The Upper Power Limit is always 3 dBm greater than the chosen Power Range value. The resolution is always 40 dBm less than the chosen Power Range value.

For example, if you set the Power Range to -30 dBm, you can measure power between -27 dBm

and -70 dBm, at a resolution of -70 dBm (0.1 nW).

Data Type: ViReal64
Input/Output: IN

Return Value:

VI_SUCCESS: No error
Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_PWM_powerRange_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_PWM_powerRange_Q(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*, ViPBoolean *rangeMode*, ViPReal64 *powerRange*);

The Get PWM Range function (hp816x_get_PWM_powerRange_Q) returns the power range setting of a Power Meter.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

rangeMode

Range Mode returns one of the following range modes:

1, the auto-ranging mode (hp816x_PWM_RANGE_AUTO), ensures that the result has a displayed value between 9% and 100% of full scale.

0, manual-ranging mode (hp816x_PWM_RANGE_MANUAL), allows you to set the range.

Data Type: ViPBoolean

Input/Output: OUT

powerRange

Power Range returns the power range in dBm.

The Power Range determines the range of power levels that can be measured.

If the measured power is less than 3 dBm greater than the Power Range, measured power can be returned.

If the measured power is less than 40 dBm less than the Power Range, measured power can be returned.

For example, if you set the Power Range to -30 dBm, you can measure power between -27 dBm and -70 dBm, at a resolution of -70 dBm (0.1 nW).

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_PWM_powerUnit

Syntax:

ViStatus _VI_FUNC hp816x_set_PWM_powerUnit(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*, ViInt32 *powerUnit*);

The set PWM Power Unit function (hp816x_set_PWM_powerUnit) sets the power unit of a Power Meter.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

powerUnit

Use the Power Unit control to choose between power units of Watts (hp816x_PU_WATT) and dBm (hp816x_PU_DBM).

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_PU_WATT	1	Watt
hp816x_PU_DBM	0	dBm

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_PWM_powerUnit_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_PWM_powerUnit_Q(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*, ViPInt32 *powerUnit*);

The Get PWM Power Unit function (hp816x_get_PWM_powerUnit_Q) returns the power units of a Power Meter.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32
Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

powerUnit

Power Unit returns the selected power units:

0, if dBm (hp816x_PU_DBM) are selected, or

1, if Watts (hp816x_PU_WATT) are selected.

Data Type: ViPInt32
Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_PWM_referenceSource

Syntax:

ViStatus _VI_FUNC hp816x_set_PWM_referenceSource(ViSession *ihandle*, **ViInt32** *PWMSlot*, **ViInt32** *channelNumber*, **ViInt32** *measureMode*, **ViInt32** *referenceSource*, **ViInt32** *slot*, **ViInt32** *channel*);

The Set PWM Reference Source function (hp816x_set_PWM_referenceSource) sets the reference control parameters.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13

hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

measureMode

Use the Measure Mode control to set power measurement to one of the following:

Relative measurement (hp816x_PWM_REF_RELATIV); decibels, dB or

Absolute measurement (hp816x_PWM_REF_ABSOLUTE); dBm or Watts.

If Relative is chosen using the Measure control, the given reference source will be checked. The function returns an error if the selected reference source does not exist.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_PWM_REF_RELATIV	1	Relative
hp816x_PWM_REF_ABSOLUTE	0	Absolute

referenceSource

Use the Reference Source control to choose between the Power Meter's Internal reference and referencing the power reading of a Channel.

Use the Set PWM Reference Value function (**hp816x_set_PWM_referenceValue**) to set the internal reference value.

If you reference a Channel you must select this channel using the Slot and Channel inputs. The power reading from this channel updates the power reference continuously.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_PWM_TO_REF	0	Internal
hp816x_PWM_TO_MOD	1	Channel

slot

Use the Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module) as your reference for Relative power measurement.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channel

Use the Channel control to select the channel that you want to reference.

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel control to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_PWM_referenceSource_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_PWM_referenceSource_Q(ViSession *ihandle*, **ViInt32** *PWMSlot*, **ViInt32** *channelNumber*, **ViInt32** *measureMode*, **ViInt32** *referenceSource*, **ViInt32** *slot*, **ViInt32** *channel*);

The Get PWM Reference Source function (hp816x_get_PWM_referenceSource_Q) gets the reference control parameters.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13

hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

measureMode

The Measure Mode control returns the power measurement setting according to one of the following:

Relative measurement (hp816x_PWM_REF_RELATIV); decibels, dB or

Absolute measurement (hp816x_PWM_REF_ABSOLUTE); dBm or Watts.

If Relative is chosen using the Measure control, the given reference source will be checked. The function returns an error if the selected reference source does not exist.

Data Type: ViInt32

Input/Output: OUT

Values:

Name	Value	Description
hp816x_PWM_REF_RELATIV	1	Relative
hp816x_PWM_REF_ABSOLUTE	0	Absolute

referenceSource

The Reference Source control returns the selection between the Power Meter's Internal reference and referencing the power reading of a Channel.

Use the Set PWM Reference Value function

(**hp816x_set_PWM_referenceValue**) to set the internal reference value.

If a Channel is referenced, this channel is selected using the Slot and Channel values. The power reading from this channel updates the power reference continuously.

Data Type: ViInt32

Input/Output: OUT

Values:

Name	Value	Description
hp816x_PWM_TO_REF	0	Internal
hp816x_PWM_TO_MOD	1	Channel

slot

The Slot control returns the installed Power Meter (a Power Sensor or Optical Head Interface Module) indicated as reference for Relative power measurement.

Data Type: ViInt32

Input/Output: OUT

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channel

The Channel control returns the channel that is referenced.

If you choose a Dual Power Sensor Module or a

Dual Optical Head Interface Module, you can use the Channel control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel control to choose Channel 1.

Data Type: ViInt32

Input/Output: OUT

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_PWM_referenceValue

Syntax:

ViStatus _VI_FUNC hp816x_set_PWM_referenceValue(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*, ViReal64 *internalReference*, ViReal64 *referenceChannel*);

The Set PWM Reference Value function (hp816x_set_PWM_referenceValue) sets the internal reference value and the reference channel.

These two reference values are independent and have separate memory locations.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3

hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

internalReference

Use Internal Reference to set the internal reference value to a value in dBm or Watts, depending on the units you choose using the Set PWM Power Unit function (**hp816x_set_PWM_powerUnit**).

For this value to be chosen as the reference, use the Set PWM Reference Source function (**hp816x_set_PWM_referenceSource**) to choose Internal as the Reference Source.

Data Type: ViReal64

Input/Output: IN

referenceChannel

Use Reference Channel to set an additional reference value in decibels (dB), depending on the units you choose using the Set PWM Power Unit function (**hp816x_set_PWM_powerUnit**) that will be used if you measure relative to a second channel by using the Set PWM Reference Source function (**hp816x_set_PWM_referenceSource**) to choose Channel as the Reference Source.

The power reading from a chosen channel is continuously added to the value you set here and the sum is used to calculate the power ratio value in decibels (dB) using one of the following equations:

P(dB) = Pa(dBm) - Pb(dBm) - Prc(dB)

P(dB) = 10log(Pa(W)/Pb(W)) - Prc(dB)

where:

P is the power returned in decibels (dB),

Pa is the absolute power in dBm or Watts measured by the current Power Meter channel (this value includes any calibration factor set for the channel),

Pb is the absolute power in dBm or Watts measured by the chosen reference Power Meter channel (this value includes any calibration factor set for the channel), and

Prc is the absolute power in decibels (dB) you set using the Reference Channel input.

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_PWM_referenceValue_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_PWM_referenceValue_Q(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*, ViPInt32 *referenceMode*, ViPReal64 *internalReference*, ViPReal64 *referenceChannel*);

The Get PWM Reference Value function (hp816x_get_PWM_referenceValue_Q) returns the current internal, external reference values and the reference mode.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

referenceMode

Reference Mode returns the following:

0 (hp816x_PWM_REF_ABSOLUTE), for absolute measurement mode, units of dBm or Watts (W) are chosen, or

1 (hp816x_PWM_REF_RELATIV), for relative measurement mode, units of decibels (dB) are chosen.

Data Type: ViPInt32

Input/Output: OUT

internalReference

Internal Reference returns the internal reference value in dBm.

Data Type: ViPReal64

Input/Output: OUT

referenceChannel

Reference Channel returns the additional reference value in dBm that will be used if you measure relative to a second channel.

The power reading from a chosen channel is continuously added to the value you set here and the sum is used to calculate the power ratio value in decibels (dB).

For this channel to be chosen as the reference, use the Set PWM Reference Source function (**hp816x_set_PWM_referenceSource**) to choose Channel as the Reference Source.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_PWM_triggerConfiguration

Syntax:

ViStatus _VI_FUNC hp816x_set_PWM_triggerConfiguration(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *triggerIn*, ViInt32 *triggerOut*);

The Set PWM Trigger Configuration (hp816x_set_PWM_triggerConfiguration) function configures input trigger handling and trigger output generation.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

triggerIn

Use the Trigger In control to select one of the following input trigger configurations:

Ignore (hp816x_PWM_TRIGIN_IGN), input triggers are ignored,

Single Measurement (hp816x_PWM_TRIGIN_SME), an input trigger triggers a single power measurement, or

Complete Measurement (hp816x_PWM_TRIGIN_CME), an input trigger triggers the data acquisition function that was started most recently to start. The Power Meter data acquisition functions are: the Set PWM Logging (**hp816x_set_PWM_logging**) function, the Set PWM Stability (**hp816x_set_PWM_stability**) function, and the Set PWM MinMax (**hp816x_set_PWM_minMax**) function.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_PWM_TRIGIN_IGN	0	Ignore
hp816x_PWM_TRIGIN_SME	1	Single Measurement
hp816x_PWM_TRIGIN_CME	2	Complete Measurement

triggerOut

Use the Trigger Out control to select one of the

following output trigger configurations:

Disabled (hp816x_PWM_TRIGOUT_NONE), no output trigger will be generated, or

End of Averaging (hp816x_PWM_TRIGOUT_AVG), an output trigger is generated at the end of the averaging time for a power measurement, or

Beginning of Averaging (hp816x_PWM_TRIGOUT_MEAS), an output trigger is generated at the beginning of the averaging time for a power measurement.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_PWM_TRIGOUT_NONE	0	Disabled
hp816x_PWM_TRIGOUT_AVG	1	End of Averaging
hp816x_PWM_TRIGOUT_MEAS	2	Beginning of Averaging

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_PWM_triggerConfiguration

Syntax:

ViStatus _VI_FUNC hp816x_get_PWM_triggerConfiguration(ViSession *ihandle*, ViInt32 *PWMSlot*, ViPInt32 *triggerIn*, ViPInt32 *triggerOut*);

The Get PWM Trigger Configuration (hp816x_get_PWM_triggerConfiguration) function returns the input trigger handling configuration and the output trigger generation configuration for a Power Meter.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

triggerIn

Trigger In returns one of the following input trigger configurations:

0 (hp816x_PWM_TRIGIN_IGN), input triggers are ignored,

1 (hp816x_PWM_TRIGIN_SME), an input trigger triggers a single power measurement, or

2 (hp816x_PWM_TRIGIN_CME), an input trigger triggers the data acquisition function that was started most recently to start. The Power Meter data acquisition functions are: the Set PWM Logging (**hp816x_set_PWM_logging**) function, the Set PWM Stability (hp81 6x_set_PWM_stability) function, and the Set PWM MinMax (**hp816x_set_PWM_minMax**) function.

Data Type: ViPInt32

Input/Output: OUT

triggerOut

Trigger Out returns one of the following output trigger configurations:

0 (hp816x_PWM_TRIGOUT_NONE), no output trigger will be generated, or

1 (hp816x_PWM_TRIGOUT_AVG), an output trigger is generated at the end of the averaging time for a power measurement, or

2 (hp816x_PWM_TRIGOUT_MEAS), an output trigger is generated at the beginning of the averaging time for a power measurement.

Data Type: ViPInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_PWM_displayToReference

Syntax:

ViStatus _VI_FUNC hp816x_PWM_displayToReference(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*);

Use the PWM Display to Reference function (hp816x_PWM_displayToReference) to choose the currently displayed power value as the new internal reference.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single--channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_PWM_internalTrigger

Syntax:

ViStatus _VI_FUNC hp816x_set_PWM_internalTrigger(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*, ViBoolean *internalTrigger*);

The Set PWM Internal Trigger function (hp816x_set_PWM_internalTrigger) determine how the Power Meter starts measuring.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12

hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

internalTrigger

Use the Internal Trigger control to choose between Continuous triggering (hp816x_PWM_CONTINUOUS) and Immediate triggering (hp816x_PWM_IMMEDIATE).

Selecting Immediate triggering performs one measurement cycle. When the averaging time is over, the result is available. No further measurement cycles are started. If the result is read with the Fetch PWM Value function, the value will always be the same .

Selecting Continuous triggering will produce different results, because after finishing one measure cycle, a new cycle is immediately triggered.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
hp816x_PWM_CONTINUOUS	1	Continuous
hp816x_PWM_IMMEDIATE	0	Immediate

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_start_PWM_internalTrigger

Syntax:

ViStatus _VI_FUNC hp816x_start_PWM_internalTrigger(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*);

The Start PWM Internal Trigger function (hp816x_start_PWM_internalTrigger) starts a Power Meter measurement cycle.

If you have previously chosen Continuous Triggering mode, see the Set PWM Internal Trigger function, the function chooses Immediate Triggering mode automatically. You need to use the Start PWM Trigger function to start another Power Meter measurement cycle.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2

hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_PWM_readValue

Syntax:

ViStatus _VI_FUNC hp816x_PWM_readValue(ViSession *ihandle*, ViInt32 *PWMSlot*, ViUInt32 *channelNumber*, ViPReal64 *measuredValue*);

The Read PWM Value function (hp816x_PWM_readValue) forces the instrument to start a measurement cycle. The evaluated power value will not be returned until the averaging time period is finished.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViUInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

measuredValue

Measured Value returns the measured power value in Watts (W) or dBm.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_PWM_fetchValue

Syntax:

ViStatus _VI_FUNC hp816x_PWM_fetchValue(ViSession *ihandle*, ViUInt32 *PWMSlot*, ViUInt32 *channelNumber*, ViPReal64 *measuredValue*);

The Fetch PWM Value function (hp816x_PWM_fetchValue) immediately returns a power value without averaging power measurements over the averaging time period.

REMARKS: If this function is called more than once within an averaging cycle, the returned values will be identical.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

measuredValue

Measured Value returns the measured power value in Watts (W) or dBm.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine hp816x_error_message

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hp816x_PWM_readAll

Syntax:

ViStatus _VI_FUNC hp816x_PWM_readValue(ViSession *Ihandle*, ViUInt32 * *Number_of_Channels*, ViInt32 *Slots*[ ], ViInt32 *Channels*[ ], ViPReal64 *Values*[ ]);

The Read All PWM Values function (hp816x_PWM_readAll) forces the instrument to begin a measurement cycle. Power values are not returned until the averaging time required to read every powermeter channel hosted by the mainframe is complete.

The returned information is stored in three arrays. Each element in the "Values" array (value[index]) was read from the powermeter channel identified by the corresponding element in the "Slots" array (slot[index]) and "Channel" array (channel[index]).

Note:

The caller of this function is responsible for allocating enough memory for the passed arrays. (The maximum array size is 34 for an 8166 mainframe)

Every powermeter module read by this function must have firmware version 3.0 or higher.

Parameter	Description
Ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
Number_of_Channels	Returns the number of powermeter channels that were read. **Data Type:** ViUInt32 * **Input/Output:** OUT

Slots[ ]

Returns an array containing the mainframe slot number for each powermeter read.

Data Type: ViInt32[ ]

Input/Output: OUT

Channels[ ]

Returns an array containing the channel number for each powermeter read.

Data Type: ViInt32[ ]

Input/Output: OUT

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

Values[ ]

Returns an array containing the measured power value (in Watts (W) or dBm) for each powermeter read.

Data Type: ViReal64[ ]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_PWM_zeroing

Syntax:

ViStatus _VI_FUNC hp816x_PWM_zeroing(ViSession *ihandle*, ViPInt32 *zeroingResult*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*);

The PWM Zeroing function (hp816x_PWM_zeroing) removes electrical offsets for one Power Meter.

The instrument measures by converting optical power to electrical power, and then measuring electrical power. An electrical offset is power that is always present, even if there is no light at the input. If this offset is not removed, it will affect powe r measurement.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
zeroingResult	Zeroing Result returns an integer value. If 0 is returned, the zeroing operation has been successful completed. If 98 is returned, the zeroing operation has failed because the Power Meter received light. The most common reason for zeroing to fail is if: - a source is connected to the Power Meter's input connector, - the fiber connected to the Power Meter's input connector is collecting light, or - the Power Meter receives ambient light because the input connector is uncovered. Any other value is an error code and can be looked up in your instrument's Programming Guide.

Data Type: ViPInt32

Input/Output: OUT

PWMSlot

Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module).

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_PWM_zeroingAll

Syntax:

ViStatus _VI_FUNC hp816x_PWM_zeroingAll(ViSession *ihandle*, ViPInt32 *summaryofZeroingAll*);

The PWM Zeroing All function (hp816x_PWM_zeroingAll) removes electrical offsets for all Power Meter modules.

The instrument measures by converting optical power to electrical power, and then measuring electrical power. An electrical offset is power that is always present, even if there is no light at the input. If this offset is not removed, it will affect powe r measurement.

NOTE: Because zeroing is a time consuming operation, the timeout is temporarily increased.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
summaryofZeroingAll	Summary of Zeroing All returns a summary of the results of every Return Loss module and Power Meter in the instrument. If 0 is returned, the zeroing operations have been successfully completed. If 98 is returned, a zeroing operation has failed because a Return Loss module or Power Meter has received light. The most common reason for zeroing to fail is if: - a source is connected to a Return Loss module's or Power Meter's input connector, - the fiber connected to a Return Loss module's or Power Meter's input connector is collecting light, or - a Return Loss module or

Power Meter received ambient light because the input connector is uncovered.

Any other value is an error code and can be looked up in your instrument's Programming Guide.

Data Type: ViPInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_PWM_ignoreError

Syntax:

ViStatus _VI_FUNC hp816x_PWM_ignoreError(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*, ViBoolean *ignoreError*, ViInt32 *instrumentErrorNumber*);

Use the Ignore Instrument Error function (hp816x_PWM_ignoreError) to trap a specific instrument error for a Power Meter.

If the specified error occurs, the returned status of a call causing this error will be VI_SUCCESS.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3

hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

ignoreError

Use the Ignore Error control to set the status of the instrument error specified by the Instrument Error Number control.

If you want the instrument error to be ignored (call returns VI-SUCCESS), select "Yes".

Otherwise, select "No".

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Yes
VI_FALSE	(0)	No

instrumentErrorNumber

Use the Instrument Error Number control to specify the instrument error number.

See your instrument's Programming Guide for a listing of the instrument error numbers and messages.

Data Type: ViInt32

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_PWM_logging

Syntax:

ViStatus _VI_FUNC hp816x_set_PWM_logging(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*, ViReal64 *averagingTime*, ViInt32 *dataPoints*, ViPInt32 *estimatedTimeout*);

The Set PWM Loggging function (hp816x_set_PWM_logging) sets the parameters for a logging operation and starts the logging.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

averagingTime

Use Averaging Time to set the length of time in seconds (s) over which a logging operation power measurement is averaged.

Data Type: ViReal64

Input/Output: IN

dataPoints

Use Data Points to set the number of values to be taken during a logging operation.

The maximum number of values is 12000.

Allocate a one-dimensional array with the same dimensions as this value and use as the Logging Result input for the Get PWM Logging Results function (**hp816x_get_PWM_loggingResults_Q**).

Data Type: ViInt32

Input/Output: IN

estimatedTimeout

Estimated Timeout returns the estimated time in milliseconds (ms), the logging operation will take; this value can be used to adjust the Timeout function (**hp816x_timeOut**).

Data Type: ViPInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_PWM_loggingResults_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_PWM_loggingResults_Q(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*, ViBoolean *waitforCompletion*, ViBoolean *resultUnit*, ViPBoolean *loggingStatus*, ViReal64*loggingResult*[ ]);

The Get PWM Logging Results function (hp816x_get_PWM_loggingResults_Q) returns the result of a previously started logging operation.

NOTE: The Wait for Completion control should only be set to Yes if the application is thoroughly tested because using this control may result in an infinite loop that blocks the calling function.

Also, if the function detects that logging is finished, the logging operation stops automatically.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1

hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

waitforCompletion

Use the Wait for Completion control to choose one of the following:

Yes (VI_TRUE), the logging operation will wait until the Total Time specified in the Set PWM Logging function has elapsed, or

No (VI_FALSE), the logging operation will stop

immediately.

Data Type: ViBoolean
Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Yes
VI_FALSE	(0)	No

resultUnit

Use the Result Unit control to choose power units of dBm (hp816x_PU_DBM) or Watts (hp816x_PU_WATT) for the logging operation result.

Data Type: ViBoolean
Input/Output: IN

Values:

Name	Value	Description
hp816x_PU_DBM	0	dBm
hp816x_PU_WATT	1	Watt

loggingStatus

Logging Status returns a Boolean representing the current status of the logging operation.

If Logging Status equals 1, the operation is completed.

If Logging Status equals 0, the operation is being performed.

Data Type: ViPBoolean
Input/Output: OUT

loggingResult

Logging Result is both an input and an output.

As an input, Logging Result accepts a one-

dimensional integer array that determines the size of the Logging Result output. Allocate a one-dimensional integer array with the same dimensions as the Data Points value you have set for the Set PWM Logging fun ction (**hp816x_set_PWM_logging**).

As an output, Logging Result returns the result of a logging operation as a one-dimensional integer array.

Data Type: ViReal64[]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_PWM_stability

Syntax:

ViStatus _VI_FUNC hp816x_set_PWM_stability(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*, ViReal64 *averagingTime*, ViReal64 *delayTime*, ViReal64 *totalTime*, ViPInt32 *estimatedResults*);

The Set PWM Stability function (hp816x_set_PWM_stability) sets up a stability operation and starts the data acquisition.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

averagingTime

Use Averaging Time to set the length of time in seconds (s) over which a stability operation power measurement is averaged.

Data Type: ViReal64

Input/Output: IN

delayTime

Use the Period time to set the time in seconds between the beginning of each averaging cycle.

Data Type: ViReal64

Input/Output: IN

totalTime

Use Total Time to how long in seconds (s) the stability operation lasts.

Data Type: ViReal64

Input/Output: IN

estimatedResults

Estimated Results returns an estimate of the number of results that the stability operation will record.

Use Estimated Results to allocate a one-dimensional array with the same dimensions as this value and use the array as the Stability Result input for the Get PWM Stability Results function (**hp816x_get_PWM_stabilityResults_Q**).

Data Type: ViPInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_PWM_stabilityResults_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_PWM_stabilityResults_Q(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*, ViBoolean *waitforCompletion*, ViBoolean *resultUnit*, ViPBoolean *stabilityStatus*, ViReal64*stabilityResult*[ ]);

The Get PWM Stability Result function (hp816x_get_PWM_stabilityResults_Q) returns the result of a previously started stability operation.

REMARK The Wait for Completion control should only be set to Yes if the application is thoroughly tested because using this control may result in an infinite loop that blocks the calling function.

Also, if the function detects that logging is finished, the stability operation stops automatically.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1

hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

waitforCompletion

Use the Wait for Completion control to choose one of the following:

Yes (VI_TRUE), the stability operation will wait until the Total Time specified in the Set PWM Stability function has elapsed, or

No (VI_FALSE), the stability operation will stop

immediately.

Data Type: ViBoolean
Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Yes
VI_FALSE	(0)	No

resultUnit

Use the Result Unit control to choose power units of dBm (hp816x_PU_DBM) or Watts (hp816x_PU_WATT) for the stability operation result.

Data Type: ViBoolean
Input/Output: IN

Values:

Name	Value	Description
hp816x_PU_DBM	0	dBm
hp816x_PU_WATT	1	Watt

stabilityStatus

Stability Status returns a Boolean representing the current status of the stability operation.

If Stability Status equals 1, the operation is completed.

If Stability Status equals 0, the operation is being performed.

Data Type: ViPBoolean
Input/Output: OUT

stabilityResult

Stability Result is both an input and an output.

As an input, Stability Result accepts a one-

dimensional integer array that determines the size of the Stability Result output. Allocate a one-dimensional integer array with the same dimensions as the Estimated Results value you have set for the Set PWM S tability function (**hp816x_set_PWM_stability**).

As an output, Stability Result returns the result of a stability operation as a one-dimensional integer array.

Data Type: ViReal64[]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_PWM_minMax

Syntax:

ViStatus _VI_FUNC hp816x_set_PWM_minMax(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*, ViInt32 *minmaxMode*, ViInt32 *dataPoints*, ViUInt32 *estimatedTime*);

The Set PWM MinMax function (hp816x_set_PWM_minMax) sets the parameters for a MinMax logging operation.

A MinMax logging operation is active as long as it is not explicitly stopped. Before you start a different Power Meter data acquisition function or different MinMax logging operation, you must call the Stop PWM Function function.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3

hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

minmaxMode

Use the MinMax control to choose one of the following MinMax modes:

Continuous mode (hp816x_MM_CONT),

Window mode (hp816x_MM_WIN), or

Refresh mode (hp816x_MM_REFRESH).

See your instrument's User's Guide for

information on MinMax modes.

Data Type: ViInt32
Input/Output: IN

Values:

Name	Value	Description
hp816x_MM_CONT	0	Continuous
hp816x_MM_WIN	1	Window
hp816x_MM_REFRESH	2	Refresh

dataPoints

Use Data Points to define the number of samples used for the MinMax logging operation. Data Points affects Refresh mode and Window mode.

See your instrument's User's Guide for information on Data Points.

Data Type: ViInt32
Input/Output: IN

estimatedTime

Estimated Time returns the time, in seconds, it will take until a minimum or maximum value is available.

This value can be used to adjust the time before you can call the Get PWM MinMax Results function (get_PWM_minMaxResults_Q). This does not apply to Continuous MinMax mode.

Data Type: ViUInt32
Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_PWM_minMaxResults_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_PWM_minMaxResults_Q(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*, ViPReal64 *minimum*, ViPReal64 *maximum*, ViPReal64 *current*);

The Get PWM MinMax function (hp816x_get_PWM_minMax) returns the minimum, maximum and current values of a previously started MinMax operation.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

minimum

Minimum returns the minimum power value in Watts for the MinMax operation.

Data Type: ViPReal64

Input/Output: OUT

maximum

Maximum returns the maximum power value in Watts for the MinMax operation.

Data Type: ViPReal64
Input/Output: OUT

current

Current returns the current power value in Watts.

Data Type: ViPReal64
Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_PWM_functionStop

Syntax:

ViStatus _VI_FUNC hp816x_PWM_functionStop(ViSession *ihandle*, ViInt32 *PWMSlot*, ViInt32 *channelNumber*);

The Stop PWM Function function (hp816x_PWM_functionStop) stops (aborts) any activated logging, stability or Min Max data aquistion operation.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMSlot	Use the PWM Slot control to choose an installed Power Meter (a Power Sensor or Optical Head Interface Module). **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_spectralCalibrationEx

Syntax:

ViStatus _VI_FUNC hp816x_spectralCalibrationEx(ViSession *ihandle,* ViInt32 *Slot*, ViInt32 *channelNumber*, ViUInt32 *Size_of_Spectrum*, ViReal64 *Wavelength*[ ], ViReal64 *Power*[ ], ViReal64 * *Wavelength_Result*, ViChar *Error_Diagnose*);

Use the Spectral Calibration function (hp816x_spectralCalibrationEx) to calibrate a power meter to an effective wavelength for a multiwavelength or broadband signal. For accurate power measurement of a multiwavelength or broadband signal, you can set the power meter to its effective wavelength (corresponding to the required weighted-averageresponsivity factor) for the signal. This extended "Ex" function allows selection of the channel number for dual-channel power meter modules.

Remark: This function only applies to power meters or Attenuator modules with power control, specifically the 81576A and 81577A. These modules use a filter technology with minimum wavelength dependence, but incorporate a power meter whose responsivity varies with wavelength.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
Slot	Use the Slot control to specify the slot used to host the Attenuator module (or powermeter). **Data Type:** ViInt32 **Input/Output:** IN **Values:** Name Value Description

hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

Size_of_Spectrum

Use the Size of Spectrum control to specifies the number of wavelength:power pairs in the multiwavelength DWDM signal.

Data Type: ViUInt32

Input/Output: IN

Wavelength

Input the array of wavelength values

Data Type: ViReal64[ ]
Input/Output: IN

Power

Input the array of power values in W

Data Type: ViReal64[ ]
Input/Output: IN

Wavelength_Result

Returns the effective wavelength corresponding to the required powermeter responsivity factor for the DWDM signal.

If 0.0 is returned, the spectral calibration failed.

Data Type: ViReal64
Input/Output: OUT

Error_Diagnose

Returns a diagnostic message, which may help you understand why a spectral calibration failed.

Data Type: ViChar
Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_getWlRespTblSizeEx

Syntax:

ViStatus _VI_FUNC hp816x_getWlRespTblSizeEx (ViSession *ihandle*, ViInt32 *Slot*, ViInt32 *channelNumber*, ViUInt32 * *Size*, ViUInt32 * *CSV_Size*);

Use the Get Wavelength Resp. Table Size function (hp816x_getWlRespTblSizeEx) to query the size of the wavelength response table used to calibrate a power meter or an Attenuator module's built-in power meter.

Remark: This function only applies to power meters or Attenuator modules with power control, specifically the 81576A and 81577A. These modules use a filter technology with minimum wavelength dependence, but incorporate a power meter whose responsivity varies with wavelength. This extended "Ex" function allows selection of the channel number for dual-channel power meter modules.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
Slot	Use the Slot control to specify the slot used to host the power meter. **Data Type:** ViInt32 **Input/Output:** IN **Values:** Name / Value / Description **hp816x_SLOT_1** / 1 / Slot 1 **hp816x_SLOT_2** / 2 / Slot 2 **hp816x_SLOT_3** / 3 / Slot 3

hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

Size

Returns the size of the power meter's wavelength response calibration table.

Remark: Use this value to allocate memory prior to using the Read Wavelength Response Table function (**hp816x_readWlRespTableEx**).

Data Type: ViUInt32

Input/Output: OUT

CSV_Size

Returns the number of bytes required by the power meter's wavelength response calibration table in CSV format.

Remark: Use this value to allocate memory prior to using the Read Wavelength Response Table function (**hp816x_readWlRepTblCSV_Ex**).

Data Type: ViUInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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E0615, Jun 2015



hp816x_readWlRespTableEx

Syntax:

ViStatus _VI_FUNC hp816x_getWlRespTableEx (ViSession *ihandle*, ViInt32 *Slot*, ViInt32 *channelNumber*, ViReal64 *Wavelength* [ ], ViReal64 *Response_Factor* [ ]);

Use the Read Wavelength Resp. Table function (hp816x_readWlRespTableEx) to read the wavelength response table used to calibrate a power meter or an Attenuator module's built-in power meter.

Use the Get Wavelength Resp. Table Size function (hp816x_getWvlRespTblSizeEx) to query the size of this table.

Remark: This function only applies to power meters or Attenuator modules with power control, specifically the 81576A and 81577A. These modules use a filter technology with minimum wavelength dependence, but incorporate a power meter whose responsivity varies with wavelength. This extended "Ex" function allows selection of the channel number for dual-channel power meter modules.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
Slot	Use the Slot control to specify the slot used to host the power meter. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

Wavelength

Wavelength is both an input and an output.

As an input, use a Real64 array with size given by the Size result from the hp816x_getWRespTblSizeEx query.

Returns the wavelength entries from the powermeter's wavelength response calibration

table.

Data Type: ViReal64[ ]

Input/Output: OUT

Response_Factor

Response_Factor is both an input and an output.

As an input, use a Real64 array with size given by the Size result from the hp816x_getWRespTblSizeEx query.

Returns the response factor entries from the powermeter's wavelength response calibration table.

Data Type: ViReal64[ ]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_readWlRepTblCSV_Ex

Syntax:

ViStatus _VI_FUNC hp816x_getWlRepTblCSV_Ex (ViSession *ihandle*, ViInt32 *Slot*, ViInt32 *channelNumber*, ViChar *CSV_List* [ ]);

Use the Read Wl Rep. Table CSV function (hp816x_readWlRepTblCSV_Ex) to read the wavelength response table used to calibrate a power meter or an Attenuator module's built-in power meter in CSV format.

Use the Get Wavelength Resp. Table Size function (hp816x_getWvlRespTblSizeEx) to query the size of this table.

Remark: This function only applies to power meters or Attenuator modules with power control, specifically the 81576A and 81577A. These modules use a filter technology with minimum wavelength dependence, but incorporate a power meter whose responsivity varies with wavelength.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
Slot	Use the ATT Slot control to specify the slot used to host the Attenuator module. **Data Type:** ViInt32 **Input/Output:** IN **Values:** Name: **hp816x_SLOT_1** — Value: 1 — Description: Slot 1

hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

channelNumber

If you choose a Dual Power Sensor Module or a Dual Optical Head Interface Module, you can use the Channel Number control to choose between Channel 1 and Channel 2 of your module.

If you choose a single-channel Power Meter, use the Channel Number to choose Channel 1.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CHAN_1	0	Channel 1
hp816x_CHAN_2	1	Channel 2

CSV_List

CSV_List is both an input and an output.

As an input, use a text string with length given by the CSV_Size result from the hp816x_getWRespTblSizeEx query.

Returns the wavelength, response_factor entries from the powermeter's wavelength response calibration table in CSV format.

Data Type: ViChar[ ]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816xerror_message"

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hp816x_set_FLS_parameters

Syntax:

ViStatus _VI_FUNC hp816x_set_FLS_parameters(ViSession *ihandle*, ViInt32 *FLSSlot*, ViInt32 *wavelengthSource*, ViBoolean *turnLaser*, ViInt32 *modulationLowerSource*, ViInt32 *modulationUpperSource*, ViReal64 *modulationFreqLowerSource*, ViReal64 *modulationFreqUpperSource*, ViReal64 *attenuationLowerSource*, ViReal64 *attenuationUpperSource*);

The Set FLS Parameters function (hp816x_set_FLS_parameters) enables you to set up Laser Sources quickly. In case of a Single-Wavelength Laser Source Module, the inputs for the lower source will be used.

Remark: For dual-wavelength Laser Sources with part numbers less than 81600, the modulation settings for the upper source will be ignored; only the lower source modulation settings are valid.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
FLSSlot	Use the FLS Slot control to choose an installed Fixed Laser Source Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3

hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

wavelengthSource

For a Laser Source Module, you can use the Wavelength Source control to select one of the following:

Lower (hp816x_LOWER_SRC), the shorter wavelength,

Upper (hp816x_UPPER_SRC), the longer wavelength,or

Both (hp816x_BOTH_SRC), both the Upper and Lower wavelengths.

This control is ignored for a Single-Wavelength Laser Source Module.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_LOWER_SRC	0	Lower
hp816x_UPPER_SRC	1	Upper
hp816x_BOTH_SRC	2	Both

turnLaser

Use the Turn Laser On/Off control to turn the Laser Source On (VI_TRUE) or Off (VI_FALSE).

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

modulationLowerSource

Use the Modulation Lower Source control to select the modulation source for a single-wavelength Fixed Laser Source. You can choose one of the following options:

Off (hp816x_AM_OFF) turns the modulation off;

Internal (hp816x_AM_INT), internal digital modulation, or

Coherence Control (hp816x_AM_CC), coherence control.

See your instrument's User's Guide for more information on modulation sources.

NOTE:

Coherence control is not available for for modules with product IDs less than 81600.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_AM_OFF	0	Off
hp816x_AM_INT	1	Internal
hp816x_AM_CC	2	Coherence Control

modulationUpperSource

Use the Modulation Upper Source control to select the modulation source for the upper wavelength source:

Off (hp816x_AM_OFF) turns the modulation off,

Internal (hp816x_AM_INT), internal digital modulation, or

Coherence Control (hp816x_AM_CC), coherence control.

See your instrument's User's Guide for more information on modulation sources.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_AM_OFF	0	Off
hp816x_AM_INT	1	Internal
hp816x_AM_CC	2	Coherence Control

modulationFreqLowerSource

Use the Modulation Freq Lower Source control to set the modulation frequency to a value in Hertz (Hz) for the lower wavelength source.

This input sets the modulation frequency for a single-wavelength Laser Source.

Data Type: ViReal64

Input/Output: IN

modulationFreqUpperSource

Use this Modulation Freq Upper Source control to set the modulation frequency to a value in Hertz (Hz) for the upper wavelength source.

Data Type: ViReal64

Input/Output: IN

attenuationLowerSource

Use the Attenuation Lower Source control to set the attenuation to a value in decibels (dB) for the lower wavelength source.

Data Type: ViReal64

Input/Output: IN

attenuationUpperSource

The Attenuation Upper Source input sets the attenuation to a value in decibels (dB) for the upper wavelength Laser Source.

This control only applies to dual-wavelength Laser Source.

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_FLS_parameters_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_FLS_parameters_Q(ViSession *ihandle*, **ViInt32** *FLSSlot*, **ViPInt32** *wavelengthSource*, **ViPBoolean** *turnLaser*, **ViPInt32** *modulationLowerSource*, **ViPInt32** *modulationUpperSource*, **ViPReal64** *modulationFreqLowerSource*, **ViPReal64** *modulationFreqUpperSource*, **ViPReal64** *attenuationLowerSource*, **ViPReal64** *attenuationUpperSource*);

The Set FLS Parameters function (**hp816x_set_FLS_parameters**) enables you to set up Laser Sources quickly. In case of a Laser Source with a single optical output, the inputs for the lower source will be used.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
FLSSlot	Use the FLS Slot control to choose an installed Fixed Laser Source Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

wavelengthSource

Wavelength Source returns the selected wavelength source as one of the following:

0 (hp816x_LOWER_SRC), the shorter wavelength, or

1 (hp816x_UPPER_SRC), the longer wavelength, or

2 (hp816x_BOTH_SRC), both the Upper and Lower wavelengths.

For a Single-Wavelength Laser Source Module, this control is ignored.

Data Type: ViPInt32

Input/Output: OUT

turnLaser

Turn Laser returns the laser state as one of the following:

0 (VI_TRUE), Laser Source On, or

1 (VI_FALSE), Laser Source Off.

Data Type: ViPBoolean

Input/Output: OUT

modulationLowerSource

Modulation Lower Source returns the one of the following modulation sources for a single-wavelength Fixed Laser Source and for both upper and lower output signals of the 81554SM dual-wavelength Fixed Laser Source.

1 (hp816x_AM_INT), internal digital modulation, or

2 (hp816x_AM_CC), coherence control.

Modulation Lower Source returns the modulation source for a single-wavelength Fixed Laser Source and for both upper and lower output signals of the 81554SM dual-wavelength Fixed Laser Source.

See your instrument's User's Guide for more information on modulation sources.

NOTE:

Coherence control is not available for for modules with product IDs less than 81600.

Data Type: ViPInt32

Input/Output: OUT

modulationUpperSource

Modulation Upper Source returns the modulation source for the upper wavelength source:

1 (hp816x_AM_INT), internal digital modulation, or

2 (hp816x_AM_CC), coherence control.

See your instrument's User's Guide for more information on modulation sources.

Data Type: ViPInt32

Input/Output: OUT

modulationFreqLowerSource

Modulation Freq Lower Source returns the modulation frequency as a value in Hertz (Hz) for the lower wavelength source.

Modulation Freq Lower Source returns the modulation frequency for a single-wavelength Fixed Laser Source and for both upper and lower output signals of than 81554SM dual-wavelength Fixed Laser Source.

Data Type: ViPReal64

Input/Output: OUT

modulationFreqUpperSource

Modulation Freq Upper Source returns the modulation frequency as a value in Hertz (Hz) for the upper wavelength Source.

This control only applies to dual-wavelength Fixed Laser Source product numbers greater than 81600.

Data Type: ViPReal64

Input/Output: OUT

attenuationLowerSource

Attenuation Lower Source returns the attenuation as a value in decibels (dB) for the lower wavelength source.

Data Type: ViPReal64

Input/Output: OUT

attenuationUpperSource

Attenuation Upper Source returns the attenuation level as a value in decibels (dB) for the upper wavelength Laser Source.

This control only applies to dual-wavelength Laser Source.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_FLS_modulation

Syntax:

ViStatus _VI_FUNC hp816x_set_FLS_modulation(ViSession *ihandle*, ViInt32 *FLSSlot*, ViInt32 *laserSource*, ViInt32 *modulationFrequency*, ViInt32 *modulationSource*, ViReal64 *manualFrequency*);

The Setup FLS Modulation function (hp816x_set_FLS_modulation) adjusts the internal modulation of a Fixed Laser Source.

Internal Modulation is described in your instrument's User's Guide.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
FLSSlot	Use the FLS Slot control to choose an installed Fixed Laser Source Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

laserSource

For a dual-wavelength Laser Source Module, you can use the Wavelength Source control to select the longer wavelength, Upper (hp816x_UPPER_SRC), or the shorter wavelength, Lower (hp816x_LOWER_SRC).

For a single-wavelength Laser Source Module, the default is Lower.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_LOWER_SRC	0	Lower
hp816x_UPPER_SRC	1	Upper

modulationFrequency

You can use the Modulation Frequency control to select one of the following:

Minimum (hp816x_AM_MIN), which is the minimum frequency,

Maximum (hp816x_AM_MAX), which is the maximum frequency,

Default (hp816x_AM_DEFAULT), which is the sum of the minimum frequency and the maximum frequency divided by two (this is not the same value that the Reset function, the Preset function, or the Preset hardkey sets the frequency to), and

Manual Input (hp816x_AM_MANUAL), which is the value entered in the Manual Frequency input.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_AM_MIN	0	Minimum
hp816x_AM_MAX	1	Maximum
hp816x_AM_DEFAULT	2	Default
hp816x_AM_MANUAL	3	Manual Input

modulationSource

Use Modulation Source to choose one of the following modulation sources:

Off (hp816x_AM_OFF) turns the modulation off;

Internal (hp816x_AM_INT), internal digital modulation, or

Coherence Control (hp816x_AM_CC), coherence control.

See your instrument's User's Guide for more information on modulation sources.

ATTENTION:

Coherence Control is only supported by modules with a product number greater than 81600.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_AM_OFF	0	Off
hp816x_AM_INT	1	Internal
hp816x_AM_CC	2	Coherence Control

manualFrequency

If you choose Manual Input as the Modulation Frequency input, you can use the Manual Frequency input to set the modulation frequency to a value in Hertz (Hz).

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_FLS_modulation_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_FLS_modulationSettings_Q(ViSession *ihandle*, **ViInt32** *FLSSlot*, **ViInt32** *wavelengthSource*, **ViPBoolean** *modulationState*, **ViPInt32** *modulationSource*, **ViPReal64** *minimumFrequency*, **ViPReal64** *maximumFrequency*, **ViPReal64** *defaultFrequency*, **ViPReal64** *currentFrequency*);

The Get FLS Modulation Settings function (hp816x_get_FLS_modulationSettings_Q) returns the current values of the following: Modulation State, Minimum Frequency, Maximum Frequency, Default Frequency, and Current Frequency.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
FLSSlot	Use the FLS Slot control to choose an installed Fixed Laser Source Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2

hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

wavelengthSource

For a dual-wavelength Laser Source Module, you can use the Wavelength Source control to select the longer wavelength, Upper (hp816x_UPPER_SRC), or the shorter wavelength, Lower (hp816x_LOWER_SRC).

For a single-wavelength Laser Source Module, the default is Lower.

An error is returned for modules with part numbers lower than 81600 if you choose Upper.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_LOWER_SRC	0	Lower
hp816x_UPPER_SRC	1	Upper

*modulation*State

Modulation returns whether the output signal is modulated:

1 (hp816x_MOD_ENABLED), if modulation is enabled, or

0 (hp816x_MOD_DISABLED), if modulation is disabled.

Data Type: ViPBoolean

Input/Output: OUT

modulationSource

Modulation Source returns the chosen modulation source as one of the following:

1 (hp816x_MOD_INT): Internal digital modulation is chosen, or

2 (hp816x_MOD_CC): Coherence Control is chosen.

See your instrument's User's Guide for more information on modulation sources.

Data Type: ViPInt32

Input/Output: OUT

minimumFrequency

Minimum Frequency returns the lowest frequency that your Fixed Laser Source can be set to.

Frequency is returned in units of Hertz (Hz).

Data Type: ViReal64

Input/Output: OUT

maximumFrequency

Maximum Frequency returns the highest frequency that your Fixed Laser Source can be set to.

Frequency is returned in units of Hertz (Hz).

Data Type: ViReal64

Input/Output: OUT

defaultFrequency

Default Frequency returns the default frequency of your Fixed Laser Source. This is identical to the default value set by setting the Modulation Frequency control of the Set FLS Modulation function to Default; it is the sum of the Minimum Frequency and the Maximum Frequency divided by two. This is not the same value that the Reset function, the Preset function, or the Preset hardkey sets the frequency to.

Frequency is returned in units of Hertz (Hz).

Data Type: ViReal64

Input/Output: OUT

currentFrequency

Current Frequency returns the modulation frequency that your Fixed Laser Source can be set to..

Frequency is returned in units of Hertz (Hz).

Data Type: ViReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_FLS_laserSource

Syntax:

ViStatus _VI_FUNC hp816x_set_FLS_laserSource(ViSession *ihandle*, ViInt32 *FLSSlot*, ViInt32 *laserSource*);

The Set FLS Laser Source function (hp816x_set_FLS_laserSource) chooses, which laser output, of a dual-wavelength Fixed Laser Source, is output.

This function is irrelevant for single-wavelength Fixed Laser Sources.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
FLSSlot	Use the FLS Slot control to choose an installed Fixed Laser Source Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

laserSource

For a Laser Source Module, you can use the Laser Source control to select one of the following:

Upper (hp816x_UPPER_SRC), the longer wavelength, Lower (hp816x_LOWER_SRC), the shorter wavelength, or Both (hp816x_BOTH_SRC), both the Upper and Lower wavelengths.

For a Single-Wavelength Laser Source Module, the default is Lower.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_LOWER_SRC	0	Lower
hp816x_UPPER_SRC	1	Upper
hp816x_BOTH_SRC	2	Both

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_FLS_laserSource_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_FLS_laserSource_Q(ViSession *ihandle*, ViInt32 *FLSSlot*, ViPInt32 *laserSource*);

The Get Laser Source function (hp816x_get_FLS_laserSource_Q) returns, for dual-wavelength Fixed Laser Sources, which wavelength is switched to the output.

This function is irrelevant for a single-wavelength Fixed Laser Source.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
FLSSlot	Use the FLS Slot control to choose an installed Fixed Laser Source Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

laserSource

Laser Source returns one of the following values:

0 (hp816x_LOWER_SRC): the shorter wavelength is the selected output.

1 (hp816x_UPPER_SRC): the longer wavelength is the selected output.

2 (hp816x_L_SRC_BOTH): both wavelengths are output simultaneously.

Data Type: ViPInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_FLS_wavelength_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_FLS_wavelength_Q(ViSession *ihandle*, ViInt32 *FLSSlot*, ViPReal64 *wavelengthLowerSource*, ViPReal64 *wavelengthUpperSource*);

The Get FLS Wavelength function (hp816x_get_FLS_wavelength_Q) returns the wavelength of a Fixed Laser Source.

REMARK:

Wavelength Upper Source returns 0.0 if a single-wavelength Fixed Laser Source is installed.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
FLSSlot	Use the FLS Slot control to choose an installed Fixed Laser Source Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2

hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

wavelengthLowerSource

Wavelength Lower Source returns the wavelength of a single-wavelength Fixed Laser Source or the shorter wavelength of a dual-wavelength Fixed Laser Source.

Wavelength is returned in meters (m).

Data Type: ViPReal64

Input/Output: OUT

wavelengthUpperSource

Wavelength Upper Source returns the upper wavelength of a dual-wavelength Fixed Laser Source.

Wavelength is returned in meters (m).

The value, 0, is returned for a single-wavelength Fixed Laser Source.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_FLS_attenuation

Syntax:

ViStatus _VI_FUNC hp816x_set_FLS_attenuation(ViSession *ihandle*, ViInt32 *FLSSlot*, ViReal64 *attenuationLowerSource*, ViReal64 *attenuationUpperSource*);

The Set FLS Attenuation function (hp816x_set_FLS_attenuation) sets the attenuation of a Fixed Laser Source.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
FLSSlot	Use the FLS Slot control to choose an installed Fixed Laser Source Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

attenuationLowerSource

Attenuation Lower Source sets the attenuation of a single-wavelength Fixed Laser Source or the attenuation of the shorter wavelength source of a dual-wavelength Fixed Laser Source.

Attenuation is set in units of decibels (dB).

Data Type: ViReal64

Input/Output: IN

attenuationUpperSource

Attenuation Upper Source sets the attenuation of the longer wavelength source of a dual-wavelength Fixed Laser Source.

Attenuation is set in units of decibels (dB).

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_FLS_attenuation_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_FLS_attenuation_Q(ViSession *ihandle*, ViInt32 *FLSSlot*, ViPReal64 *attenuationLowerSource*, ViPReal64 *attenuationUpperSource*);

The Get FLS Attenuation function (hp816x_get_FLS_attenuation_Q) returns the set attenuation of a Fixed Laser Source.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
FLSSlot	Use the FLS Slot control to choose an installed Fixed Laser Source Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

attenuationLowerSource

Attenuation Lower Source returns the set attenuation of a single-wavelength Fixed Laser Source or the attenuation of the shorter wavelength source of a dual-wavelength Fixed Laser Source.

Attenuation is returned in units of decibels (dB).

Data Type: ViPReal64

Input/Output: OUT

attenuationUpperSource

Attenuation Upper Source returns the set attenuation of the longer wavelength source of a dual-wavelength Fixed Laser Source.

0.0 is returned for a single-wavelength Fixed Laser Source.

Attenuation is returned in units of decibels (dB).

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_FLS_triggerState

Syntax:

ViStatus _VI_FUNC hp816x_set_FLS_triggerState(ViSession *ihandle*, ViInt32 *FLSSlot*, ViBoolean *outputTrigger*);

The Set FLS Trigger State (hp816x_set_FLS_triggerState) function specifies when an output trigger is generated.

You can enable an output trigger to be generated for every leading edge of a digitally-modulated TTL signal.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
FLSSlot	Use the FLS Slot control to choose an installed Fixed Laser Source Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

outputTrigger

Use Output Trigger to enable an output trigger to be generated for every leading edge of a digitally-modulated TTL signal.

Choose Enable (VI_TRUE) to enable an output trigger to be generated for every leading edge of a digitally-modulated TTL signal.

Choose Disable (VI_FALSE) to disable output trigger for the chosen Laser Source.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Enable
VI_FALSE	(0)	Disable

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_FLS_laserState_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_FLS_laserState_Q(ViSession *ihandle*, ViInt32 *FLSSlot*, ViBoolean *outputTrigger*);

The Get FLS Trigger State (hp816x_get_FLS_triggerState) function returns whether an output trigger is generated for every leading edge of a digitally-modulated TTL signal.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
FLSSlot	Use the FLS Slot control to choose an installed Fixed Laser Source Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

outputTrigger

Output Trigger returns whether an output trigger is generated for every leading edge of a digitally-modulated TTL signal.

If 0 (VI_FALSE) is returned, an output trigger is not generated for every leading edge of a digitally-modulated TTL signal.

If 1 (VI_TRUE) is returned, an output trigger is generated for every leading edge of a digitally-modulated TTL signal.

Data Type: ViBoolean

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_FLS_power

Syntax:

ViStatus _VI_FUNC hp816x_get_FLS_power(ViSession *ihandle*, ViInt32 *FLSSlot*, ViInt32 *laserSource*, ViPReal64 *laserPower*);

The Get FLS Power function (hp816x_get_FLS_power_Q) returns the output power of a fixed-wavelength Laser Source in Watts.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
FLSSlot	Use the FLS Slot control to choose an installed Fixed Laser Source Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

laserSource

For a Laser Source Module, you can use the Laser Source control to select one of the following:

Upper (hp816x_UPPER_SRC), the longer wavelength source, or Lower (hp816x_LOWER_SRC), the shorter wavelength source.

For a Single-Wavelength Laser Source Module, the default is Lower.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_LOWER_SRC	0	Lower
hp816x_UPPER_SRC	1	Upper

laserPower

Laser Power returns the the laser beam power of the current selected Laser Source in Watts.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_FLS_laserState

Syntax:

ViStatus _VI_FUNC hp816x_set_FLS_laserState(ViSession *ihandle*, ViInt32 *FLSSlot*, ViBoolean *laserState*);

The Set FLS Laser State function (hp816x_set_FLS_laserState) turns the laser on or off.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
FLSSlot	Use the FLS Slot control to choose an installed Fixed Laser Source Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

laserState

Use the Laser State control to turn the laser output On (VI_TRUE) or Off (VI_FALSE).

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_FLS_laserState_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_FLS_laserState_Q(ViSession *ihandle*, ViInt32 *FLSSlot*, ViPBoolean *laserState*);

The Get FLS Laser State function (hp816x_get_FLS_laserState_Q) returns whether the laser is turned on or off.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
FLSSlot	Use the FLS Slot control to choose an installed Fixed Laser Source Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

laserState

Laser State returns whether the laser is turned on or off.

If 0 (VI_FALSE) is returned, the laser is turned off.

If 1 (VI_TRUE) is returned, the laser is turned on.

Data Type: ViPBoolean

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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Information for N77 Series Instruments

This version of the 816x VXIplug&play instrument driver has been extended to support the N773xA, N774xA optical power meter and the N775xA and N776xA Optical Attenuator instruments. These are stand-alone instruments that do not require installation in an 816x mainframe. The SCPI command set has been designed for almost complete programming compatibility with the modular instruments of the 816x family. Please be aware of the following details and limitations, when using this driver.

- The N77-series instruments can be controlled via USB, LAN or GPIB interfaces. Especially for transferring large data volumes, the USB and usually LAN interfaces provide faster performance.
- The identification of the individual ports of these multiport instruments in the SCPI and hp816x functions corresponds to the slot number. So for example the ports of the N7745A can be addressed with the slot numbers 1 to 8. The channel number in the commands and functions is not used and can generally be omitted or set to Channel 1 (index 0).
- The N771x-series and 81950A tunable lasers are not supported by the 816x VXIplug&play driver.
- The N773x-series optical switches are supported by the 816x VXIplug&play functions for switches and relevant functions for the mainframes.
- The N774x-series power meters are supported by the 816x VXIplug&play functions for power meters, relevant functions for the mainframes like for standard trigger configuration, and the Multi Frame Lambda Scan application functions.
- The N776x-series optical attenuators are supported by the 816x VXIplug&play functions for attenuators and the relevant functions for the mainframes.
- The N775x-series optical attenuators and power meters are supported by the 816x VXIplug&play functions for attenuators and the relevant functions for the mainframes and the basic functions for the power meters, excluding the advanced functions using data power logging. The choices of averaging time is also more limited for these power meters.
- The averaging time parameter for the power meters is limited in the 816x VXIplug&play functions to a limited set of discrete values, similar to the modular 816x power meters. This set includes: 1, 2, 5, 10, 25, 100, 200, and 500 ìs, 1, 2, 5, 10, 20, 50, 100, 200, 500 ms, and 1, 2, 5, and 10s.
- The maximum number of logging data points that can be acquired is limited by the 816x VXIplug&play data acquisition functions to 100,000 per port.
- The shortest averaging time used by the Multi Frame Lambda Scan application is 25 ìs.
- For the Multi Frame Lambda Scan application, the N774xA instruments can be included simply by registering the instrumet as a mainframe,after the first

mainframe with the tunable laser has been registered.

Further details to available functions are found under the categories below:

Utility Functions

Mainframe Specific Functions

Power Sensor Specific Functions

Applications

hp816x_get_SWT_type

Syntax:

ViStatus _VI_FUNC hp816x_get_SWT_type(ViSession *ihandle*, ViInt32 *SWT_Slot*, ViPInt32 * *Switch_Type*, ViChar *Switch_Description*);

The Get Switch Type function (hp816x_get_SWT_type) returns a number and description corresponding to the switch module hosted by a specified slot.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
SWT_Slot	Use the SWT Slot control to specify the slot used to host the Switch module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Switch_Type

Switch Type returns an integer corresponding to the type of switch module hosted in SWT Slot.

Data Type: ViInt32
Input/Output: OUT

Switch_Description

Switch Description returns a textural description of the specified switch module, including:

"1:2 switch module"
"2*1:2 switch module"
"2x2 switch module"
"1:4 switch module"
"2:4 switch module"
"1:8 switch module"
"unknown"

Data Type: ViChar
Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_SWT_route

Syntax:

ViStatus _VI_FUNC hp816x_set_SWT_route(ViSession *ihandle*, ViInt32 *SWT_Slot*, ViPInt32 *Input*, ViInt32 *Output*);

Use the Set Route function (hp816x_set_SWT_route) to set the path between port A and another numbered port within a switch module and, if applicable, port B and another numbered port.

Valid paths depend on the type of switch module. Refer to the Get Switch Type function (hp816x_get_SWT_type) and the Get Route Table function (hp816x_get_SWT_routeTable).

Remarks: Although ports A and B are referred to as inputs, and numbered ports as outputs, the paths set are bi-directional.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
SWT_Slot	Use the SWT Slot control to specify the slot used to host the Switch module. **Data Type:** ViInt32 **Input/Output:** IN **Values:** **hp816x_SLOT_1** — 1 — Slot 1 **hp816x_SLOT_2** — 2 — Slot 2

hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Input

Use the Input control to choose either port A or, if applicable to the switch type, port B.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_SWT_INP_A	0	Port A
hp816x_SWT_INP_B	1	Port B

Output

Use the Output control to choose the numbered port to be connected to the port specified by the Input control.

Remarks: If the switch is a dual channel device, the channels are entirely separate. So, port A can be coonnected to port 1 and port 2, and port B can be connected to another port 1 and another port 2.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_SWT_OUT_1	0	Port 1
hp816x_SWT_OUT_2	1	Port 2
hp816x_SWT_OUT_3	2	Port 3
hp816x_SWT_OUT_4	3	Port 4
hp816x_SWT_OUT_5	4	Port 5
hp816x_SWT_OUT_6	5	Port 6
hp816x_SWT_OUT_7	6	Port 7
hp816x_SWT_OUT_8	7	Port 8

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_SWT_route

Syntax:

ViStatus _VI_FUNC hp816x_get_SWT_route(ViSession *ihandle*, ViInt32 *SWT_Slot*, ViPInt32 *Input*, ViInt32 * *Output*);

Use the Get Route function (hp816x_get_SWT_route) to query the path between port A and another numbered port within a switch module and, if applicable, port B and another numbered port.

Remarks: Although ports A and B are referred to as inputs, and numbered ports as outputs, the paths set are bi-directional.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
SWT_Slot	Use the SWT Slot control to specify the slot used to host the Switch module. **Data Type:** ViInt32 **Input/Output:** IN **Values:** Name / Value / Description **hp816x_SLOT_1** / 1 / Slot 1 **hp816x_SLOT_2** / 2 / Slot 2 **hp816x_SLOT_3** / 3 / Slot 3 **hp816x_SLOT_4** / 4 / Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Input

Use the Input control to choose either port A or, if applicable to the switch type, port B.

Use the Get Switch Type function (**hp816x_get_SWT_type**) to query the switch type hosted by a specified slot.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_SWT_INP_A	0	Port A
hp816x_SWT_INP_B	1	Port B

Output

Returns the number of the port connected to the port specified by the Input control.

Remarks: If the switch is a dual channel device, the channels are entirely separate. So, port A can be connected to port 1 and port 2, and port B can be connected to another port 1 and another port 2.

Data Type: ViInt32

Input/Output: OUT

Values:

Name	Value	Description
hp816x_SWT_OUT_1	0	Port 1
hp816x_SWT_OUT_2	1	Port 2
hp816x_SWT_OUT_3	2	Port 3
hp816x_SWT_OUT_4	3	Port 4
hp816x_SWT_OUT_5	4	Port 5
hp816x_SWT_OUT_6	5	Port 6
hp816x_SWT_OUT_7	6	Port 7
hp816x_SWT_OUT_8	7	Port 8

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_SWT_routeTable

Syntax:

ViStatus _VI_FUNC hp816x_get_SWT_routeTable (ViSession *ihandle*, ViInt32 *SWT_Slot*, ViPInt32 *Input*, ViChar *RouteTable* [ ]);

Use the Get Route Table function (hp816x_get_SWT_routeTable) to query the paths allowed between a specified switch module's ports.

Remarks: Although ports A and B are referred to as inputs, and numbered ports as outputs, the allowed paths are bi-directional.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
SWT_Slot	Use the SWT Slot control to specify the slot used to host the Switch module. **Data Type:** ViInt32 **Input/Output:** IN **Values:** Name / Value / Description **hp816x_SLOT_1** / 1 / Slot 1 **hp816x_SLOT_2** / 2 / Slot 2 **hp816x_SLOT_3** / 3 / Slot 3 **hp816x_SLOT_4** / 4 / Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Input

Use the Input control to choose either port A or, if applicable to the switch type, port B.

Use the Get Switch Type function (**hp816x_get_SWT_type**) to query the switch type hosted by a specified slot.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_SWT_INP_A	0	Port A
hp816x_SWT_INP_B	1	Port B

Route_Table

Returns the route table for the specified switch module as an array of strings.

Remarks: If the switch is a dual channel device, the channels are entirely separate. So, port A can be connected to port 1 and port 2, and port B can be connected to another port 1 and another port 2.

Data Type: ViInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_RLM_parameters

Syntax:

ViStatus _VI_FUNC hp816x_set_RLM_parameters(ViSession *ihandle*, ViInt32 *RLMSlot*, ViBoolean *internalTrigger*, ViReal64 *wavelength*, ViReal64 *averagingTime*, ViBoolean *laserSource*, ViBoolean *laserState*);

The Set RLM Parameters function (hp816x_set_RLM_parameters) sets the most important parameters for a Return Loss module, preparing the instrument to start a measurement.

If a value is not yet applied to the instrument and the Force Transaction function (**hp816x_forceTransaction**) is switched off, the new values will be set.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8

hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

internalTrigger

Use the Internal Trigger control to choose between Continuous triggering and Immediate triggering.

If you select Immediate triggering (hp816x_RLM_IMMEDIATE) one measurement cycle is performed. When the averaging time is over, the result is available. No further measurement cycles are started. If the result is read with the Fetch RLM Return Loss (hp816 x_RLM_fetchReturnLoss) function, the value will always be the same.

Selecting Continous triggering (hp816x_RLM_CONTINUOUS) will continuous measurements are performed, because after finishing one measurement cycle, a new cycle is immediately triggered.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
hp816x_RLM_IMMEDIATE	0	Immediate
hp816x_RLM_CONTINUOUS	1	Continuous

wavelength

Use Wavelength to set the measurement wavelength in meters (m).

This is the Wavelength value. The responsivity of the Power Sensor varies with wavelength. For accurate power measurement, you need to input the wavelength of the optical input.

Data Type: ViReal64

Input/Output: IN

averagingTime

Use Averaging Time to set the length of time in seconds (s) over which a signal is averaged.

Longer averaging times increase the accuracy and improve the noise rejection of the measurement. Longer averaging times also decrease sensitivity.

Data Type: ViReal64

Input/Output: IN

laserSource

For a Return Loss module, you can use the Laser Source control to select one of the following:

Upper (hp816x_LOWER_SRC), the longer Laser Source wavelength or Lower (hp816x_UPPER_SRC), the shorter Laser Source wavelength.

For a single Laser Source wavelength Return Loss module, the default is Lower.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
hp816x_UPPER_SRC	1	Upper
hp816x_LOWER_SRC	0	Lower

laserState

Use the Laser State control to turn the laser output On (VI_TRUE) or Off (VI_FALSE).

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine

"hp816x_error_message"

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hp816x_get_RLM_parameters_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_RLM_parameters_Q(ViSession *ihandle*, ViInt32 *RLMSlot*, ViPInt32 *internalTrigger*, ViPReal64 *wavelength*, ViPReal64 *averagingTime*, ViPBoolean *laserSource*, ViPBoolean *laserState*);

The Get RLM parameters function (hp816x_get_RLM_parameters_Q) returns the most important parameters for a Return Loss module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

internalTrigger

Internal Trigger returns one of the following internal trigger configurations:

0 (hp816x_RLM_IMMEDIATE), where only one power measurement is triggered, or

1 (hp816x_RLM_CONTINUOUS), where power measurements are continuously triggered.

Data Type: ViPInt32

Input/Output: OUT

wavelength

Wavelength returns the measurement wavelength in meters (m).

This is the Wavelength value. The responsivity of the Power Sensor varies with wavelength. For accurate power measurement, you need to input the wavelength of the optical input.

Data Type: ViPReal64

Input/Output: OUT

averagingTime

Averaging Time returns the length of time in seconds (s) over which a signal is averaged.

Longer averaging times increase the accuracy and improve the noise rejection of the measurement. Longer averaging times also decrease sensitivity.

Data Type: ViPReal64

Input/Output: OUT

laserSource

Laser Source returns one of the following values:

0 (hp816x_LOWER_SRC) :the shorter wavelength is the selected output.

1 (hp816x_UPPER_SRC) :the longer wavelength is the selected output.

Data Type: ViPBoolean

Input/Output: OUT

laserState

Laser State returns whether the laser is turned on or off.

If 0 (VI_FALSE) is returned, the laser is turned off.

If 1 (VI_TRUE) is returned, the laser is turned on.

Data Type: ViPBoolean

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_RLM_internalTrigger

Syntax:

ViStatus _VI_FUNC hp816x_set_RLM_internalTrigger(ViSession *ihandle*, ViInt32 *RLMSlot*, ViBoolean *internalTrigger*);

The Set RLM Internal Trigger function (hp816x_set_RLM_internalTrigger) determines how the Return Loss module starts measuring.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11

hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

internalTrigger

Use the Internal Trigger control to choose between Continuous triggering (hp816x_RLM_CONTINUOUS) and Immediate triggering (hp816x_RLM_IMMEDIATE).

Selecting Immediate triggering performs one measurement cycle. When the averaging time is over, the result is available. No further measurement cycles are started. If the result is read with the Fetch RLM Return Loss (**hp816x_RLM_fetchReturnLoss**) function , the value will always be the same.

Selecting Continuous triggering will produce different results, because after finishing one measure cycle, a new cycle is immediately triggered.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
hp816x_RLM_CONTINUOUS	1	Continuous
hp816x_RLM_IMMEDIATE	0	Immediate

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_RLM_averagingTime

Syntax:

ViStatus _VI_FUNC hp816x_set_RLM_averagingTime(ViSession *ihandle*, ViInt32 *RLMSlot*, ViReal64 *averagingTime*);

The Set RLM Averaging Time function (hp816x_set_RLM_averagingTime) sets the averaging time of a Return Loss module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

averagingTime

Use Averaging Time to set the length of time over which a signal is averaged.

Longer averaging times increase the accuracy and improve the noise rejection of the measurement. Longer averaging times also decrease sensitivity.

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_RLM_averagingTime_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_RLM_averagingTime_Q(ViSession *ihandle*, ViInt32 *RLMSlot*, ViPReal64 *averagingTime*);

The Get RLM Averaging Time (hp816x_get_RLM_averagingTime_Q) function returns the averaging time of a Return Loss module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

averagingTime

Averaging Time returns the length of time over which a signal is averaged.

Longer averaging times increase the accuracy and improve the noise rejection of the measurement. Longer averaging times also decrease sensitivity.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_RLM_wavelength

Syntax:

ViStatus _VI_FUNC hp816x_set_RLM_wavelength(ViSession *ihandle*, ViInt32 *RLMSlot*, ViReal64 *wavelength*);

The Set RLM Wavelength function (hp816x_set_RLM_wavelength) sets the wavelength of a Return Loss module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

wavelength

Use Wavelength to set the measurement wavelength in meters (m).

The responsivity of the Power Sensor varies with wavelength. For accurate power measurement, you need to input the wavelength of the optical input.

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_RLM_wavelength_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_RLM_wavelength_Q(ViSession *ihandle*, ViInt32 *RLMSlot*, ViPReal64 *minWavelength*, ViPReal64 *maxWavelength*, ViPReal64 *defaultWavelength*, ViPReal64 *currentWavelength*);

The Get RLM Wavelength function (hp816x_get_RLM_wavelength_Q) returns the minimum, maximum, and current wavelength of a Return Loss module.

The responsivity of the Power Sensor varies with wavelength. For accurate power measurement, you need to input the wavelength of the optical input.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3

hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

minWavelength

Min Wavelength returns the lowest measurement wavelength that can be set in meters (m).

Data Type: ViPReal64

Input/Output: OUT

maxWavelength

Max Wavelength returns the highest measurement wavelength that can be set in meters (m).

Data Type: ViPReal64

Input/Output: OUT

defaultWavelength

Default Wavelength returns the default measurement wavelength of your Return Loss module. This is the sum of the Minimum Wavelength and the Maximum Wavelength divided by two. This is not the same value that the Reset function, the Preset function, or the Preset hardkey sets the wavelength to.

Wavelength is returned in meters (m).

Data Type: ViPReal64

Input/Output: OUT

currentWavelength

Current Wavelength returns the measurement wavelength in meters (m) that is currently set.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_RLM_powerRange

Syntax:

ViStatus _VI_FUNC hp816x_set_RLM_powerRange(ViSession *ihandle*, **ViInt32** *RLMSlot*, **ViBoolean** *rangeMode*, **ViReal64** *powerRange*, **ViReal64** *powerRangeSecondSensor*);

The Set RLM Range function (hp816x_set_RLM_powerRange) sets the power range of a Return Loss module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15

hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

rangeMode

Use the Range Mode control to choose one of the following range modes:

Auto, the auto-ranging mode, ensures that the result has a displayed value between 9% and 100% of full scale. The default state is for automatic ranging to be enabled.

Manual, which allows you to set a user-defined range.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
hp816x_PWM_RANGE_AUTO	1	Auto
hp816x_PWM_RANGE_MANUAL	0	Manual

powerRange

If you choose Manual using the Range Mode control, you must set a Power Range in dBm.

The Power Range determines the range of power levels that can be measured.

The Upper Power Limit is always 3 dBm greater than the chosen Power Range value. The resolution is always 40 dBm less than the chosen Power Range value.

For example, if you set the Power Range to -30 dBm, you can measure power between -27 dBm and -70 dBm, at a resolution of -70 dBm (0.1 nW).

Data Type: ViReal64

Input/Output: IN

powerRangeSecondSensor

If you choose Manual using the Range Mode control, you must set a Power Range in dBm.

The Power Range determines the range of power levels that can be measured.

The Upper Power Limit is always 3 dBm greater than the chosen Power Range value. The resolution is always 40 dBm less than the

chosen Power Range value.

For example, if you set the Power Range to -30 dBm, you can measure power between -27 dBm and -70 dBm, at a resolution of -70 dBm (0.1 nW).

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_RLM_powerRange_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_RLM_powerRange_Q(ViSession *ihandle*, ViInt32 *RLMSlot*, ViPBoolean *rangeMode*, ViPReal64 *powerRange*, ViPReal64 *powerRangeSecondSensor*);

The Get RLM Range function (hp816x_get_RLM_powerRange_Q) returns the power range setting of a Return Loss module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

rangeMode

Range Mode returns one of the following range modes:

1, the auto-ranging mode (hp816x_RLM_RANGE_AUTO), ensures that the result has a displayed value between 9% and 100% of full scale.

0, manual-ranging mode (hp816x_RLM_RANGE_MANUAL), allows you to set the range.

Data Type: ViPBoolean

Input/Output: OUT

powerRange

Power Range returns the power range in dBm.

The Power Range determines the range of power levels that can be measured.

If the measured power is less than 3 dBm greater than the Power Range, measured power can be returned.

If the measured power is less than 40 dBm less than the Power Range, measured power can be returned.

For example, if you set the Power Range to -30 dBm, you can measure power between -27 dBm and -70 dBm, at a resolution of -70 dBm (0.1 nW).

Data Type: ViPReal64

Input/Output: OUT

powerRangeSecondSensor

Power Range returns the power range in dBm.

The Power Range determines the range of power levels that can be measured.

If the measured power is less than 3 dBm greater than the Power Range, measured power can be returned.

If the measured power is less than 40 dBm less than the Power Range, measured power can be returned.

For example, if you set the Power Range to -30 dBm, you can measure power between -27 dBm and -70 dBm, at a resolution of -70 dBm (0.1 nW).

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_RLM_triggerConfiguration

Syntax:

ViStatus _VI_FUNC hp816x_set_RLM_triggerConfiguration(ViSession *ihandle*, ViInt32 *RLMSlot*, ViInt32 *triggerIn*, ViInt32 *triggerOut*);

The Set RLM Trigger Configuration (hp816x_set_RLM_triggerConfiguration) function configures input trigger handling and trigger output generation.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13

hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

triggerIn

Use the Trigger In control to select one of the following input trigger configurations:

Ignore (hp816x_RLM_TRIGIN_IGN), input triggers are ignored,

Single Measurement (hp816x_RLM_TRIGIN_SME), an input trigger triggers a single power measurement, or

Continuous Measurement (hp816x_RLM_TRIGIN_CME), an input trigger triggers continuous power measurement.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_RLM_TRIGIN_IGN	0	Ignore
hp816x_RLM_TRIGIN_SME	1	Single Measurement
hp816x_RLM_TRIGIN_CME	2	Continuous Measurement

triggerOut

Use the Trigger Out control to select one of the following output trigger configurations:

Disabled (hp816x_RLM_TRIGOUT_NONE), no output trigger will be generated, or

End of Averaging (hp816x_RLM_TRIGOUT_AVG), at the end of the averaging time for a power measurement an output trigger is generated.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_RLM_TRIGOUT_NONE	0	Disabled
hp816x_RLM_TRIGOUT_AVG	1	End of Averaging

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_start_RLM_internalTrigger

Syntax:

ViStatus _VI_FUNC hp816x_start_RLM_internalTrigger(ViSession *ihandle*, ViInt32 *RLMSlot*);

The Start RLM Trigger function (hp816x_start_RLM_internalTrigger) starts a Return Loss measurement cycle.

If you have precviously chosen Continuous Triggering mode, see the Set RLM Internal Trigger function, the function chooses Immediate Triggering mode automatically. You need to use the Start RLM Trigger function to start another Return Loss measurement cy cle.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_RLM_readReturnLoss

Syntax:

ViStatus _VI_FUNC hp816x_RLM_readReturnLoss(ViSession *ihandle*, ViInt32 *RLMSlot*, ViPReal64 *returnLoss*);

The Read RLM Return Loss function (hp816x_RLM_readReturnLoss) forces the instrument to start a measurement cycle. The evaluated return loss value will not be returned, until the averaging time period is finished.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

returnLoss

Measured Value returns the measured return loss in dB

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_RLM_fetchReturnLoss

Syntax:

ViStatus _VI_FUNC hp816x_RLM_fetchReturnLoss(ViSession *ihandle*, ViInt32 *RLMSlot*, ViPReal64 *returnLoss*);

The Fetch RLM Return Loss function (hp816x_RLM_fetchReturnLoss) immediately returns a return loss value without averaging return loss measurements over the averaging time period.

REMARKS: If this function is called more than once within an averaging cycle, the returned values will be identical.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

returnLoss

Measured Value returns the measured return loss in dB

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_RLM_readValue

Syntax:

ViStatus _VI_FUNC hp816x_RLM_readValue(ViSession *ihandle*, ViInt32 *RLMSlot*, ViBoolean *monitorDiode*, ViPReal64 *powerValue*);

The Read RLM Power Value function (hp816x_RLM_readValue) forces the instrument to start a measurement cycle. The Power Value will not be returned until the averaging time period is finished. You may choose whether you want to read a power value for the i nternal monitor diode, or for the return loss diode.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

monitorDiode

Use the Monitor Diode control to choose whether you want to read a power value for the internal monitor diode (Yes), or for the return loss diode (No).

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Yes
VI_FALSE	(0)	No

powerValue

Power Value returns the measured power value in Watts (W).

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_RLM_fetchValue

Syntax:

ViStatus _VI_FUNC hp816x_RLM_fetchValue(ViSession *ihandle*, ViInt32 *RLMSlot*, ViBoolean *monitorDiode*, ViPReal64 *powerValue*);

The Fetch RLM Power Value function (hp816x_RLM_fetchValue) immediately returns a power value for either the internal monitor diode, or the return loss diode, without averaging power measurements over the averaging time period.

REMARKS: If this function is called more than once within an averaging cycle, the returned values will be identical.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

monitorDiode

Use the Monitor Diode control to choose whether you want to fetch a power value for the internal monitor diode (Yes), or for the return loss diode (No).

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Yes
VI_FALSE	(0)	No

powerValue

Power Value returns the measured power value in Watts (W)

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_RLM_rlReference

Syntax:

ViStatus _VI_FUNC hp816x_set_RLM_rlReference(ViSession *ihandle*, ViInt32 *RLMSlot*, ViReal64 *returnLossReference*);

The Set RLM Return Loss Reference function (hp816x_set_RLM_rlReference) sets the return loss reference for a Return Loss module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

returnLossReference

Use Return Loss Reference to set the return loss value of your reference reflector in decibels (dB).

For example, the 81610CC reference cable provides a calibrated and stable reference near 14.8 dB.

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_RLM_rlReference_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_RLM_rlReference_Q(ViSession *ihandle*, ViInt32 *RLMSlot*, ViPReal64 *returnLossReference*);

The Get RLM Return Loss Reference function (hp816x_get_RLM_rlReference_Q) returns the return loss reference for a Return Loss module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

returnLossReference

Return Loss Reference returns the return loss value that has been set for your reference reflector in decibels (dB).

For example, the 81610CC reference cable provides a calibrated and stable reference near 14.8 dB return loss.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_RLM_FPDelta

Syntax:

ViStatus _VI_FUNC hp816x_set_RLM_FPDelta(ViSession *ihandle*, ViInt32 *RLMSlot*, ViReal64 *frontPanelDelta*);

The Set RLM Front Panel Delta function (hp816x_set_RLM_FPDelta) sets the loss variation value due to the front panel connector.

Follow the instructions in the User's Guide to measure this value.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

frontPanelDelta

Use Front Panel Delta to set the correction value for losses that are caused by the front panel connector and other losses.

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_RLM_FPDelta_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_RLM_FPDelta_Q(ViSession *ihandle*, ViInt32 *RLMSlot*, ViPReal64 *frontPanelDelta*);

The Get RLM Front Panel Delta function (hp816x_get_RLM_FPDelta_Q) returns the loss variation value due to the front panel connector.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

frontPanelDelta

Front Panel Delta returns the correction value for losses that are caused by the front panel connector and other losses.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_calibrate_RLM

Syntax:

ViStatus _VI_FUNC hp816x_calibrate_RLM(ViSession *ihandle*, ViInt32 *RLMSlot*, ViInt32 *calibrate*);

The Calibrate RLM function (hp816x_calibrate_RLM) allows you to set the Reflection Reference calibration values and the Termination Reference calibration values. You may set calibration values specific to your setup or choose the factory-set calibration values.

Calibration helps to eliminate the affect of wavelength dependencies, coupler directivity, insertion losses, backscattering, and other non-ideal system characteristics.

Performing a Reflection Reference calibration and a Termination Reference calibration will help you perform more accurate return loss measurements than if you use the factory-set calibration values. If you wish to perform a calibration, set up your instr umentation as explained in the mainframe's User's Guide and then call this function.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10

hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

calibrate

Use Calibrate to select one of the following options:

Reflection Reference (hp816x_CAL_REFL) - sets the Reflection Reference calibration values to the values currently measured by the chosen Return Loss module,

Termination Reference (hp816x_CAL_TERM) - sets the Termination Reference calibration values to the values currently measured by the chosen Return Loss module, or

Factory-Set Calibration (hp816x_CAL_FACTORY) - chooses the factory-set calibration for Reflection Reference calibration values and the Termination Reference calibration values.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_CAL_REFL	0	Reflection Reference
hp816x_CAL_TERM	1	Termination Reference
hp816x_CAL_FACTORY	2	Factory-Set Calibration

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_RLM_zeroing

Syntax:

ViStatus _VI_FUNC hp816x_RLM_zeroing(ViSession *ihandle*, ViInt32 *RLMSlot*, ViPInt32 *zeroingResult*);

The RLM Zeroing function (hp816x_RLM_zeroing) removes electrical offsets for one Return Loss module.

The instrument measures by converting optical power to electrical power, and then measuring electrical power. An electrical offset is power that is always present, even if there is no light at the input. If this offset is not removed, it will affect powe r measurement.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

zeroingResult

Zeroing Result returns an integer value.

If 0 is returned, the zeroing operation has been successful completed.

If 98 is returned, the zeroing operation has failed because the Power Meter received light. The most common reason for zeroing to fail is if:

- a source is connected to the Power Meter's input connector,

- the fiber connected to the Power Meter's input connector is collecting light, or

- the Power Meter receives ambient light because the input connector is uncovered.

Any other value is an error code and can be looked up in your instrument's Programming Guide.

Data Type: ViPInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine

"hp816x_error_message"

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hp816x_RLM_zeroingAll

Syntax:

ViStatus _VI_FUNC hp816x_RLM_zeroingAll(ViSession *ihandle*, ViPInt32 *summaryofZeroingAll*);

The RLM Zeroing All function (hp816x_RLM_zeroingAll) removes electrical offsets for all Return Loss modules and Power Meter modules and Power Meter modules.

The instrument measures by converting optical power to electrical power, and then measuring electrical power. An electrical offset is power that is always present, even if there is no light at the input. If this offset is not removed, it will affect powe r measurement.

NOTE: Because zeroing is a time consuming operation, the timeout is temporarily increased.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
summaryofZeroingAll	Summary of Zeroing All returns a summary of the results of every Return Loss module and Power Meter in the instrument. If 0 is returned, the zeroing operations have been successfully completed. If 98 is returned, a zeroing operation has failed because a Return Loss module or Power Meter has received light. The most common reason for zeroing to fail is if: - a source is connected to a Return Loss

module's or Power Meter's input connector,

- the fiber connected to a Return Loss module's or Power Meter's input connector is collecting light, or

- a Return Loss module or Power Meter received ambient light because the input connector is uncovered.

Any other value is an error code and can be looked up in your instrument's Programming Guide.

Data Type: ViPInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_enable_RLM_sweep

Syntax:

ViStatus _VI_FUNC hp816x_enable_RLM_sweep(ViSession *ihandle*, ViBoolean *enableRLMLambdaSweep*);

The Enable RLM Lambda Sweep function (hp816x_enable_RLM_sweep) includes or excludes all Return Loss modules from a lambda scan operation.

A Return Loss module included in a lambda scan appears in the channel list as if it were a dual-channel power meter whose first channel addresses the power measured at its power diode, and whose second channel addresses the power measured at its monitor diode.

For example, if slot 1 is occupied by a single-channel power meter, and slot 2 by a Return Loss module:

- channel 0 addresses the power meter's power diode,
- channel 1 addresses the Return Loss module's power diode,
- channel 3 addresses the Return Loss module's monitor diode.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
enableRLMLamdaSweep	Use the Enable RLM Lambda Sweep control to include (VI_TRUE) or exclude (VI_FALSE) Return Loss modules from a lambda scan operation. **Data Type:** ViBoolean **Input/Output:** IN

Values:

Name	Value	Description
VI_TRUE	(1)	Include in sweep
VI_FALSE	(0)	Exclude from sweep

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_RLM_reflectanceValues_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_RLM_reflectanceValues_Q(ViSession *ihandle*, ViInt32 *RLMSlot*, ViPReal64 *mref*, ViPReal64 *pref*);

The Get RLM Reflectance Values function (get_RLM_reflectanceValues_Q) returns Reflection Reference calibration values. These calibration values are required to calculate the return loss.

If you wish to perform a calibration, set up your instrumentation for a Reflection Reference calibration as explained in the mainframe's User's Guide and then call the Calibrate RLM function (**hp816x_calibrate_RLM**).

You can also use the Calibrate RLM function (**hp816x_calibrate_RLM**) to choose factory-set calibration values.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2

hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

mref

Mref returns the Reflection Reference calibration value for the internal monitor diode.

Data Type: ViPReal64

Input/Output: OUT

pref

Pref returns the Reflection Reference calibration value for the internal power diode

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_RLM_terminationValues_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_RLM_terminationValues_Q(ViSession *ihandle*, ViInt32 *RLMSlot*, ViPReal64 *mpara*, ViPReal64 *ppara*);

The Get RLM Termination Values function (get_RLM_terminationValues_Q) returns Termination Reference calibration values. These calibration values are required to calculate the return loss.

If you wish to perform a calibration, set up your instrumentation for a Termination Reference calibration as explained in the mainframe's User's Guide and then call the Calibrate RLM function (**hp816x_calibrate_RLM**).

You can also use the Calibrate RLM function (**hp816x_calibrate_RLM**) to choose factory-set calibration values.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1

hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

mpara

Mpara returns the Termination Reference calibration value for the internal monitor diode.

Data Type: ViPReal64

Input/Output: OUT

ppara

Ppara returns the Termination Reference calibration value for the internal power diode

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_RLM_dutValues_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_RLM_dutValues_Q(ViSession *ihandle*, ViInt32 *RLMSlot*, ViPReal64 *mdut*, ViPReal64 *pdut*);

The Get RLM DUT Values function (hp816x_get_RLM_dutValues_Q) reads the power values at the internal monitor and power diode. Set up your instrumentation to measure the Return Loss of the DUT as explained in the mainframe's User's Guide and then call this function.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

mdut

Mdut returns the power value measured by the internal monitor diode.

Data Type: ViPReal64

Input/Output: OUT

pdut

Pdut returns the power value measured by the internal power diode.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_calculate_RL

Syntax:

ViStatus _VI_FUNC hp816x_calculate_RL(ViSession *ihandle*, ViInt32 *RLMSlot*, ViReal64 *mref*, ViReal64 *mpara*, ViReal64 *pref*, ViReal64 *ppara*, ViReal64 *mdut*, ViReal64 *pdut*, ViReal64 *frontPanelDelta*, ViPReal64 *returnLoss*);

The Calculate Return loss function (hp816x_calculate_RL) returns the Return Loss and the Return Loss of the DUT.

This function is intended to be used in more complex test set-ups. Use this function in combination with the following functions:

- the Calibrate RLM function (**hp816x_calibrate_RLM**),

- the Get Front Panel Delta function (**hp816x_get_RLM_FPDelta_Q**),

- the Get RLM Reflectance Values function (**hp816x_get_RLM_reflectanceValues_Q**),

- the Get RLM Termination Values function (**hp816x_get_RLM_terminationValues_Q**), and

- the Get RLM DUT Values function (**hp816x_get_RLM_dutValues_Q**).

The last four functions return the input values that enable this function to calculate the Return Loss .

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed

Return Loss Module.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

mref

Use Mref to input the Reflection Reference calibration value for the internal monitor diode. This value is returned by the Get RLM Reflectance Values function (**hp816x_get_RLM_reflectanceValues_Q**).

Data Type: ViReal64

Input/Output: IN

mpara

Use Mpara to input the Termination Reference calibration value for the internal monitor diode. This value is returned by the Get RLM Reference Values function (**hp816x_get_RLM_terminationValues_Q**).

Data Type: ViReal64

Input/Output: IN

pref

Use Pref to input the Reflection Reference calibration value for the internal power diode. This value is returned by the Get RLM Reflectance Values function (**hp816x_get_RLM_reflectanceValues_Q**).

Data Type: ViReal64

Input/Output: IN

ppara

Use Ppara to input the Termination Reference calibration value for the internal power diode. This value is returned by the Get RLM Reference Values function (**hp816x_get_RLM_terminationValues_Q**).

Data Type: ViReal64

Input/Output: IN

mdut

Use Mdut to input the power value for the internal monitor diode. This value is returned by the Get RLM DUT Values function (**hp816x_get_RLM_dutValues_Q**).

Data Type: ViReal64

Input/Output: IN

pdut

Use Pdut to input the power value for the internal power diode. This value is returned by the Get RLM DUT Values function (**hp816x_get_RLM_dutValues_Q**).

Data Type: ViReal64

Input/Output: IN

frontPanelDelta

Use Front Panel Delta to input the power value for the internal power diode. This value is returned by the Get Front Panel Delta function (hp816x_get_FPDelta_Q).

Data Type: ViReal64

Input/Output: IN

returnLoss

Return Loss returns the return loss.

This value includes the loss variation due to the front panel connector.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_RLM_srcWavelength_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_RLM_srcWavelength_Q(ViSession *ihandle*, ViInt32 *RLMSlot*, ViPReal64 *wavelengthLowerSource*, ViPReal64 *wavelengthUpperSource*);

The Get RLM Source Wavelength function (hp816x_get_RLM_srcWavelength) returns the wavelength of a laser source that is built into a Return Loss module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

wavelengthLowerSource

If the Returm Loss Module has one built-in laser source, Wavelength Lower Source returns its wavelength.

If there are two built-in laser sources, Wavelength Lower Source returns the wavelength value for the source with the shorter wavelength.

If the Return Loss Module does not have a built-in laser source, or Laser Source is set to External, the value returned is 0.0

Data Type: ViPReal64

Input/Output: OUT

wavelengthUpperSource

If there are two built-in laser sources, Wavelength Upper Source returns the wavelength value for the source with the longer wavelength.

If the Return Loss Module does not have a second built-in laser source, or Laser Source is set to External, the value returned is 0.0

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_RLM_laserSourceParameters

Syntax:

ViStatus _VI_FUNC hp816x_set_RLM_laserSourceParameters(ViSession *ihandle*, ViInt32 *RLMSlot*, ViInt32 *laserSource*, ViBoolean *turnLaser*);

The Set RLM Laser Source Parameters function (hp816x_set_RLM_laserSourceParameters) sets all parameters for a Return Loss module with an internal Laser Source.

In case of a single frequency Laser Source, the inputs for the lower source will be used.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

laserSource

For a Return Loss module, you can use the Laser Source control to select one of the following:

Upper (hp816x_LOWER_SRC), the longer wavelength Laser Source or

Lower (hp816x_UPPER_SRC), the shorter wavelength Laser Source.

For a single-wavelength Return Loss module, this control is ignored.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_LOWER_SRC	0	Lower
hp816x_UPPER_SRC	1	Upper

turnLaser

Use the Turn Laser On/Off control to turn the Laser Source On (VI_TRUE) or Off (VI_FALSE).

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_RLM_laserSourceParameters_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_RLM_laserSourceParameters_Q(ViSession *ihandle*, ViInt32 *RLMSlot*, ViPInt32 *laserSource*, ViPBoolean *laserState*);

The Set RLM Laser Source Parameters function (**hp816x_set_RLM_laserSourceParameters**) returns all parameters for a Return Loss module internal Laser Source.

In case of a single frequency Laser Source, the inputs for the lower source will be used.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

laserSource

Laser Source returns one of the following values:

0 (hp816x_LOWER_SRC): the shorter wavelength is the selected output.

1 (hp816x_UPPER_SRC): the longer wavelength is the selected output.

Data Type: ViPInt32

Input/Output: OUT

laserState

Laser State returns whether the laser is turned on or off.

If 0 (VI_FALSE) is returned, the laser is turned off.

If 1 (VI_TRUE) is returned, the laser is turned on.

Data Type: ViPBoolean

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine

"hp816x_error_message"

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hp816x_set_RLM_modulationState

Syntax:

ViStatus _VI_FUNC hp816x_set_RLM_modulationState(ViSession *ihandle*, ViInt32 *RLMSlot*, ViInt32 *laserSource*, ViBoolean *lowFrequencyControl*);

The Set RLM Modulation State function (hp816x_set_RLM_modulationState) adjusts the internal modulation of a laser source that is built into a Return Loss module.

Internal Modulation is described in your instrument's User's Guide.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

laserSource

For a Return Loss module that inclues two built-in laser sources, you can use the Laser Source control to select one of the following:

Lower (hp816x_RLM_LOWER_SRC), the shorter wavelength built-in Laser Source; or Upper (hp816x_RLM_UPPER_SRC), the longer wavelength built-in Laser Source.

For a Return Loss module with one built-in laser source, this control is ignored.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_LOWER_SRC	0	Lower
hp816x_UPPER_SRC		

lowFrequencyControl

Use the Low Frequency Control to turn internal low frequency modulation On (VI_TRUE) or Off (VI_FALSE).

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_RLM_modulationState_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_RLM_modulationState_Q(ViSession *ihandle*, ViInt32 *RLMSlot*, ViInt32 *laserSource*, ViPBoolean *lowFrequencyControl*);

The Get RLM Modulation function (hp816x_get_RLM_modulationState_Q) returns the internal modulation state of a laser source that is built into a Return Loss module.

Internal Modulation is described in your instrument's User's Guide.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

laserSource

For a Return Loss module that inclues two built-in laser sources, you can use the Laser Source control to select one of the following:

Lower (hp816x_RLM_LOWER_SRC), the shorter wavelength built-in Laser Source; or Upper (hp816x_RLM_UPPER_SRC), the longer wavelength built-in Laser Source.

For a Return Loss module with one built-in laser source, this control is ignored.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_LOWER_SRC	0	Lower
hp816x_UPPER_SRC	1	Upper

lowFrequencyControl

Low Frequenct Control returns the low frequency modulation mode.

If 1 (VI_TRUE) is returned, low frequency modulation is turned on.

If 0 (VI_FALSE) is returned, low frequency modulation is turned off.

Data Type: ViPBoolean

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_RLM_laserState

Syntax:

ViStatus _VI_FUNC hp816x_set_RLM_laserState(ViSession *ihandle*, ViInt32 *RLMSlot*, ViBoolean *laserState*);

The Set RLM Laser State function (hp816x_set_RLM_laserState) turns the laser on or off.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

laserState

Use the Laser State control to turn the laser output On (VI_TRUE) or Off (VI_FALSE).

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_RLM_laserState_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_RLM_laserState_Q(ViSession *ihandle*, ViInt32 *RLMSlot*, ViPBoolean *laserState*);

The Get RLM Laser State function (hp816x_get_RLM_laserState) turns the laser on or off.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

laserState

Laser State returns whether the laser is turned on or off.

If 0 (VI_FALSE) is returned, the laser is turned off.

If 1 (VI_TRUE) is returned, the laser is turned on.

Data Type: ViPBoolean

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_RLM_logging

Syntax:

ViStatus _VI_FUNC hp816x_set_RLM_logging(ViSession *ihandle*, ViInt32 *RLMSlot*, ViReal64 *averagingTime*, ViInt32 *dataPoints*, ViPInt32 *estimatedTimeout*);

The Set RLM Loggging function (hp816x_set_RLM_logging) sets the parameters for a logging operation and starts the logging.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

averagingTime

Use Averaging Time to set the length of time in seconds (s) over which a logging operation power measurement is averaged.

Data Type: ViReal64

Input/Output: IN

dataPoints

Use Data Points to set the number of values to be taken during a logging operation.

The maximum number of values is 12000.

Allocate a one-dimensional array with the same dimensions as this value and use as the Logging Result input for the Get RLM Logging Results function (**hp816x_get_RLM_loggingResults_Q**).

Data Type: ViInt32

Input/Output: IN

estimatedTimeout

Estimated Timeout returns the estimated time in milliseconds (ms), the logging operation will take; this value can be used to adjust the Timeout function (**hp816x_timeOut**).

Data Type: ViPInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_RLM_loggingResults_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_RLM_loggingResults_Q(ViSession *ihandle*, ViInt32 *RLMSlot*, ViBoolean *waitforCompletion*, ViBoolean *resultUnit*, ViPBoolean *loggingStatus*, ViReal64*loggingResult*[ ]);

The Get RLM Logging Results function (hp816x_get_RLM_loggingResults_Q) returns the result of a previously started logging operation.

NOTE: The Wait for Completion control should only be set to Yes if the application is thoroughly tested because using this control may result in an infinite loop that blocks the calling function.

Also, if the function detects that logging is finished, the logging operation stops automatically.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1

hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

waitforCompletion

Use the Wait for Completion control to choose one of the following:

Yes (VI_TRUE), the stability operation will wait until the Total Time specified in the Set PWM Stability function has elapsed, or

No (VI_FALSE), the stability operation will stop immediately.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Yes
VI_FALSE	(0)	No

resultUnit

Use the Result Unit control to choose whether to log the return loss value in dB (hp816x_PU_DBM) or in Watts (hp816x_PU_WATT).

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
hp816x_PU_DBM	0	dB
hp816x_PU_WATT	1	Watts

loggingStatus

Logging Status returns a Boolean representing the current status of the logging operation.

If Logging Status equals 1, the operation is completed.

If Logging Status equals 0, the operation is being performed.

Data Type: ViPBoolean

Input/Output: OUT

loggingResult

Logging Result is both an input and an output.

As an input, Logging Result accepts a one-dimensional integer array that determines the size of the Logging Result output. Allocate a one-dimensional integer array with the same dimensions as the Data Points value you have set for the Set RLM Logging fun ction (**hp816x_set_RLM_logging**).

As an output, Logging Result returns the result of a logging operation as a one-dimensional integer array.

Data Type: ViReal64[]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine

"hp816x_error_message"

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hp816x_set_RLM_stability

Syntax:

ViStatus _VI_FUNC hp816x_set_RLM_stability(ViSession *ihandle*, ViInt32 *RLMSlot*, ViReal64 *averagingTime*, ViReal64 *periodTime*, ViReal64 *totalTime*, ViPInt32 *estimatedResults*);

The Set RLM Stability function (hp816x_set_RLM_stability) sets up a stability operation and starts the data acquisition.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

averagingTime

Use Averaging Time to set the length of time in seconds (s) over which a stability operation power measurement is averaged.

Data Type: ViReal64

Input/Output: IN

periodTime

Use the Period time to set the time in seconds between the beginning of each averaging cycle.

Data Type: ViReal64

Input/Output: IN

totalTime

Use Total Time to how long in seconds (s) the stability operation lasts.

Data Type: ViReal64

Input/Output: IN

estimatedResults

Estimated Results returns an estimate of the number of results that the stability operation will record.

Use Estimated Results to allocate a one-dimensional array with the same dimensions as this value and use the array as the Stability Result input for the Get RLM Stability function (**hp816x_get_RLM_stabilityResults_Q**).

Data Type: ViPInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_RLM_stabilityResults_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_RLM_stabilityResults_Q(ViSession *ihandle*, ViInt32 *RLMSlot*, ViBoolean *waitforCompletion*, ViBoolean *resultUnit*, ViPBoolean *stabilityStatus*, ViReal64*stabilityResult*[ ]);

The Get RLM Stability Results function (hp816x_get_RLM_stabilityResults_Q) returns the result of a previously started stability operation.

REMARK The Wait for Completion control should only be set to Yes if the application is thoroughly tested because using this control may result in an infinite loop that blocks the calling function.

Also, if the function detects that logging is finished, the stability operation stops automatically.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1

hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

waitforCompletion

Use the Wait for Completion control to choose one of the following:

Yes (VI_TRUE), the stability operation will wait until the Total Time specified in the Set PWM Stability function has elapsed, or

No (VI_FALSE), the stability operation will stop immediately.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Yes
VI_FALSE	(0)	No

resultUnit

Use the Result Unit control to choose whether to log the return loss value in dB (hp816x_PU_DBM) or in Watts (hp816x_PU_WATT).

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
hp816x_PU_DBM	0	dB
hp816x_PU_WATT	1	Watts

stabilityStatus

Stability Status returns a Boolean representing the current status of the stability operation.

If Stability Status equals 1, the operation is completed.

If Stability Status equals 0, the operation is being performed.

Data Type: ViPBoolean

Input/Output: OUT

stabilityResult

Stability Result is both an input and an output.

As an input, Stability Result accepts a one-dimensional integer array that determines the size of the Stability Result output. Allocate a one-dimensional integer array with the same dimensions as the Estimated Results value you have set for the Set RLM S tability function (**hp816x_set_RLM_stability**).

As an output, Stability Result returns the result of a stability operation as a one-dimensional integer array.

Data Type: ViReal64[]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine

"hp816x_error_message"

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hp816x_set_RLM_minMax

Syntax:

ViStatus _VI_FUNC hp816x_set_RLM_minMax(ViSession *ihandle*, ViInt32 *RLMSlot*, ViInt32 *minmaxMode*, ViInt32 *dataPoints*, ViUInt32 *estimatedTime*);

The Set RLM MinMax function (hp816x_set_RLM_minMax) sets the parameters for a MinMax logging operation.

A MinMax logging operation is active as long as it is not explicitly stopped. Before you start a different Return Loss module data acquisition function or different MinMax logging operation, you must call the Stop RLM Function (**hp816x_RLM_functionStop**) f unction.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

minmaxMode

Use the MinMax control to choose one of the following MinMax modes:

Continuous mode (hp816x_MM_CONT),

Window mode (hp816x_MM_WIN), or

Refresh mode (hp816x_MM_REFRESH).

See your instrument's User's Guide for information on MinMax modes.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_MM_CONT	0	Continuous
hp816x_MM_WIN	1	Window
hp816x_MM_REFRESH	2	Refresh

dataPoints

Use Data Points to define the number of samples used for the MinMax logging operation. Data Points affects Refresh mode and Window mode.

See your instrument's User's Guide for information on Data Points.

Data Type: ViInt32

Input/Output: IN

estimatedTime

Estimated Time returns the time, in seconds, it will take until a minimum or maximum value is available.

This value can be used to adjust the time before you can call the Get RLM MinMax Results function (get_RLM_minMaxResults_Q). This does not apply to Continuous mode.

Data Type: ViUInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_RLM_minMaxResults_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_RLM_minMaxResults_Q(ViSession *ihandle*, ViInt32 *RLMSlot*, ViPReal64 *minimum*, ViPReal64 *maximum*, ViPReal64 *current*);

The Get RLM MinMax Results function (hp816x_get_RLM_minMaxResults_Q) returns the minimum, maximum and current values of a previously started MinMax operation.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

minimum

Minimum returns the minimum power value in Watts for the MinMax operation.

Data Type: ViPReal64

Input/Output: OUT

maximum

Maximum returns the maximum power value in Watts for the MinMax operation.

Data Type: ViPReal64

Input/Output: OUT

current

Current returns the current power value in Watts.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_RLM_functionStop

Syntax:

ViStatus _VI_FUNC hp816x_RLM_functionStop(ViSession *ihandle*, ViInt32 *RLMSlot*);

The Stop RLM Function function (hp816x_RLM_functionStop) stops (aborts) any activated logging, stability or Min Max data acquistion operation.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
RLMSlot	Use the RLM Slot control to choose an installed Return Loss Module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7

hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_enable_RLM_sweep

Syntax:

ViStatus _VI_FUNC hp816x_enable_RLM_sweep(ViSession *ihandle*, ViBoolean *enableRLMLambdaSweep*);

The Enable RLM Lambda Sweep function (hp816x_enable_RLM_sweep) includes or excludes all Return Loss modules from a lambda scan operation.

A Return Loss module included in a lambda scan appears in the channel list as if it were a dual-channel power meter whose first channel addresses the power measured at its power diode, and whose second channel addresses the power measured at its monitor diode.

For example, if slot 1 is occupied by a single-channel power meter, and slot 2 by a Return Loss module:

- channel 0 addresses the power meter's power diode,
- channel 1 addresses the Return Loss module's power diode,
- channel 3 addresses the Return Loss module's monitor diode.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
enableRLMLamdaSweep	Use the Enable RLM Lambda Sweep control to include (VI_TRUE) or exclude (VI_FALSE) Return Loss modules from a lambda scan operation. **Data Type:** ViBoolean **Input/Output:** IN

Values:

Name	Value	Description
VI_TRUE	(1)	Include in sweep
VI_FALSE	(0)	Exclude from sweep

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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Information for N77 Series Instruments

This version of the 816x VXIplug&play instrument driver has been extended to support the N773xA, N774xA optical power meter and the N775xA and N776xA Optical Attenuator instruments. These are stand-alone instruments that do not require installation in an 816x mainframe. The SCPI command set has been designed for almost complete programming compatibility with the modular instruments of the 816x family. Please be aware of the following details and limitations, when using this driver.

- The N77-series instruments can be controlled via USB, LAN or GPIB interfaces. Especially for transferring large data volumes, the USB and usually LAN interfaces provide faster performance.
- The identification of the individual ports of these multiport instruments in the SCPI and hp816x functions corresponds to the slot number. So for example the ports of the N7745A can be addressed with the slot numbers 1 to 8. The channel number in the commands and functions is not used and can generally be omitted or set to Channel 1 (index 0).
- The N771x-series and 81950A tunable lasers are not supported by the 816x VXIplug&play driver.
- The N773x-series optical switches are supported by the 816x VXIplug&play functions for switches and relevant functions for the mainframes.
- The N774x-series power meters are supported by the 816x VXIplug&play functions for power meters, relevant functions for the mainframes like for standard trigger configuration, and the Multi Frame Lambda Scan application functions.
- The N776x-series optical attenuators are supported by the 816x VXIplug&play functions for attenuators and the relevant functions for the mainframes.
- The N775x-series optical attenuators and power meters are supported by the 816x VXIplug&play functions for attenuators and the relevant functions for the mainframes and the basic functions for the power meters, excluding the advanced functions using data power logging. The choices of averaging time is also more limited for these power meters.
- The averaging time parameter for the power meters is limited in the 816x VXIplug&play functions to a limited set of discrete values, similar to the modular 816x power meters. This set includes: 1, 2, 5, 10, 25, 100, 200, and 500 ìs, 1, 2, 5, 10, 20, 50, 100, 200, 500 ms, and 1, 2, 5, and 10s.
- The maximum number of logging data points that can be acquired is limited by the 816x VXIplug&play data acquisition functions to 100,000 per port.
- The shortest averaging time used by the Multi Frame Lambda Scan application is 25 ìs.
- For the Multi Frame Lambda Scan application, the N774xA instruments can be included simply by registering the instrumet as a mainframe,after the first

mainframe with the tunable laser has been registered.

Further details to available functions are found under the categories below:

Utility Functions

Mainframe Specific Functions

Power Sensor Specific Functions

Applications

hp816x_WaitForOPC

Syntax:

ViStatus _VI_FUNC hp816x_WaitForOPC(ViSession *IHandle*, ViInt32 *TLS_Slot*, ViBoolean *Wait_for_Operation_Complete*);

The Wait for Operation Complete function (hp816x_WaitForOPC) can be useful if DFB Laser Sources are installed.

Since changeing the wavelength of a DFB source is a time consuming operation, this function may be used to return immediately from a wavelength setting operation.

Parameter	Description
IHandle	IHandle is the session handle returned from the Initialize Mainframe function (hp816x_init). A new session handle is created every time the mainframe is initialized. For LabView Programmers This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (hp816x_init). **Data Type:** ViSession **Input/Output:** IN
TLS_Slot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN
Wait_for_ Operation_Complete	Use the Wait_for_Operation_Complete control, a boolean input, to set whether to "Wait for OPC" or not.

If "Wait for OPC" is set to "No", the VXI-PnP driver will return immediately without checking the OPC flag. The desired parameter adjustment may be incomplete.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Wait for OPC
VI_FALSE	(0)	Do not wait for OPC

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_TLS_parameters

Syntax:

ViStatus _VI_FUNC hp816x_set_TLS_parameters(ViSession *ihandle*, ViInt32 *TLSSlot*, ViInt32 *powerUnit*, ViInt32 *opticalOutput*, ViBoolean *turnLaser*, ViReal64 *power*, ViReal64 *attenuation*, ViReal64 *wavelength*);

The Set TLS Parameters (hp816x_set_TLS_parameters) function sets the most important parameters of a Tunable Laser module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

powerUnit

You can use Power Unit to choose either Watts (hp816x_PU_WATT) or dBm (hp816x_PU_DBM) as the power unit.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_PU_WATT	1	Watt
hp816x_PU_DBM	0	dBm

opticalOutput

Use the Optical Output control to choose the optical output for a TLS module with two outputs. You can choose one of the following:

High Power (hp816x_HIGHPOW), where Output 2, the High Power output is the regulated output.

Low SSE (hp816x_LOWSSE), where Output 1, the Low SSE output is the regulated output.

BHR (hp816x_BHR), is the same as choosing Both (Master 2) from the instrument's front panel, Output 2, the High Power output is the regulated output and Output 1, the Low SSE output is the unregulated output.

BLR (hp816x_BLR), is the same as choosing

Both (Master 1) from the instrument's front panel, the Low SSE output is the regulated output and Output 2, the High Power output is the unregulated output.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_HIGHPOW	0	High Power
hp816x_LOWSSE	1	Low SSE
hp816x_BHR	2	BHR
hp816x_BLR	3	BLR

turnLaser

You can use the Turn Laser On/Off control to turn the Laser Source On (VI_TRUE) or Off (VI_FALSE).

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

power

Use Power to set the output power.

You must choose the power unit using the Power Unit control.

Data Type: ViReal64

Input/Output: IN

attenuation

If your Tunable Laser module has an in-built optical attenuator, use Attenuation to set the attenuation to a value in decibels (dB).

Data Type: ViReal64

Input/Output: IN

wavelength

Use Wavelength to set the output wavelength in meters (m) of the Tunable Laser module.

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_parameters_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_parameters_Q(ViSession *ihandle*, ViInt32 *TLSSlot*, ViPInt32 *powerUnit*, ViPBoolean *laserState*, ViPInt32 *opticalOutput*, ViPReal64 *power*, ViPReal64 *attenuation*, ViPReal64 *wavelength*);

The Get TLS Parameters (hp816x_get_TLS_parameters_Q) function returns the most important parameters of a Tunable Laser module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

powerUnit

Power Unit returns the following:

0 (hp816x_PU_DBM) units of dBm or

1 (hp816x_PU_WATT) units of Watts.

Data Type: ViPInt32
Input/Output: OUT

laserState

Laser State returns the following:

0 (VI_FALSE), the laser is turned off.

1 (VI_TRUE), the laser is turned on.

Data Type: ViPBoolean
Input/Output: OUT

opticalOutput

Optical Output returns the regulated output of the Tunable Laser module, whereby the one the following values is returned:

0 (hp816x_HIGHPOW); when Output 2, the High

Power output is the regulated output.

1 (hp816x_LOWSSE); when Output 1, the Low SSE output is the regulated output.

2 (hp816x_BHR); when Output 2, the High Power output is the regulated output and Output 1, the Low SSE output is the unregulated output.

3 (hp816x_BLR); when Output 1, the Low SSE output is the regulated output and Output 2, the High Power output is the unregulated output.

Data Type: ViPInt32

Input/Output: OUT

power

Power returns the output power of the TLS in the chosen power unit.

Use the Unit control of the Set TLS Power function (**hp816x_set_TLS_power**) to set the units to dBm or Watts.

Use the value returned by Power Unit to find out whether Power is returned in Watts or dBm.

Data Type: ViPReal64

Input/Output: OUT

attenuation

Attenuation returns the attenuation value in decibels (dB).

Check if your Tunable Laser module has an in-built optical attenuator.

Data Type: ViPReal64

Input/Output: OUT

wavelength

Wavelength returns the wavelength value of the Tunable Laser module in meters (m).

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_TLS_wavelength

Syntax:

ViStatus _VI_FUNC hp816x_set_TLS_wavelength(ViSession *ihandle*, ViInt32 *TLSSlot*, ViInt32 *wavelengthSelection*, ViReal64 *wavelength*);

The Set TLS Wavelength (hp816x_set_TLS_wavelength) function sets the wavelength of a Tunable Laser module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

wavelengthSelection

You can use the Wavelength Selection control to select one of the following:

Minimum (hp816x_INP_MIN), which is the minimum wavelength,

Maximum (hp816x_INP_MAX), which is the maximum wavelength,

Default (hp816x_INP_DEF), which is the sum of the minimum wavelength and the maximum wavelength divided by two (this is not the same value that the Reset function, the Preset function, or the Preset hardkey sets the frequency to), and

Manual Input (hp816x_INP_MAN), which is the value entered in the Wavelength input.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_INP_MIN	0	Minimum
hp816x_INP_DEF	1	Default
hp816x_INP_MAX	2	Maximum
hp816x_INP_MAN	3	Manual Input

wavelength

If you choose Manual Input as the Wavelength

Selection input, you can use the Wavelength input to set the wavelength to a value in meters (m).

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_wavelength_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_wavelength_Q(ViSession *ihandle*, ViInt32 *TLSSlot*, ViPReal64 *minimumWavelength*, ViPReal64 *defaultWavelength*, ViPReal64 *maximumWavelength*, ViPReal64 *currentWavelength*);

The Get TLS Wavelength (hp816x_get_TLS_wavelenght_Q) function returns the current values of the following: Minimum Wavelength, Maximum Wavelength, Default Wavelength, and Current Wavelength.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

minimumWavelength

Minimum Wavelength returns the lowest wavelength that your Tunable Laser module can be set to.

Wavelength is returned in meters (m).

Data Type: ViPReal64

Input/Output: OUT

defaultWavelength

Default Wavelength returns the default wavelength of your Tunable Laser module. This is identical to the default value set by setting the Wavelength Selection control of the Set TLS Wavelength function to Default; it is the sum of the Minimum Wavelength and the Maximum Wavelength divided by two. This is not the same value that the Reset function, the Preset function, or the Preset hardkey sets the wavelength to.

Wavelength is returned in meters (m).

Data Type: ViPReal64

Input/Output: OUT

maximumWavelength

Maximum Wavelength returns the highest wavelength that your Tunable Laser module can

be set to.

Wavelength is returned in meters (m).

Data Type: ViPReal64
Input/Output: OUT

currentWavelength

Current Wavelength returns the wavelength that your Tunable Laser module is set to.

Wavelength is returned in meters (m).

Data Type: ViPReal64
Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_TLS_power

Syntax:

ViStatus _VI_FUNC hp816x_set_TLS_power(ViSession *ihandle*, ViInt32 *TLSSlot*, ViInt32 *unit*, ViInt32 *powerSelection*, ViReal64 *manualPower*);

The Set TLS Power (hp816x_set_TLS_power) function sets the power of a Tunable Laser module.

REMARK If you choose Manual Input as the Power Selection input, you can use the Manual Power input to set the power to a value in dBm or Watts, depending on how you set the Unit control.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

unit

You can use the Unit control to choose either Watts (hp816x_PU_WATT) or dBm (hp816x_PU_DBM) as the power unit.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_PU_DBM	0	dBm
hp816x_PU_WATT	1	Watt

powerSelection

You can use the Power Selection control to select one of the following:

Minimum (hp816x_INP_MIN), which is the minimum power,

Maximum (hp816x_INP_MAX), which is the maximum power,

Default (hp816x_INP_DEF), which is the sum of the minimum power and the maximum power divided by two (this is not the same value that the Reset function, the Preset function, or the Preset hardkey sets the frequency to), and

Manual Power (hp816x_INP_MAN), which is the

value entered in the Wavelength input.

Data Type: ViInt32
Input/Output: IN

Values:

Name	Value	Description
hp816x_INP_MIN	0	Minimum
hp816x_INP_DEF	1	Default
hp816x_INP_MAX	2	Maximum
hp816x_INP_MAN	3	Manual Input

manualPower

If you choose Manual Input as the Power Selection input, you can use the Manual Power input to set the power to a value in dBm or Watts, depending on how you set the Unit control.

Data Type: ViReal64
Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_power_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_power_Q(ViSession *ihandle*, ViInt32 *TLSSlot*, ViPInt32 *powerUnit*, ViPReal64 *minimumPower*, ViPReal64 *defaultPower*, ViPReal64 *maximumPower*, ViPReal64 *currentPower*);

The Get TLS Power (hp816x_get_TLS_power_Q) function returns the current values of the following: the Power Units, Minimum Power, Maximum Power, Default Power, and Current Power. The power limits depend on the current optical output.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

powerUnit

Power Units returns the currently selected power units for your Tunable Laser module.

0 (hp816x_PU_DBM) is returned if the units are dBm.

1 (hp816x_PU_WATT) is returned if the units are Watts (W).

Data Type: ViPInt32

Input/Output: OUT

minimumPower

Minimum Power returns the lowest power that your Tunable Laser module can be set to.

Power is returned in units of dBm or Watts.

Use the Unit control of the Set TLS Power function (**hp816x_set_TLS_power**) to set the units to dBm or Watts.

Use the value returned by Power Unit to find out whether Power is returned in Watts or dBm.

Data Type: ViPReal64

Input/Output: OUT

defaultPower

Default Power returns the default power of your Tunable Laser module. This is identical to the default value set by setting the Power Selection control of the Set TLS Power function to Default; it is the sum of the Minimum Power and the Maximum Power div ided by two. This is not the same value that the Reset function, the Preset function, or the Preset hardkey sets the wavelength to.

Power is returned in units of dBm or Watts. Use the Unit control of the Set TLS power function (**hp816x_set_TLS_power**) to set the units to dBm or Watts.

Data Type: ViPReal64

Input/Output: OUT

maximumPower

Maximum Power returns the highest power that your Tunable Laser module can be set to.

Power is returned in units of dBm or Watts. Use the Unit control of the Set TLS power function (**hp816x_set_TLS_power**) to set the units to dBm or Watts.

Data Type: ViPReal64

Input/Output: OUT

currentPower

Current Power returns the power that your Tunable Laser module is set to.

Power is returned in units of dBm or Watts. Use the Unit control of the Set TLS power function (**hp816x_set_TLS_power**) to set the units to dBm or Watts.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_TLS_opticalOutput

Syntax:

ViStatus _VI_FUNC hp816x_set_TLS_opticalOutput(ViSession *ihandle*, ViInt32 *TLSSlot*, ViInt32 *setOpticalOutput*);

The Set TLS Optical Output (hp816x_set_TLS_opticalOutput) function configures the optical output of Tunable Laser module with two optical outputs.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

setOpticalOutput

Use the Optical Output control to choose the optical output for a TLS module with two outputs. You can choose one of the following:

High Power (hp816x_HIGHPOW), where Output 2, the High Power output is the regulated output.

Low SSE (hp816x_LOWSSE), where Output 1, the Low SSE output is the regulated output.

BHR (hp816x_BHR) is the same as choosing Both (Master 2) from the instrument's front panel, Output 2, the High Power output is the regulated output and Output 1, the Low SSE output is the unregulated output.

BLR (hp816x_BLR) is the same as choosing Both (Master 1) from the instrument's front panel, the Low SSE output is the regulated output and Output 2, the High Power output is the unregulated output.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_HIGHPOW	0	High Power
hp816x_LOWSSE	1	Low SSE
hp816x_BHR	2	BHR
hp816x_BLR	3	BLR

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_opticalOutput_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_opticalOutput_Q(ViSession *ihandle*, ViInt32 *TLSSlot*, ViPInt32 *opticalOutput*);

The Get TLS Optical Output (hp816x_get_TLS_opticalOutput_Q) function returns the optical output configuration of a Tunable Laser module with two optical outputs.

REMARK

If this function addresses a module with only one optical output, an error is generated.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2

hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

opticalOutput

Optical Output returns the regulated output of the Tunable Laser module, whereby the one the following values is returned:

0 (hp816x_HIGHPOW); when Output 2, the High Power output is the regulated output.

1 (hp816x_LOWSSE); when Output 1, the Low SSE output is the regulated output.

2 (hp816x_BHR); when Output 2, the High Power output is the regulated output and Output 1, the Low SSE output is the unregulated output.

3 (hp816x_BLR); when Output 1, the Low SSE output is the regulated output and Output 2, the High Power output is the unregulated output.

Data Type: ViPInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_powerMaxInRange_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_powerMaxInRange_Q(ViSession *ihandle*, ViInt32 *TLSSlot*, ViReal64 *startofRange*, ViReal64 *endofRange*, ViPReal64 *maximumPower*);

The Get TLS Max Power in Range (hp816x_get_TLS_powerMaxInRange_Q) function returns the maximum permitted power for a given wavelength interval.

This function can be used to calculate the maximum permitted power for a wavelength sweep.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3

hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

startofRange

The Start of Range input is the lowest value of the wavelength interval in meters (m).

Data Type: ViReal64

Input/Output: IN

endofRange

The End of Range input is the highest value of the wavelength interval in meters (m).

Data Type: ViReal64

Input/Output: IN

maximumPower

Maximum Power returns the maximum permitted power for your Tunable Laser module in the range between the Start of Range and End of Range inputs you have entered.

Maximum Power is returned in dBm or Watts depending on the units you choose using the Set TLS Power function.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_TLS_laserState

Syntax:

ViStatus _VI_FUNC hp816x_set_TLS_laserState(ViSession *ihandle*, ViInt32 *TLSSlot*, ViBoolean *laserState*);

The Set TLS Laser State (hp816x_set_TLS_laserState) function turns the laser on or off.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

laserState

Use the Laser State control to turn the laser output On (VI_TRUE) or Off (VI_FALSE).

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_laserState_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_laserState_Q(ViSession *ihandle*, ViInt32 *TLSSlot*, ViPBoolean *laserState*);

The Get TLS laser State (hp816x_get_TLS_laserState_Q) function returns whether the laser is turned on or off.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

laserState

Laser State returns whether the laser is turned on or off.

If 0 (VI_FALSE) is returned, the laser is turned off.

If 1 (VI_TRUE) is returned, the laser is turned on.

Data Type: ViPBoolean

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_TLS_laserRiseTime

Syntax:

ViStatus _VI_FUNC hp816x_set_TLS_laserRiseTime(ViSession *ihandle*, ViInt32 *TLSSlot*, ViReal64 *laserRiseTime*);

The Set TLS Laser Rise Time (hp816x_set_TLS_laserRiseTime) function adjusts the laser rise time.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

laserRiseTime

Use Laser Rise Time to input the laser rise time in meters (0-3).

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_laserRiseTime

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_laserRiseTime(ViSession *ihandle*, ViInt32 *TLSSlot*, ViPReal64 *laserRiseTime*);

The Get TLS Laser Rise Time (hp816x_get_TLS_laserRiseTime) function returns the laser rise time.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

laserRiseTime

Laser Rise Time returns the laser rise time in meters.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_TLS_triggerConfiguration

Syntax:

ViStatus _VI_FUNC hp816x_set_TLS_triggerConfiguration(ViSession *ihandle*, ViInt32 *TLSSlot*, ViInt32 *triggerIn*, ViInt32 *triggerOut*);

The Set TLS Trigger Configuration (hp816x_set_TLS_triggerConfiguration) function configures:

- how the Tunable Laser module processes input triggers, and

- how an output trigger is generated.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9

hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

triggerIn

You can use the Trigger In control to choose one of the following input trigger configurations:

Ignore (hp816x_TLS_TRIGIN_IGN), where input triggers are ignored,

Next Step (hp816x_TLS_TRIGIN_NEXTSTEP), where an input trigger will cause the next step of a sweep to be performed, and

Run Sweep (hp816x_TLS_TRIGIN_SWEEPSTART), where an input trigger will cause a sweep to run.

Trigger In only applies to Tunable Laser modules.

Data Type: ViInt32
Input/Output: IN

Values:

Name	Value	Description
hp816x_TLS_TRIGIN_IGN	0	Ignore
hp816x_TLS_TRIGIN_NEXTSTEP	1	Next Step
hp816x_TLS_TRIGIN_SWEEPSTART	2	Run Sweep

triggerOut

You can use the Trigger Out to choose one of the following output trigger configurations:

Disabled (hp816x_TLS_TRIGOUT_DISABLED), where output trigger mode is disabled,

Modulation (hp816x_TLS_TRIGOUT_MOD), where the output trigger connector outputs a TTL signal at the frequency of the modulation (this signal is output whether the laser is switched on or off),

Step Finished (hp816x_TLS_TRIGOUT_STEPEND), where a trigger is output after every step of a sweep finishes,

Sweep Finished

(hp816x_TLS_TRIGOUT_SWEND), where a trigger is output after a sweep finishes, and

Sweep Started (hp816x_TLS_TRIGOUT_SWSTART), where a trigger is output after a sweep starts.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_TLS_TRIGOUT_DISABLED	0	Disabled
hp816x_TLS_TRIGOUT_MOD	1	Modulation
hp816x_TLS_TRIGOUT_STEPEND	2	Step Finished
hp816x_TLS_TRIGOUT_SWSTART	3	Sweep Started
hp816x_TLS_TRIGOUT_SWEND	4	Sweep Finished

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_triggerConfiguration

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_triggerConfiguration(ViSession *ihandle*, ViInt32 *TLSSlot*, ViPInt32 *triggerIn*, ViPInt32 *triggerOut*);

The Get TLS Trigger Configuration (hp816x_get_TLS_triggerConfiguration) function returns:

- the selected Tunable Laser module's input triggers configuration and

- the selected Tunable Laser module's output triggers configuration.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3

hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

triggerIn

You can use the Trigger In to choose one of the following input trigger configurations:

0 (hp816x_TLS_TRIGIN_IGN), where input triggers are ignored,

1 (hp816x_TLS_TRIGIN_NEXTSTEP), where an input trigger will cause the next step of a sweep to be performed, and

2 (hp816x_TLS_TRIGIN_SWEEPSTART), where an input trigger will cause a sweep to be run.

Data Type: ViPInt32

Input/Output: OUT

triggerOut

You can use the Trigger Out to choose one of the following output trigger configurations:

0 (hp816x_TLS_TRIGOUT_DISABLED), where output trigger mode is disabled,

1 (hp816x_TLS_TRIGOUT_MOD), where the output trigger connector outputs a TTL signal at the frequency of the internal modulation (this signal is output whether the laser is switched on or off),

2 (hp816x_TLS_TRIGOUT_STEPEND), where a trigger is output after every step of a sweep finishes,

3 (hp816x_TLS_TRIGOUT_SWSTART), where a trigger is output after a sweep starts, and

4 (hp816x_TLS_TRIGOUT_SWEND), where a trigger is output after a sweep finishes.

Data Type: ViPInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_TLS_attenuation

Syntax:

ViStatus _VI_FUNC hp816x_set_TLS_attenuation(ViSession *ihandle*, ViInt32 *TLSSlot*, ViBoolean *powerMode*, ViBoolean *darkenLaser*, ViReal64 *attenuation*);

The Set TLS Attenuation (hp816x_set_TLS_attenuation) function adjusts the attenuation of a Tunable Laser module.

NOTE If this function addresses a module without a built-in attenuator, an error is generated.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

powerMode

For the Power Mode control, you can choose the following:

Auto (VI_TRUE), where the module automatically regulates the optimum combination between built-in attenuator and the laser current to achieve lowest noise, or

Manual (VI_FALSE), where you can enter the attenuation.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Auto
VI_FALSE	(0)	Manual

darkenLaser

Use the Darken Laser control to darken the laser output.

You use this control when you want to turn the laser On (VI_TRUE) and Off (VI_FALSE) without any settling delays when you turn the laser on.

Your Tunable Laser module must have built-in attenuation.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

attenuation

If you choose Manual as the Power Mode, you can use the Attenuation input to set the attenuation level in decibels (dB).

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_attenuationSettings_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_attenuationSettings_Q(ViSession *ihandle*, **ViInt32** *TLSSlot*, **ViPBoolean** *powerMode*, **ViPBoolean** *dark*, **ViPReal64** *attenuation*);

The Get TLS Attenuation Settings (hp816x_get_TLS_attenuation_Q) function returns the attenuation level, power mode and dark position of a Tunable Laser module.

NOTE:

If this function addresses a module that does not have an built-in attenuator, an error is generated.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3

hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

powerMode

Power Mode returns one of the following values for a Tunable Laser module with built-in attenuation:

1 (VI_TRUE), when the module automatically regulates the optimum combination between built-in attenuator and the laser current to achieve lowest noise, or

0 (VI_FALSE), when the instrument uses a manually-entered attenuation value.

Data Type: ViPBoolean

Input/Output: OUT

dark

Dark returns whether the laser output your Tunable Laser module with built-in attenuation is darkened.

1 (VI_TRUE), when the optical output is darkened, or

0 (VI_FALSE), when the optical output is not darkened.

Data Type: ViPBoolean

Input/Output: OUT

attenuation

Attenuation returns the attenuation level of your Tunable Laser module with built-in attenuation in decibels (dB).

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_TLS_dark

Syntax:

ViStatus _VI_FUNC hp816x_set_TLS_dark(ViSession *ihandle*, ViInt32 *TLSSlot*, ViBoolean *darkenLaser*);

The Set TLS Dark (hp816x_set_TLS_dark) function sets the attenuation of your Tunable Laser module with built-in attenuation to the dark position.

No light is emitted from the optical output if the TLS is set to the dark position.

Use this function to turn the laser on or off without any settling delays.

NOTE If this function addresses a module without a built-in attenuator, an error is generated.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0

hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

darkenLaser

Use the Darken Laser control to darken the laser output.

You use this control when you want to turn the laser On (VI_TRUE) and Off (VI_FALSE) without any settling delays when you turn the laser on.

Your Tunable Laser module must have built-in attenuation.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_darkState_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_darkState_Q(ViSession *ihandle*, ViInt32 *TLSSlot*, ViPBoolean *dark*);

The Get TLS Dark State (hp816x_get_TLS_darkState_Q) function returns whether the optical output of your Tunable Laser module with built-in attenuation is set to the dark position.

NOTE If this function addresses a module without a built-in attenuator, an error is generated.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

dark

Dark returns whether the laser output your Tunable Laser module with built-in attenuation is darkened. One of the folowing values is returned:

1 (VI_TRUE), when the optical output is darkened, or

0 (VI_FALSE), when the optical output is not darkened.

Data Type: ViPBoolean

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_temperatures

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_temperatures(ViSession *Ihandle*, ViReal64 * *Actual_Temperature*, ViReal64 * *Temperature_Difference*, ViReal64 * *Temperature_Last _Zero*);

The Get TLS Temperatures (hp816x_get_temperatures) function returns temperatures related to the auto zeroing of backloadable tuneable laser sources.

Parameter	Description
Ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
Actual_Temperature	Returns the current lamda zero temperature. **Data Type:** ViReal64 * **Input/Output:** OUT
TemperatureDifference	Returns the temperature delta when an auto lambda zero will take place. **Data Type:** ViReal64. *. **Input/Output:** OUT

Temperature_Last_Zero

Returns the temperature at which last auto lambda zero took place.

Data Type: ViReal64.*

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_temperaturesEx

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_temperaturesEx(ViSession *Ihandle*, ViInt32 *TLSSlot*, ViReal64 * *Actual_Temperature*, ViReal64 * *Temperature_Difference*, ViReal64 * *Temperature_Last _Zero*);

The Get TLS Temperatures Ex (hp816x_get_temperaturesEx) function returns temperatures related to the auto zeroing of a selected tuneable laser source.

Parameter	Description
Ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Actual_Temperature

Returns the current lamda zero temperature.

Data Type: ViReal64 *
Input/Output: OUT

TemperatureDifference

Returns the temperature delta when an auto lambda zero will take place.

Data Type: ViReal64. *.
Input/Output: OUT

Temperature_Last_Zero

Returns the temperature at which last auto lambda zero took place.

Data Type: ViReal64.*
Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_TLS_zeroing

Syntax:

ViStatus _VI_FUNC hp816x_TLS_zeroing (ViSession *ihandle*);

The TLS Zeroing (hp816x_TLS_zeroing) function starts a lambda zeroing operation for a Tunable Laser module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_TLS_zeroingAll

Syntax:

ViStatus _VI_FUNC hp816x_TLS_zeroingAll (ViSession *ihandle*);

The TLS Zeroing All (hp816x_TLS_zeroingAll) function starts a lambda zeroing operation for all Tunable Laser modules within the mainframe.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_TLS_zeroingEx

Syntax:

ViStatus _VI_FUNC hp816x_TLS_zeroingEx (ViSession *ihandle*, ViInt32 *TLSSlot*);

The TLS Zeroing Ex (hp816x_TLS_zeroingEx) function starts a lambda zeroing operation for a selected Tunable Laser module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_TLS_autoCalibration

Syntax:

ViStatus _VI_FUNC hp816x_set_TLS_autoCalibration(ViSession *ihandle*, ViInt32 *TLS_Slot*, ViBoolean *Autocalibration*);

Use the Set Auto Calibration function (hp816x_set_TLS_autoCalibration) to choose whether a backloadable Tunable Laser module's autocalibration function is ON or OFF.

A backloadable TLS automatically performs a lambda zeroing routine when the instrument boots.

Setting autocalibration ON allows the TLS to perform lamba zeroing from time to time, to compensate for drift caused, for example, by changes in temperature or other enviromental factors.

Setting autocalibration OFF allows the TLS to operate continuously without interruption by lambda zeroing.

You can force the TLS to perform lambda zeroing at a time of your choice using the TLS Zeroing function (hp816x_TLS_zeroing)

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLS_Slot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Autocalibration

Use the Autocalibration control to choose whether a backloadable Tunable Laser module's autocalibration function is ON or OFF.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Autocalibration ON
VI_FALSE	(0)	Autocalibration OFF

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_autoCalState

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_autoCalState(*ViSession ihandle*, ViInt32 *TLS_Slot*, ViBoolean * *Auto_Calibration_State*);

Use the Get Auto Calibration Sta function (hp816x_get_TLS_autoCalState) to query whether a backloadable Tunable Laser module's autocalibration function is ON or OFF.

If autocalibration is ON, the TLS performs lamba zeroing from time to time, to compensate for drift caused, for example, by changes in temperature or other enviromental factors.

If autocalibration is OFF, the TLS operates continuously without interruption by lambda zeroing.

You can force the TLS to perform lambda zeroing at a time of your choice using the TLS Zeroing function (hp816x_TLS_zeroing)

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLS_Slot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Auto_Calibration_State

Auto Calibration State returns whether a backloadable Tunable Laser module's autocalibration function is ON or OFF.

If 0 (VI_FALSE) is returned, autocalibration is turned off.

If 1 (VI_TRUE) is returned, autocalibration is turned on.

Data Type: ViBoolean

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_accClass

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_accClass(*ViSession ihandle*, ViInt32 *TLS_Slot*, Vint32 **Accuracy_Class*);

Use the Get Accuracy Class function (hp816x_get_TLS_autoCalState) to query the accuracy class of a backloadable Tunable Laser module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLS_Slot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Accuracy_Class

This control returns the accuracy class of backloadable tunable laser source.

If the TLS is of the highest accuracy, 1 is returned.

Data Type: ViInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_displayToLambdaZero

Syntax:

ViStatus _VI_FUNC hp816x_displayToLambdaZero(ViSession *ihandle*, ViInt32 *TLSSlot*);

The TLS Display to Lambda Zero function (**hp816x_get_TLS_lambdaZero_Q**) transfers the actual displayed wavelength to Lambda0, the base wavelength for calculating the frequency offset.

See your instrument's User's Guide for more information on the frequency offset.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_lambdaZero_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_lambdaZero_Q(ViSession *ihandle*, ViInt32 *TLSSlot*, ViPReal64 *lambda0*);

The Get Lambda Zero function (hp816x_get_TLS_lambdaZero_Q) returns the base wavelength (Lambda0) of a Tunable Laser module.

See your instrument's User's Guide for more information on the frequency offset.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

lambda0

Lamda0 outputs the base wavelength of your Tunable Laser module in meters (m).

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_TLS_frequencyOffset

Syntax:

ViStatus _VI_FUNC hp816x_set_TLS_frequencyOffset(ViSession *ihandle*, ViInt32 *TLSSlot*, ViReal64 *offset*);

The Set TLS Frequency Offset function (hp816x_set_TLS_frequencyOffset) sets the reference wavelength in Hertz (Hz).

See your instrument's User's Guide for more information on the frequency offset.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

offset

You can use the Offset input to set the frequency offset of your Tunable Laser module in Hertz (Hz).

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_frequencyOffset_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_frequencyOffset_Q(ViSession *ihandle*, ViInt32 *TLSSlot*, ViPReal64 *offset*);

The Get TLS Frequency Offset function (hp816x_get_TLS_frequencyOffset_Q) function returns the frequency offset in Hertz (Hz).

See your instrument's User's Guide for more information on the frequency offset.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4

hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

offset

Offset returns the frequency offset of your Tunable Laser module in Hertz (Hz).

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_TLS_sweep

Syntax:

ViStatus _VI_FUNC hp816x_set_TLS_sweep(ViSession *ihandle*, **ViInt32** *TLSSlot*, **ViInt32** *sweepMode*, **ViInt32** *repeatMode*, **ViUInt32** *cycles*, **ViReal64** *dwellTime*, **ViReal64** *startWavelength*, **ViReal64** *stopWavelength*, **ViReal64** *stepSize*, **ViReal64** *sweepSpeed*);

The Set TLS Sweep (hp816x_set_TLS_Sweep) function sets up a sweep operation for a Tunable Laser module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10

hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

sweepMode

Use the Sweep Mode control to choose one of the following sweep modes:

Stepped (hp816x_SWEEP_STEP), which stops at wavelengths that are separated by a certain step size,

Manual (hp816x_SWEEP_MAN), which you can run each step manually, and

Continuous (hp816x_SWEEP_CONT), which sweeps continually at the speed you set.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_SWEEP_STEP	0	Stepped
hp816x_SWEEP_MAN	1	Manual
hp816x_SWEEP_CONT	2	Continuous

repeatMode

Use the Repeat Mode control to choose between:

Oneway (hp816x_ONEWAY), where every cycle of a sweep starts at the Start Wavelength, and

Twoway (hp816x_TWOWAY), where every odd cycle of the sweep starts at the start wavelength and every even cycle starts at the Stop Wavelength.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description

hp816x_TWOWAY	1	Twoway
hp816x_ONEWAY	0	Oneway

cycles

Use Cycles to set the number of times a sweep is repeated.

Data Type: ViUInt32
Input/Output: IN

Values:

Cycles_MIN	1
Cycles_MAX	65535

dwellTime

Use Dwell time to enter the amount of time in seconds spent at each step, for a stepped sweep.

Data Type: ViReal64
Input/Output: IN

startWavelength

Use Start Wavelength to enter the wavelength in meters (m) at which the sweep begins.

Data Type: ViReal64
Input/Output: IN

stopWavelength

Use Stop Wavelength to enter the wavelength in meters (m) at which the sweep ends.

Data Type: ViReal64
Input/Output: IN

stepSize

Use Step Size in "Stepped" sweep mode to enter the size of the change in the wavelength in meters (m) for each wavelength step.

Use Step Size in "Continuous" sweep mode to set the wavelength interval that determines when the instrument triggers. This allows you to take power

measurements at specific wavelength intervals using a logging or stability data acquisition function.

Data Type: ViReal64

Input/Output: IN

sweepSpeed

Use Sweep Speed to select the speed of a continuous sweep. You can choose one of the following values:

This parameter is only used if you use continuous sweep mode.

Data Type: ViReal64

Input/Output: IN

Values:

Name	Value	Description
hp816x_SWEEP_SPEED_LOW	0.5e-9	0.5 nm/s
hp816x_SWEEP_SPEED_MEDIUM	5.0e-9	5 nm/s
	10.0e-9	10 nm/s
	20.0e-9	20 nm/s
hp816x_SWEEP_SPEED_HIGH	40.0e-9	40 nm/s
	80.0e-9	80 nm/s

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_TLS_sweepControl

Syntax:

ViStatus _VI_FUNC hp816x_TLS_sweepControl(ViSession *ihandle*, ViInt32 *TLSSlot*, ViInt32 *action*);

The TLS Sweep Control (hp816x_TLS_sweepControl) function controls a wavelength sweep operation.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9

hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

action

Use the Action control to perform one of the following wavelength sweep actions:

Start (hp816x_SW_CMD_START) , start a wavelength sweep,

Stop (hp816x_SW_CMD_STOP), stop a wavelength sweep,

Pause (hp816x_SW_CMD_PAUSE) , pause a wavelength sweep,

Continue (hp816x_SWEEP_CONT), continue a paused wavelength sweep.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_SW_CMD_STOP	0	Stop
hp816x_SW_CMD_START	1	Start
hp816x_SW_CMD_PAUSE	2	Pause
hp816x_SW_CMD_CONT	3	Continue

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_sweepState_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_sweepState_Q(ViSession *ihandle*, ViInt32 *TLSSlot*, ViPInt32 *sweepState*);

The Get TLS Sweep State (hp816x_get_TLS_sweepState_Q) function returns the current state of a sweep.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

sweepState

Sweep State returns the one of the following sweep states:

0 (hp816x_SW_CMD_STOP), no sweep is being performed,

1 (hp816x_SW_CMD_START), sweeping has been started, or

2 (hp816x_SW_CMD_PAUSE), sweeping has been paused.

Data Type: ViPInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_TLS_sweepNextStep

Syntax:

ViStatus _VI_FUNC hp816x_TLS_sweepNextStep(ViSession *ihandle*, ViInt32 *TLSSlot*);

The TLS Next Step (hp816x_TLS_sweepNextStep) function sweeps one step forward.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_TLS_sweepPreviousStep

Syntax:

ViStatus _VI_FUNC hp816x_TLS_sweepPreviousStep(ViSession *ihandle*, ViInt32 *TLSSlot*);

The TLS Previous Step (hp816x_TLS_sweepPreviousStep) function sweeps one step backward.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_TLS_sweepWait

Syntax:

ViStatus _VI_FUNC hp816x_TLS_sweepWait(ViSession *ihandle*, ViInt32 *TLSSlot*);

The Wait Sweep Finished (hp816x_TLS_sweepWait) function forces the program to wait until the sweep finishes.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_TLS_modulation

Syntax:

ViStatus _VI_FUNC hp816x_set_TLS_modulation(ViSession *ihandle*, **ViInt32** *TLSSlot*, **ViInt32** *modulationSource*, **ViInt32** *modulationOutput*, **ViBoolean** *modulation*, **ViReal64** *modulationFrequency*);

The Set TLS Modulation (hp816x_setup_TLS_modulation) function sets the modulation of your Tunable Laser module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13

hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

modulationSource

Use Modulation Source to choose one of the following modulation sources:

Internal (hp816x_MOD_INT), internal digital modulation,

Coherence Control (hp816x_MOD_CC), coherence control,

External Analog (hp816x_MOD_AEXT), external analog modulation,

External Digital (hp816x_MOD_DEXT), external digital modulation,

Wavelength Locking (hp816x_MOD_VWLOCK), wavelength locking,

Backplane (hp816x_MOD_BACKPL), external digital modulation using the Input Trigger connector, or

ATTENTION Not all modulation sources are available for all Tunable Laser modules.

See your instrument's User's Guide for more information on modulation sources.

Data Type: ViInt32
Input/Output: IN

Values:

Name	Value	Description
hp816x_MOD_INT	0	Internal
hp816x_MOD_CC	1	Coherence Control
hp816x_MOD_AEXT	2	External Analog
hp816x_MOD_DEXT	3	External Digital
hp816x_MOD_VWLOCK	4	Wavelength Locking
hp816x_MOD_BACKPL	5	Backplane

modulationOutput

Use the Modulation Output control to specify whether the modulation output signal is only active when the laser is active, a Laser Ready signal (hp816x_MOD_LREADY), or if the modulation output is always active regardless of laser state, Always (hp816x_MO D_ALWAYS).

Modulation Output only applies to backloadable

Tunable Laser modules.

Data Type: ViInt32
Input/Output: IN

Values:

Name	Value	Description
hp816x_MOD_LREADY	1	Laser Ready
hp816x_MOD_ALWAYS	0	Always

modulation

Use the Modulation control to turn modulation On (hp816x_MOD_ENABLED) or Off (hp816x_MOD_DISABLED).

Data Type: ViBoolean
Input/Output: IN

Values:

Name	Value	Description
hp816x_MOD_ENABLED	1	On
hp816x_MOD_DISABLED	0	Off

modulationFrequency

Use Modulation Frequency to set the modulation frequency of an internally modulated signal in Hertz (Hz).

Data Type: ViReal64
Input/Output: IN

Return Value:

VI_SUCCESS: No error
Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_modulationSettings_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_modulationSettings_Q(ViSession *ihandle*, ViInt32 *TLSSlot*, ViPInt32 *modulationSource*, ViPBoolean *modulationOutput*, ViPBoolean *modulationState*, ViPReal64 *frequency*);

The Get TLS Modulation Setting (hp816x_get_TLS_modulationSettings_Q) function returns the current modulation setting of a Tunable Laser module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

modulationSource

Modulation Source returns the one of the following modulation sources:

0 (hp816x_MOD_INT), internal digital modulation,

1 (hp816x_MOD_CC), coherence control,

2 (hp816x_MOD_AEXT), external analog modulation,

3 (hp816x_MOD_DEXT), external digital modulation using the front connector,

4 (hp816x_MOD_VWLOCK), wavelength locking,

5 (hp816x_MOD_BACKPL), external digital modulation using the Input Trigger connector at the backplane, or

See your instrument's User's Guide for more information on modulation sources.

Data Type: ViPInt32

Input/Output: OUT

modulationOutput

Modulation Output returns one of the following:

1 (hp816x_MOD_LREADY), the modulation output signal is only active when the laser is active, or

0 (hp816x_MOD_ALWAYS), the modulation output is always active regardless of laser state.

Data Type: ViPBoolean

Input/Output: OUT

modulationState

Modulation State returns whether the output signal is modulated:

1 (hp816x_MOD_ENABLED), if modulation is enabled, or

0 (hp816x_MOD_DISABLED), if modulation is disabled.

Data Type: ViPBoolean

Input/Output: OUT

frequency

Frequency returns the frequency in Hertz (Hz) for internally modulated signals.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_TLS_configureBNC

Syntax:

ViStatus _VI_FUNC hp816x_TLS_configureBNC(ViSession *ihandle*, ViInt32 *TLSSlot*, ViInt32 *BNCOutput*);

The TLS Configure BNC Output (hp816x_TLS_configureBNC) function configures the BNC output connector on the front panel of some Tunable Laser modules.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

BNCOutput

You can use the BNC Output control to choose between the following:

Modulation (hp816x_BNC_MOD), where the BNC output signal is a TTL signal at the same modulation frequency as the optical output signal (if the optical output signal is digitally modulated), and

VPP (hp816x_BNC_VPP), where the BNC output signal is proportional to the power of the optical signal.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_BNC_MOD	0	Modulation
hp816x_BNC_VPP	1	VPP

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_BNC_config_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_BNC_config_Q(ViSession *ihandle*, ViInt32 *TLSSlot*, ViPInt32 *BNCOutput*);

The Get TLS BNC Configuration (hp816x_get_TLS_BNC_config_Q) function returns the configuration of the BNC output connector on the front panel of some Tunable Laser modules.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

BNCOutput

BNC Output returns the following BNC output connector configurations:

0 (hp816x_BNC_MOD), when the BNC output signal is a TTL signal at modulation frequency if your optical output signal is digitally modulated, or

1 (hp816x_BNC_VPP), when the BNC output signal is proportional to the power of the optical signal.

Data Type: ViPInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_TLS_ccLevel

Syntax:

ViStatus _VI_FUNC hp816x_set_TLS_ccLevel(ViSession *ihandle*, ViInt32 *TLS_Slot*, ViReal64 *CC_Level*);

Use the Set Coherence Control Lev (DFB) function (hp816x_set_TLS_ccLevel) to specify the coherence control level (1-99.98%) of a DFB laser source.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLS_Slot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

CC_Level

Use this control to set the coherence control level in percent.

Allowed values are between 1 and 99.98

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_ccLevel_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_ccLevel_Q(ViSession *ihandle*, ViInt32 *TLS_Slot*, ViReal64 * *CC_Level*);

Use the Get Coherence Control Lev (DFB) function (hp816x_set_TLS_ccLevel) to query the coherence control level set for a DFB laser source.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLS_Slot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

CC_Level

Returns the coherence control level set for a DFB source, in percent.

Allowed values are between 1 and 99.98

Data Type: ViReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_TLS_SBSLevel

Syntax:

ViStatus _VI_FUNC hp816x_set_TLS_SBSLevel(ViSession *ihandle*, ViInt32 *TLS_Slot*, ViReal64 *SBSLevel*);

Use the Set SBS Level function (hp816x_set_TLS_SBSLevel) to specify the SBS Suppression level (1-99.98%) of a tunable laser source.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLS_Slot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

SBSLevel

Use this control to set the SBS Suppression level in percent.

Allowed values are between 1 and 99.98

Data Type: ViReal64
Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_SBSLevel_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_SBSLevel_Q(ViSession *ihandle*, ViInt32 *TLS_Slot*, ViReal64 * *SBSLevel*);

Use the Get SBS Level function (hp816x_set_TLS_SBSLevel) to query the SBS Suppression level set for a tunable laser source.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLS_Slot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

SBSLevel

Returns the SBS Suppression level set for a tunable laser source, in percent.

Allowed values are between 1 and 99.98

Data Type: ViReal64
Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_TLS_SBS_control

Syntax:

ViStatus _VI_FUNC hp816x_set_TLS_SBS_control(ViSession *ihandle*, ViInt32 *TLSSlot*, ViBoolean *mod_state*, ViReal64 *modulationFrequency*);

The Set TLS SBS Control (hp816x_set_TLS_SBS_control) function enables or disables the SBS Suppression feature of an 8194xA or 8198xA compact Tunable Laser Source module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8

hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

mod_state

Use the Modulation State control to enable SBS suppression (hp816x_SBS_CONTROL_ENABLED), or disable SBS suppression (hp816x_SBS_CONTROL_DISABLED)

SBS Suppression is a feature of 8194xA and 8196xA compact TLS modules.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
hp816x_SBS_CONTROL_ENABLED	1	On
hp816x_SBS_CONTROL_DISABLED	0	Off

modulationFrequency

Use Modulation Frequency to set the modulation frequency of the SBS Suppression in Hertz (Hz).

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_SBS_control_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_SBS_control_Q(ViSession *ihandle*, ViInt32 *TLSSlot*,ViPBoolean **modState*, ViPReal64 **frequency*);

The Get TLS SBS Control (hp816x_get_TLS_SBS_control_Q) function returns the current SBS suppression settings for an 8194xA or 8198xA compact Tunable Laser Source module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6

hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

**mod_state*

Modulation State returns:

1 (hp816x_SBS_CONTROL_ENABLED), if SBS suppression is on, or

0 (hp816x_SBS_CONTROL_DISABLED), if SBS suppression is off.

Data Type: ViPBoolean

Input/Output: OUT

**frequency*

Frequency returns the frequency of SBS suppression in Hertz (Hz).

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_powerPoints_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_powerPoints_Q(ViSession *ihandle*, ViInt32 *TLSSlot*, ViPInt32 *numberofPowerPoints*);

The Get Number of Power Points (hp816x_get_TLS_powerPoints_Q) function returns the maximum number of data items (wavelength, power) which will be returned from the Get Logged Power Data (hp816x_get_TLS_power_data_Q) function.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

numberofPowerPoints

Number of Power Points returns the maximum number of data items (wavelength, power) which will be returned from the Get Logged Power Data (hp816x_get_TLS_power_data_Q) function.

Use to generate the following 3 inputs for the Get Logged Power Data (hp816x_get_TLS_power_data_Q) function:

- set the Number of Data Items input to the same value as this output,

- allocate a one-dimensional ViReal64 array of the same dimension as this output and use as the Wavelength Data input, and

- allocate a one-dimensional ViReal64 array of the same dimension as this output and use as the Power Data input.

Data Type: ViPInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_powerData_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_powerData_Q(ViSession *ihandle*, ViInt32 *TLSSlot*, ViInt32 *numberofDataItems*, ViReal64*wavelengthData*[ ], ViReal64*powerData*[ ]);

The Get Logged Power Data (hp816x_get_TLS_power_data_Q) function returns two arrays, the first array returns wavelength values and the second array returns corresponding power values.

The position of each wavelength and power value in each array corresponds to describe the maximum power available at the corresponding wavelength. In this way the two arrays describe the maximum power curve for the Tunable Laser module.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2

hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

numberofDataItems

Data Type: ViInt32

Input/Output: IN

wavelengthData

Wavelength Data is both an input and a output.

As an input, allocate Wavelength Data as a one-dimensional ViReal64 array of the same dimension as returned by the Number of Power Values output of the Get Number of Power Values (**hp816x_get_TLS_powerPoints_Q**) function.

As a return, Wavelength Data returns a one dimensional array. Each component contains a wavelength. The wavelength corresponds to a power value at the same component index in the Power Data array.

Data Type: ViReal64[]

Input/Output: OUT

powerData

Power Data is both an input and an output.

As an input, allocate Power Data as a one-dimensional ViReal64 array of the same dimension as returned by the Number of Power Values output of the Get Number of Power Values (**hp816x_get_TLS_powerPoints_Q**) function.

As a return, Power Data returns a one-dimensional array. The power value corresponds to a wavelength at the same component index in the Wavelength Data array.

Data Type: ViReal64[]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_TLS_lambdaLoggingState

Syntax:

ViStatus _VI_FUNC hp816x_set_TLS_lambdaLoggingState(ViSession *ihandle*, ViBoolean *lambdaLoggingState*);

The Set TLS Lambda Logging State (hp816x_set_TLS_lambdaLoggingState) function turns Lambda Logging On or Off.

Lambda Logging is a feature that records the exact wavelength of a Tunable Laser module when a trigger is generated during a continuous sweep.

Lambda Logging is disabled at the end of a sweep. If you want to start a new wavelength sweep with Lambda Logging activated, you must be reset Lambda Logging to On.

NOTE: Lambda Logging is not available if your Tunable Laser module firmware revision is lower than 2.0.

NOTE: After this function completes, Trigger Out mode of the Set TLS Trigger Configuration (**hp816x_set_TLS_triggerConfiguration**) function will be set to Step Finished (hp816x_TRIGOUT_STEP_FINISHED). Call the Set TLS Trigger Configuration (**hp816x_set_TLS_triggerConfiguration**) function to readjust the Trigger Out mode.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
lambdaLoggingState	Use the Lambda Logging State control to turn the Lambda Logging On (VI_TRUE) or Off (VI_FALSE). **Data Type:** ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_lambdaLoggingState_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_lambdaLoggingState_Q(ViSession *ihandle*, ViPBoolean *lambdaLoggingState*);

The Get TLS Lambda Logging State (hp816x_get_TLS_lambdaLoggingState_Q) function returns whether Lambda Logging is turned On or Off.

Lambda Logging is a feature that records the exact wavelength of a Tunable Laser module when a trigger is generated during a continuous sweep.

You can use this function to determine when a Lambda Logging operation is completed. This is useful because the Get Number of Wavelength Values Extended(**hp816x_get_TLS_wavelengthPoints_Q**) function and the Get Logged Wavelength Data **(hp816x_get_TLS_wavelengthData_Q)** function return meaningless values unless the Lambda Logging operation is completed.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
lambdaLoggingState	Lambda Logging State returns whether the Lambda Logging is turned on or off. If 0 (VI_FALSE) is returned, the Lambda Logging is turned off. If 1 (VI_TRUE) is returned, the Lambda Logging is turned on. **Data Type:** ViPBoolean

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_TLS_lambdaLoggingStateEx

Syntax:

ViStatus _VI_FUNC hp816x_set_TLS_lambdaLoggingStateEx(ViSession *ihandle*, ViInt32 *TLSSlot*, ViBoolean *lambdaLoggingState*);

The Set TLS Lambda Logging State Ex (hp816x_set_TLS_lambdaLoggingStateEx) function turns Lambda Logging On or Off for a selected Tunable Laser Module.

Lambda Logging is a feature that records the exact wavelength of a Tunable Laser module when a trigger is generated during a continuous sweep.

Lambda Logging is disabled at the end of a sweep. If you want to start a new wavelength sweep with Lambda Logging activated, you must be reset Lambda Logging to On.

NOTE: Lambda Logging is not available if your Tunable Laser module firmware revision is lower than 2.0.

NOTE: After this function completes, Trigger Out mode of the Set TLS Trigger Configuration (**hp816x_set_TLS_triggerConfiguration**) function will be set to Step Finished (hp816x_TRIGOUT_STEP_FINISHED). Call the Set TLS Trigger Configuration (**hp816x_set_TLS_triggerConfiguration**) function to readjust the Trigger Out mode.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

lambdaLoggingState

Use the Lambda Logging State control to turn the Lambda Logging On (VI_TRUE) or Off (VI_FALSE).

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_lambdaLoggingStateEx_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_lambdaLoggingStateEx_Q(ViSession *ihandle*, ViInt32 *TLSSlot*, ViPBoolean *lambdaLoggingState*);

The Get TLS Lambda Logging State Extended (hp816x_get_TLS_lambdaLoggingStateEx_Q) function returns, for a selected Tunable Laser module, whether Lambda Logging is turned On or Off.

Lambda Logging is a feature that records the exact wavelength of a Tunable Laser module when a trigger is generated during a continuous sweep.

You can use this function to determine when a Lambda Logging operation is completed. This is useful because the Get Number of Wavelength Values Ex (hp816x_get_TLS_wavelengthPointsEx_Q) function and the Get Logged Wavelength Data Ex **(hp816x_get_TLS_wavelengthDataEx_Q)** function return meaningless values unless the Lambda Logging operation is completed.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0

hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

lambdaLoggingState

Lambda Logging State returns whether the Lambda Logging is turned on or off.

If 0 (VI_FALSE) is returned, the Lambda Logging is turned off.

If 1 (VI_TRUE) is returned, the Lambda Logging is turned on.

Data Type: ViPBoolean

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_wavelengthPoints_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_wavelengthPoints_Q(ViSession *ihandle*, ViPReal64 *numberofWavelengthValues*);

The Get Number of Wavelength Values (hp816x_get_TLS_wavelengthPoints_Q) function returns the number of wavelength points that will be used in a Lambda Logging operation.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
numberofWavelengthValues	Number of Wavelength Values returns the number of wavelength values that were returned by the last Lambda Logging operation. If you call this function before the Lambda Logging operation completes, the returned value is meaningless. Use the Get Lambda Logging State function (**hp816x_get_TLS_lambdaLoggingState_Q**) to confirm that the Lambda Logging operation is completed. **Data Type:** ViPReal64 **Input/Output:** OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_wavelengthPointsEx_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_wavelengthPointsEx_Q(ViSession *ihandle*, ViInt32 *TLSSlot*, ViPReal64 *numberofWavelengthValues*);

The Get Number of Wavelength Values Ex (hp816x_get_TLS_wavelengthPointsEx_Q) function returns, for a selected Tunable Laser Module, the number of wavelength points that will be used in a Lambda Logging operation.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3
hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5

hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

numberofWavelengthValues

Number of Wavelength Values returns the number of wavelength values that were returned by the last Lambda Logging operation.

If you call this function before the Lambda Logging operation completes, the returned value is meaningless. Use the Get Lambda Logging State Ex function (**hp816x_get_TLS_lambdaLoggingStateEx_Q**) to confirm that the Lambda Logging operation is completed.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_wavelengthData_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_wavelengthData_Q(ViSession *ihandle*, ViInt32 *arraySize*, ViReal64*wavelengthData*[ ]);

The Get Logged Wavelength Data (hp816x_get_TLS_wavelengthData_Q) function returns the results of a Lambda Logging operation.

If you call this function before the Lambda Logging operation completes, the returned value is meaningless. Use the Get Lambda Logging State function (**hp816x_get_TLS_lambdaLoggingState_Q**) to confirm that the Lambda Logging operation is completed.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
arraySize	Use Array Size to set the size of the memory buffer in bytes needed for the function to complete. Use the Number of Wavelength Values output of the Get Number of Wavelength Values (**hp816x_get_TLS_wavelengthPoints_Q**) to set this value. You should multiply this value by 8 to ensure that the function completes. **Data Type:** ViInt32 **Input/Output:** IN

wavelengthData

Wavelength Data is both an input and an output.

As an input, Wavelength Data accepts a one-dimensional real array that determines the size of the Wavelength Data output. Allocate a one-dimensional real array using the the Number of Wavelength Values output of the Get Number of Wavelength Values (hp816 x_get_TLS_wavelengthPoints_Q) to set the dimension of the array.

As an output, Wavelength Data returns a one dimensional array containing the exact wavelength values of the Tunable Laser module when triggers were generated during the continuous sweep.

Data Type: ViReal64[]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_get_TLS_wavelengthDataEx_Q

Syntax:

ViStatus _VI_FUNC hp816x_get_TLS_wavelengthDataEx_Q(ViSession *ihandle*, ViInt32 *TLSSlot*, ViInt32 *arraySize*, ViReal64*wavelengthData*[ ]);

The Get Logged Wavelength Data Extended (hp816x_get_TLS_wavelengthDataEx_Q) function returns, for a selected Tunable Laser Module, the results of a Lambda Logging operation.

If you call this function before the Lambda Logging operation completes, the returned value is meaningless. Use the Get Lambda Logging State Ex function (**hp816x_get_TLS_lambdaLoggingStateEx_Q**) to confirm that the Lambda Logging operation is completed.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
TLSSlot	Use the TLS Slot control to choose an installed Tunable Laser module. **Data Type:** ViInt32 **Input/Output:** IN

Values:

Name	Value	Description
hp816x_SLOT_0	0	Slot 0
hp816x_SLOT_1	1	Slot 1
hp816x_SLOT_2	2	Slot 2
hp816x_SLOT_3	3	Slot 3

hp816x_SLOT_4	4	Slot 4
hp816x_SLOT_5	5	Slot 5
hp816x_SLOT_6	6	Slot 6
hp816x_SLOT_7	7	Slot 7
hp816x_SLOT_8	8	Slot 8
hp816x_SLOT_9	9	Slot 9
hp816x_SLOT_10	10	Slot 10
hp816x_SLOT_11	11	Slot 11
hp816x_SLOT_12	12	Slot 12
hp816x_SLOT_13	13	Slot 13
hp816x_SLOT_14	14	Slot 14
hp816x_SLOT_15	15	Slot 15
hp816x_SLOT_16	16	Slot 16
hp816x_SLOT_17	17	Slot 17

arraySize

Use Array Size to set the size of the memory buffer in bytes needed for the function to complete.

Use the Number of Wavelength Values output of the Get Number of Wavelength Values Extended (**hp816x_get_TLS_wavelengthPointsEx_Q**) to set this value. You should multiply this value by 8 to ensure that the function completes.

Data Type: ViInt32

Input/Output: IN

wavelengthData

Wavelength Data is both an input and an output.

As an input, Wavelength Data accepts a one-dimensional real array that determines the size of the Wavelength Data output. Allocate a one-dimensional real array using the the Number of Wavelength Values output of the Get Number of Wavelength Values (hp816 x_get_TLS_wavelengthPoints_Q) to set the dimension of the array.

As an output, Wavelength Data returns a one dimensional array containing the exact wavelength values of the Tunable Laser module when triggers were generated during the

continuous sweep.

Data Type: ViReal64[]
Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_set_LambdaScan_wavelength

Syntax:

ViStatus _VI_FUNC hp816x_set_LambdaScan_wavelength(ViSession *ihandle*, ViReal64 *powerMeterWavelength*);

The Set Lambda Scan Wavelength (hp816x_set_LambdaScan_wavelength) function allows you to use a different wavelength than 1550 nm during a Lambda Scan operation. All Power Meters taking part in the Lambda Scan operation will be set to the chosen wavelengt h.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
powerMeterWavelength	Use Power Meter Wavelength to change the wavelength for the Lambda Scan operation. **Data Type:** ViReal64 **Input/Output:** IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_enableHighSweepSpeed

Syntax:

ViStatus _VI_FUNC hp816x_enableHighSweepSpeed(ViSession *ihandle*, ViBoolean *fastSweepSpeed*);

The Enable High Sweep Speed (hp816x_enableHighSweepSpeed) function enables/disables the highest available sweep speed (40 nanometers per second) for Lambda Scan operations. The Lambda Scan operation chooses the highest possible sweep speed for the chosen step size.

If you choose Enable, the highest sweep speed possible will be used. This may lead to less accurate measurements.

If you choose Disable, the highest sweep speed will not be used.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
fastSweepSpeed	Use the Fast Sweep Speed control to enable/disable the highest available sweep speed (40 nanometers per second) for Lambda Scan operations. If you choose Enable (VI_TRUE), the highest sweep speed possible will be used. This may lead to less accurate measurements. If you choose Disable (VI_FALSE), the highest sweep speed will not be used. **Data Type:** ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Enable
VI_FALSE	(0)	Disable

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_prepareLambdaScan

Syntax:

ViStatus _VI_FUNC hp816x_prepareLambdaScan(ViSession *ihandle*, ViInt32 *powerUnit*, ViReal64 *power*, ViInt32 *opticalOutput*, ViInt32 *numberofScans*, ViInt32 *PWMChannels*, ViReal64 *startWavelength*, ViReal64 *stopWavelength*, ViReal64 *stepSize*, ViUInt32 *numberofDatapoints*, ViUInt32 *numberofChannels*);

The Prepare Lambda Scan (hp816x_prepareLambdaScan) function prepares a Lambda Scan operation.

That is, it prepares an operation where a 8164A/B Lightwave Measurement System with a back-loadable Tunable Laser module and up to four Power Sensors installed, performs a wavelength sweep where the Tunable Laser module and Power Sensors are coordinated with each other.

The Prepare Lambda Scan (hp816x_prepareLambdaScan) function must be called before a Lambda Scan operation is executed. Use the return values of this function (Number of Datapoints and Number of Power Arrays) to allocate arrays for the Execute Lambda Scan (**hp816x_executeLambdaScan**) function.

To obtain a higher precision, the Tunable Laser Source is set 1 nm before the Start Wavelength, this means, you have to choose a Start Wavelength 1 nm greater than the minimum possible wavelength. Also, the wavelength sweep is actually started 90 pm befo re the Start Wavelength and ends 90 pm after the Stop Wavelength, this means, you have to choose a Stop Wavelength 90 pm less than the maximum possible wavelength.

Triggers coordinate the Tunable Laser module with all Power Meters. The function sets for the lowest possible averaging time available for the installed Power Meters and, then, sets the highest possible sweep speed for the selected Tunable Laser module s weep.

If one of the following circumstances occurs, the "parameter mismatch" error will be returned:

1. If one Power Meter is out of the specification at 1550 nm, the error "powermeter wavelength does not span 1550nm" will be returned. For example, the 81530A Power Sensor and the 81520A Optical Head are out of specification at 1550 nm. Remove the Power Meter that is out of specification at 1550 nm from the mainframe.

2. If the Step Size is too small and results in a trigger frequency that is to high for the installed Power Meters, the error "could not calculate a sweep speed!" will be returned. Increase the Step Size.

3. If the chosen wavelength range is too large and Step Size is too small, the error "too many datapoints to log!" will be returned. In this case, reduce the wavelength range and/or increase the Step Size.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**).

Data Type: ViSession

Input/Output: IN

powerUnit

You can use the Unit control to choose either Watts (hp816x_PU_WATT) or dBm (hp816x_PU_DBM) as the power unit.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_PU_DBM	0	dBm
hp816x_PU_WATT	1	Watt

power

You can use the Power input to set the output power to a value in dBm.

Data Type: ViReal64

Input/Output: IN

opticalOutput

Use the Optical Output control to choose the optical output for a TLS module with two outputs. You can choose one of the following:

High Power (hp816x_HIGHPOW), where Output 2, the High Power output is the regulated output.

Low SSE (hp816x_LOWSSE), where Output 1, the Low SSE output is the regulated output.

BHR (hp816x_BHR) is the same as choosing Both (Master 2) from the instrument's front panel, Output 2, the High Power output is the regulated output and Output 1, the Low SSE output is the unregulated output.

BLR (hp816x_BLR) is the same as choosing Both (Master 1) from the instrument's front panel, the Low SSE output is the regulated output and Output 2, the High Power output is the unregulated output.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_HIGHPOW	0	High Power
hp816x_LOWSSE	1	Low SSE
hp816x_BHR	2	BHR
hp816x_BLR	3	BLR

numberofScans

Number of Scans sets the number of sweep cycles that will be performed.

To perform 1 scan enter 0 (hp816x_NO_OF_SCANS_1).

To perform 2 scans enter 1 (hp816x_NO_OF_SCANS_2).

To perform 3 scans enter 2 (hp816x_NO_OF_SCANS_3).

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_NO_OF_SCANS_1	0	1
hp816x_NO_OF_SCANS_2	1	2
hp816x_NO_OF_SCANS_3	2	3

PWMChannels

Use the PWM Channels control to specify how many Power Meter channels should be used.

Data Type: ViInt32

Input/Output: IN

startWavelength

Use Start Wavelength to enter the wavelength in meters (m) at which the sweep begins.

Data Type: ViReal64

Input/Output: IN

stopWavelength

Use Stop Wavelength to enter the wavelength in meters (m) at which the sweep ends.

Data Type: ViReal64

Input/Output: IN

stepSize

Use Step Size to enter the size of the change in the wavelength in meters (m) for each wavelength step.

Data Type: ViReal64

Input/Output: IN

numberofDatapoints

Number of Datapoints returns the number of wavelength steps that the Lambda Scan will perform.

Each Power Meter channel will make the same number of measurements, use the returned value for allocating the size of power value arrays.

Data Type: ViUInt32

Input/Output: OUT

numberofChannels

Number of Channels returns the number of Power Meter channel that a Lambda Scan operation will use.

Use this value to allocate the number of power value arrays for the Execute Lambda Scan function (**hp816x_executeLambdaScan**).

Data Type: ViUInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_getLambdaScanParameters_Q

Syntax:

ViStatus _VI_FUNC hp816x_getLambdaScanParameters_Q(ViSession *ihandle*, ViPReal64 *startWavelength*, ViPReal64 *stopWavelength*, ViPReal64 *averagingTime*, ViPReal64 *sweepSpeed*);

The Get Lambda Scan Parameters function (hp816x_getLambdaScanParameters_Q) returns all parameters that the Prepare Lambda Scan (**hp816x_prepareLambdaScan**) function adjusts or automatically calculates.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
startWavelength	Start Wavelength returns the adjusted wavelength value in meters that the Prepare Lambda Scan (**hp816x_prepareLambdaScan**) function generates. This Start Wavelength is used as the Tunable Laser module start wavelength for the Lambda Scan operation. **Data Type:** ViPReal64 **Input/Output:** OUT
stopWavelength	Stop Wavelength returns the adjusted wavelength value in meters that the Prepare Lambda Scan

(**hp816x_prepareLambdaScan**) function generates.

This Stop Wavelength is used as the Tunable Laser module stop wavelength for the Lambda Scan operation.

Data Type: ViPReal64

Input/Output: OUT

averagingTime

Averaging Time returns the averaging time value in seconds that the Prepare Lambda Scan (**hp816x_prepareLambdaScan**) function generates.

This Averaging Time is used for all installed Power Meters for the Lambda Scan operation.

Data Type: ViPReal64

Input/Output: OUT

sweepSpeed

Sweep Speed returns the sweep speed in meters per second (m/s) that the Prepare Lambda Scan (**hp816x_prepareLambdaScan**) function generates.

This Sweep Speed is used as the Tunable Laser module sweep speed for the Lambda Scan operation.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_executeLambdaScan

Syntax:

ViStatus _VI_FUNC hp816x_executeLambdaScan(ViSession *ihandle*, ViReal64*wavelengthArray*[ ], ViReal64*powerArray1*[ ], ViReal64*powerArray2*[ ], ViReal64*powerArray3*[ ], ViReal64*powerArray4*[ ], ViReal64*powerArray5*[ ], ViReal64*powerArray6*[ ], ViReal64*powerArray7*[ ], ViReal64*powerArray8*[ ]);

The Execute Lambda Scan (hp816x_executeLambdaScan) function runs and returns the results of a Lambda Scan operation.

That is, it executes an operation where a 8164A/B Lightwave Measurement System with a back-loadable Tunable Laser module and up to four Power Sensors installed, performs a wavelength sweep where the Tunable Laser module and Power Sensors are coordi nated with each other.

The Prepare Lambda Scan (**hp816x_prepareLambdaScan**) function must be called before a Lambda Scan operation is executed. Use the return values of this function (Number of Datapoints and Number of Power Arrays) to allocate arrays for the Execute Lambda Sca n (hp816x_executeLambdaScan) function.

Equally Spaced Datapoints is enabled as part of this function and cannot be disabled, see the Windows Help file or your Programming Guide for more details. Use Multi Frame Lambda Scan if you need to have inequally spaced datapoints.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
wavelengthArray	Wavelength Array returns the wavelength points at which power values were measured. The wavelength points returned by this array are always equally spaced.

Data Type: ViReal64[]

Input/Output: OUT

powerArray1

Power Array 1 returns the measured power values for the first Power Meter channel.

Equally Spaced Datapoints is enabled as part of this function, see the Windows Help file or your Programming Guide for more details.

Data Type: ViReal64[]

Input/Output: OUT

powerArray2

Power Array 2 returns the measured power values for the second Power Meter channel.

Equally Spaced Datapoints is enabled as part of this function, see the Windows Help file or your Programming Guide for more details.

Data Type: ViReal64[]

Input/Output: OUT

powerArray3

Power Array 3 returns the measured power values for the third Power Meter channel.

Equally Spaced Datapoints is enabled as part of this function, see the Windows Help file or your Programming Guide for more details.

Data Type: ViReal64[]

Input/Output: OUT

powerArray4

Power Array 4 returns the measured power values for the fourth Power Meter channel.

Equally Spaced Datapoints is enabled as part of this function, see the Windows Help file or your Programming Guide for more details.

Data Type: ViReal64[]

Input/Output: OUT

powerArray5

Power Array 5 returns the measured power values for the fifth Power Meter channel.

Equally Spaced Datapoints is enabled as part of this function, see the Windows Help file or your Programming Guide for more details.

Data Type: ViReal64[]

Input/Output: OUT

powerArray6

Power Array 6 returns the measured power values for the sixth Power Meter channel.

Equally Spaced Datapoints is enabled as part of this function, see the Windows Help file or your Programming Guide for more details.

Data Type: ViReal64[]

Input/Output: OUT

powerArray7

Power Array 7 returns the measured power values for the seventh Power Meter channel.

Equally Spaced Datapoints is enabled as part of this function, see the Windows Help file or your Programming Guide for more details.

Data Type: ViReal64[]

Input/Output: OUT

powerArray8

Power Array 8 returns the measured power values for the eighth Power Meter channel.

Equally Spaced Datapoints is enabled as part of this function, see the Windows Help file or your Programming Guide for more details.

Data Type: ViReal64[]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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Information for N77 Series Instruments

This version of the 816x VXIplug&play instrument driver has been extended to support the N773xA, N774xA optical power meter and the N775xA and N776xA Optical Attenuator instruments. These are stand-alone instruments that do not require installation in an 816x mainframe. The SCPI command set has been designed for almost complete programming compatibility with the modular instruments of the 816x family. Please be aware of the following details and limitations, when using this driver.

- The N77-series instruments can be controlled via USB, LAN or GPIB interfaces. Especially for transferring large data volumes, the USB and usually LAN interfaces provide faster performance.
- The identification of the individual ports of these multiport instruments in the SCPI and hp816x functions corresponds to the slot number. So for example the ports of the N7745A can be addressed with the slot numbers 1 to 8. The channel number in the commands and functions is not used and can generally be omitted or set to Channel 1 (index 0).
- The N771x-series and 81950A tunable lasers are not supported by the 816x VXIplug&play driver.
- The N773x-series optical switches are supported by the 816x VXIplug&play functions for switches and relevant functions for the mainframes.
- The N774x-series power meters are supported by the 816x VXIplug&play functions for power meters, relevant functions for the mainframes like for standard trigger configuration, and the Multi Frame Lambda Scan application functions.
- The N776x-series optical attenuators are supported by the 816x VXIplug&play functions for attenuators and the relevant functions for the mainframes.
- The N775x-series optical attenuators and power meters are supported by the 816x VXIplug&play functions for attenuators and the relevant functions for the mainframes and the basic functions for the power meters, excluding the advanced functions using data power logging. The choices of averaging time is also more limited for these power meters.
- The averaging time parameter for the power meters is limited in the 816x VXIplug&play functions to a limited set of discrete values, similar to the modular 816x power meters. This set includes: 1, 2, 5, 10, 25, 100, 200, and 500 ìs, 1, 2, 5, 10, 20, 50, 100, 200, 500 ms, and 1, 2, 5, and 10s.
- The maximum number of logging data points that can be acquired is limited by the 816x VXIplug&play data acquisition functions to 100,000 per port.
- The shortest averaging time used by the Multi Frame Lambda Scan application is 25 ìs.
- For the Multi Frame Lambda Scan application, the N774xA instruments can be included simply by registering the instrumet as a mainframe,after the first

mainframe with the tunable laser has been registered.

Further details to available functions are found under the categories below:

Utility Functions

Mainframe Specific Functions

Power Sensor Specific Functions

Applications

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hp816x_returnEquidistantData

Syntax:

ViStatus _VI_FUNC hp816x_returnEquidistantData(ViSession *ihandle*, ViBoolean *equallySpacedDatapoints*);

The Equally Spaced Datapoints (hp816x_returnEquidistantData) function performs a linear interpolation on the wavelength point and power measurement data and returns an equidistant wavelength array.

This function is used because Lambda Scan functions make use of Lambda Logging to log the exact wavelength that measurements were triggered at. This results in Lambda Array wavelength points that are not equally spaced.

NOTE: Lambda Logging is not available if your Tunable Laser module firmware revision is lower than 2.0.

Equally Spaced Datapoints is enabled as a default.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
equallySpacedDatapoints	The Equally Spaced Datapoints control enables/disables the Equally Spaced Datapoints (hp816x_returnEquidistantData) function. If you choose Enable (VI_TRUE), a linear interpolation is performed on the wavelength point and power measurement data and an equidistant wavelength array is returned.

If you choose Disable (VI_FALSE), the results are returned directly from the instrument.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Enable
VI_FALSE	(0)	Disable

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_registerMainframe

Syntax:

ViStatus _VI_FUNC hp816x_registerMainframe(ViSession *ihandle*);

Use the Register Mainframe (hp816x_registerMainframe) function to register your mainframe as a participant in a Multi Frame Lambda Scan operation. The mainframe must be connected to the GPIB bus and have its Input Trigger Connector connected to the Output Trigger Connector of the 8164A/B Lightwave Measurement System mainframe that the Tunable Laser module is installed in.

The IHandle for the session is internally registered and is used to identify the instrument for the Multi Frame Lambda Scan operation.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_unregisterMainframe

Syntax:

ViStatus _VI_FUNC hp816x_unregisterMainframe(ViSession *ihandle*);

Use the Unregister Mainframe function (hp816x_unregisterMainframe) to remove a mainframe from a Multi Frame Lambda Scan operation and clear the driver's internal data structures.

If you use LabView 5.0 the following items should be noted:

- All multi frame functions are not re-entrant, if the driver is running and initialized more than once, results may be unpredictable.

- To avoid wrong results, call the Unregister Mainframe function prior to the Initialize function (**hp816x_init**). This is especially necessary during program debugging, if the Close function (**hp816x_close**) is not called.

Parameter	**Description**
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). You can unregister all mainframes if you and call this function with IHandle set to zero. **Data Type:** ViSession **Input/Output:** IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_setSweepSpeed

Syntax:

ViStatus _VI_FUNC hp816x_setSweepSpeed(ViSession *ihandle*, ViInt32 *Sweep_Speed*);

Use the Set Sweep Speed function (hp816x_setSweepSpeed) to specify a sweep speed for a multi frame lambda scan.

Call this function before calling the Prepare Multi Frame Lambda Scan function (hp816x_prepareMfLambdaScan).

The Prepare Multi Frame Lambda Scan function checks whether your desired Sweep Speed meets the following condition:

Averageing Time (of all the powermeters) < (Step Size / Sweep Speed)

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
Sweep_Speed	Use Sweep Speed to select the speed of a sweep. Auto Select: Calculates, and sets, the highest possible sweep speed, which depends on the step size and the minimum averaging time for all powermeters. Remark: If it is not possible to use a specified

sweep speed, the Prepare Multi Frame Lambda Scan function (hp816x_prepareMfLambdaScan) will return an error.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_SPEED_80NM	-1	80 nm/s
hp816x_SPEED_40NM	0	40 nm/s
hp816x_SPEED_20NM	1	20 nm/s
hp816x_SPEED_10NM	2	10 nm/s
hp816x_SPEED_5NM	3	5 nm/s
hp816x_SPEED_05NM	4	0.5 nm/s
hp816x_SPEED_AUTO	5	Auto Select

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_prepareMfLambdaScan

Syntax:

ViStatus _VI_FUNC hp816x_prepareMfLambdaScan(ViSession *ihandle*, **ViInt32** *powerUnit*, **ViReal64** *power*, **ViInt32** *opticalOutput*, **ViInt32** *numberofScans*, **ViInt32** *PWMChannels*, **ViReal64** *startWavelength*, **ViReal64** *stopWavelength*, **ViReal64** *stepSize*, **ViUInt32** *numberofDatapoints*, **ViUInt32** *numberofChannels*);

The Prepare Multi Frame Lambda Scan (hp816x_prepareMfLambdaScan) function prepares a Lambda Scan operation for multiple Mainframes.

That is, it prepares an operation where an 8164A/B Lightwave Measurement System with a sweepable Tunable Laser module and up to 1000 Power Sensor Channels located in different Mainframes are installed. The function performs a wavelength sweep where the Tunable Laser module and Power Sensors are co-ordinated with each other.

The Prepare Multi Frame Lambda Scan (hp816x_prepareMfLambdaScan) function must be called before a Multi Frame Lambda Scan is executed. Use the return values of this function (Number of Datapoints and Number of Power Arrays) to allocate arrays for the Execute Multi Frame Lambda Scan (**hp816x_executeMfLambdaScan**) function.

The function scans all mainframes to find swept-wavelength Tunable Laser Sources. The function scans each mainframe in the order that they were originally registered by the Register Mainframe function (**hp816x_registerMainframe**). The first Tunable Laser Source found will perform the sweep operation.

To obtain a higher precision, the wavelength sweep is started 90 pm before the Start Wavelength (or, if the Step Size is > 90 pm, one step before the Start Wavelength) and ends 90 pm after the Stop Wavelength (or, if the Step Size is > 90 pm, one step after the Stop Wavelength). This means that you have to choose a Stop Wavelength 90 pm less than the maximum possible wavelength (or, if the Step Size is > 90 pm, one step less than the maximum possible wavelength).

Triggers coordinate the Tunable Laser module with all Power Meters. The function sets for the lowest possible averaging time available for the installed Power Meters and, then, sets the highest possible sweep speed for the selected Tunable Laser module sweep. All mainframes must be connected to the GPIB bus and have their Input Trigger Connector connected to the Output Trigger Connector of the 8164A/B Lightwave Measurement System mainframe that the Tunable Laser module is installed in.

If one of the following circumstances occurs, the "parameter mismatch" error will be returned:

1. If one Power Meter is out of the specification at 1550 nm, the error "powermeter wavelength does not span 1550nm" will be returned. For example, the 81530A Power Sensor and the 81520A Optical Head are out of specification at 1550 nm. Remove the Power Meter that is out of specification at 1550 nm from the mainframe.

2. If the Step Size is too small and results in a trigger frequency that is to high for the installed Power Meters, the error "could not calculate a sweep speed!" will be returned. Increase the Step Size.

3. If the chosen wavelength range is too large and Step Size is too small, the error "too many datapoints to log!" will be returned. In this case, reduce the wavelength range and/or increase the Step Size.

Parameter	**Description**
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new

session handle is created every time the mainframe is initialized.

For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**).

Data Type: ViSession

Input/Output: IN

powerUnit

You can use the Unit control to choose either Watts (hp816x_PU_WATT) or dBm (hp816x_PU_DBM) as the power unit.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_PU_DBM	0	dBm
hp816x_PU_WATT	1	Watt

power

You can use the Power input to set the output power to a value in dBm.

Data Type: ViReal64

Input/Output: IN

opticalOutput

Use the Optical Output control to choose the optical output for a TLS module with two outputs. You can choose one of the following:

High Power (hp816x_HIGHPOW), where Output 2, the High Power output is the regulated output.

Low SSE (hp816x_LOWSSE), where Output 1, the Low SSE output is the regulated output.

BHR (hp816x_BHR) is the same as choosing Both (Master 2) from the instrument's front panel, Output 2, the High Power output is the regulated output and Output 1, the Low SSE output is the

unregulated output.

BLR (hp816x_BLR) is the same as choosing Both (Master 1) from the instrument's front panel, the Low SSE output is the regulated output and Output 2, the High Power output is the unregulated output.

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_HIGHPOW	0	High Power
hp816x_LOWSSE	1	Low SSE
hp816x_BHR	2	BHR
hp816x_BLR	3	BLR

numberofScans

Number of Scans sets the number of sweep cycles that will be performed.

To perform 1 scan enter 0 (hp816x_NO_OF_SCANS_1).

To perform 2 scans enter 1 (hp816x_NO_OF_SCANS_2).

To perform 3 scans enter 2 (hp816x_NO_OF_SCANS_3).

Data Type: ViInt32

Input/Output: IN

Values:

Name	Value	Description
hp816x_NO_OF_SCANS_1	0	1
hp816x_NO_OF_SCANS_2	1	2
hp816x_NO_OF_SCANS_3	2	3

PWMChannels

Use the PWM Channels input to specify how many Power Meter channels should be used.

If some channels have been excluded, the PWMChannels control must be high enough to include the original index of the highest included channel index. See: hp816x_excludeChannel

The maximum number of channels that may be specified is 1000.

Data Type: ViInt32

Input/Output: IN

startWavelength

Use Start Wavelength to enter the wavelength in meters (m) at which the sweep begins.

Data Type: ViReal64

Input/Output: IN

stopWavelength

Use Stop Wavelength to enter the wavelength in meters (m) at which the sweep ends.

Data Type: ViReal64

Input/Output: IN

stepSize

Use Step Size to enter the size of the change in the wavelength in meters (m) for each wavelength step.

Data Type: ViReal64

Input/Output: IN

numberofDatapoints

Number of Datapoints returns the number of wavelength steps that the Multi Frame Lambda Scan will perform.

Each Power Meter channel will make the same number of measurements, use the returned value for allocating power value arrays.

Data Type: ViUInt32

Input/Output: OUT

numberofChannels

Number of Channels returns the number of Power Meter channel that a Lambda Scan operation will use.

Use this value to allocate the number of power value arrays for the Execute Multi Frame Lambda Scan function (hp816x_executemfLambdaScan).

Data Type: ViUInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_getMFLambdaScanParameters_Q

Syntax:

ViStatus _VI_FUNC hp816x_getMFLambdaScanParameters_Q(ViSession *ihandle*, ViPReal64 *startWavelength*, ViPReal64 *stopWavelength*, ViPReal64 *averagingTime*, ViPReal64 *sweepSpeed*);

The Get MF Lambda Scan Parameters (hp816x_getMFLambdaScanParameters_Q) function returns all parameters that the Prepare Multi Frame Lambda Scan (**hp816x_prepareMfLambdaScan**) function adjusts or automatically calculates.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
startWavelength	Start Wavelength returns the adjusted wavelength value in meters that the Prepare Multi Frame Lambda Scan (**hp816x_prepareMfLambdaScan**) function generates. This Start Wavelength is used as the Tunable Laser module start wavelength for the Lambda Scan operation. **Data Type:** ViPReal64 **Input/Output:** OUT
stopWavelength	Stop Wavelength returns the adjusted wavelength

value in meters that the Prepare Multi Frame Lambda Scan (**hp816x_prepareMfLambdaScan**) function generates.

This Stop Wavelength is used as the Tunable Laser module stop wavelength for the Lambda Scan operation.

Data Type: ViPReal64

Input/Output: OUT

averagingTime

Averaging Time returns the averaging time value in seconds that the Prepare Multi Frame Lambda Scan (**hp816x_prepareMfLambdaScan**) function generates.

This Averaging Time is used for all installed Power Meters for the Lambda Scan operation.

Data Type: ViPReal64

Input/Output: OUT

sweepSpeed

Sweep Speed returns the sweep speed in meters per second (m/s) that the Prepare Multi Frame Lambda Scan (**hp816x_prepareMfLambdaScan**) function generates.

This Sweep Speed is used as the Tunable Laser module sweep speed for the Lambda Scan operation.

Data Type: ViPReal64

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_executeMfLambdaScan

Syntax:

ViStatus _VI_FUNC hp816x_executeMfLambdaScan(ViSession *ihandle*, ViReal64*wavelengthArray*[ ]);

The Execute Multi Frame Lambda Scan (hp816x_executeMfLambdaScan) function runs a Lambda Scan operation and returns an array that contains the wavelength values at which power measurements are made.

That is, it executes an operation where a 8164A/B Lightwave Measurement System with a sweepable Tunable Laser module and up to 1000 Power Sensors installed, performs a wavelength sweep where the Tunable Laser module and Power Sensors are coordin ated with each other.

Use the values returned from the Prepare Multi Frame Lambda Scan (**hp816x_prepareMfLambdaScan**) function to set the parameters of the Execute Multi Frame Lambda Scan (hp816x_executeMfLambdaScan) function.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
wavelengthArray	Wavelength Array returns the wavelength values for the Lambda Scan operation. The Get Lambda Scan Result function (**hp816x_getLambdaScanResult**) returns the corresponding power value arrays. **Data Type:** ViReal64[] **Input/Output:** OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_getLambdaScanResult

Syntax:

ViStatus _VI_FUNC hp816x_getLambdaScanResult(ViSession *ihandle*, ViInt32 *PWMChannel*, ViBoolean *cliptoLimit*, ViReal64 *clippingLimit*, ViReal64*powerArray*[ ], ViReal64*lambdaArray*[ ]);

The Get Lambda Scan Result function (hp816x_getLambdaScanResult) returns for a given Power Meter channel a power value array and a wavelength value array.

These arrays contains the results of the last Multi Frame Lambda Scan operation.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMChannel	Use the PWM Channel input to choose the Power Meter channel for which the power value array should be returned. The maximum number of channels that may be specified is 1000. **Data Type:** ViInt32 **Input/Output:** IN
cliptoLimit	The Clip to Limit control turns the clipping function

On (VI_TRUE) or Off (VI_FALSE).

If the control is turned on any power value less than the Clipping Limit value will be set to this Clipping Limit value.

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	On
VI_FALSE	(0)	Off

clippingLimit

The Clipping Limit value is the value to which a power measurement will be clipped to if a value is less than the Clipping Limit.

The Clipping Limit will be used if you turn On the Clip to Limit control.

Data Type: ViReal64

Input/Output: IN

powerArray

Power Array returns for the chosen Power Meter channel the results of the last Multi Frame Lambda Scan operation.

Data Type: ViReal64[]

Input/Output: OUT

lambdaArray

Lambda Array returns for the logged wavelength points of a channel.

Data Type: ViReal64[]

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_getNoOfRegPWMChannels_Q

Syntax:

ViStatus _VI_FUNC hp816x_getNoOfRegPWMChannels_Q(ViSession *ihandle*, ViUInt32 *numberofPWMChannels*);

The Get Number of PWM Channels (hp816x_getNoOfRegPWMChannels_Q) function returns the number of Power Meter channels in a test setup.

Only Power Meters whose mainframe was registered using the Register Mainframe (**hp816x_registerMainframe**) function are counted.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
numberofPWMChannels	Get Number of PWM Channels returns the number of available Power Meter channels. The channels are numbered from 0 to n, where n is the number of available channels minus 1. **Data Type:** ViUInt32 **Input/Output:** OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine

"hp816x_error_message"

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hp816x_getChannelLocation

Syntax:

ViStatus _VI_FUNC hp816x_getChannelLocation(ViSession *ihandle*, ViInt32 *PWMChannel*, ViPInt32 *mainframeNumber*, ViPInt32 *slotNumber*, ViPInt32 *channelNumber*);

The Get Channel Location function (hp816x_getChannelLocation_Q) returns the location of the chosen Power Meter channel as used in a Multi Frame Lambda Scan operation.

The maximum number of channels that may be specified is 1000.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMChannel	Use the PWM Channels input to select the Power Meter channel location. The maximum input may be 999. Remember that Power Meter channels are numbered from 0 to 999. **Data Type:** ViInt32 **Input/Output:** IN
mainframeNumber	

Mainframe Number returns the mainframe number for the requested channel.

This number corresponds to the order in which you registered mainframes using the Register Mainframe function (**hp816x_registerMainframe**).

Data Type: ViPInt32

Input/Output: OUT

slotNumber

Slot Number returns the slot in which the chosen Power Meter is located.

Data Type: ViPInt32

Input/Output: OUT

channelNumber

Channel Number returns the channel number of the chosen Power Meter channel.

For a dual Power Meter the number of the channel (1 or 2) is returned.

For single Power Meter the returned value is always 1.

Data Type: ViPInt32

Input/Output: OUT

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_excludeChannel

Syntax:

ViStatus _VI_FUNC hp816x_excludeChannel(ViSession *ihandle*, ViInt32 *PWMChannel*);

The Exclude Channel function (hp816x_excludeChannel) excludes a specified PWM channel from a Lambda Scan operation. This function should be called before hp816x_prepareMFLambdaScan, so the settings are not limited by excluded channels.

The original indexing of the power meter channels, resulting from the hp816x_registerMainframe calls, is not changed by the Exclude Channel function. The hp816x_getLambdaScanResult command should be used with the same values for *PWMChannel* for the included channels as if no channels were excluded. In addition, all registered channels upt to the last included channel, including those excluded, should be considered for setting *PWMChannels* in 816x_prepareMFLambdaScan.

Example: if 8 power meters are registered, they receive the indices 0 to 7. If the first two are excluded, then the Lambda Scan Result should be read for *PWMChannel* 2 to 7. And the *PWMChannels* parameter in hp816x_prepareMFLambdaScan should be set to 8.

Parameter	**Description**
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMChannel	Use the PWM Channels control to choose the Power Meter channel to be excluded from the Lambda Scan operation. The maximum input may be 999. Remember that Power Meter channels are numbered from 0 to 999. **Data Type:** ViInt32

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_setInitialRangeParams

Syntax:

ViStatus _VI_FUNC hp816x_setInitialRangeParams(ViSession *ihandle*, ViInt32 *PWMChannel*, ViBoolean *resettoDefault*, ViReal64 *initialRange*, ViReal64 *rangeDecrement*);

Use the Set Initial Range Parameters function (hp816x_setInitialRangeParams) to set the Initial Range, and the Range Decrement, for a selected PWM Channel.

Call this function after Prepare Multi Frame Lambda Scan, and before Execute Multi Frame Lambda Scan.

If this function is not called default values are used.

The default value for the Initial Range used in a lambda scan depends on the initial power.

The default value for the Range Decrement in subsequent scans is 20.0 dBm

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
PWMChannel	Use the PWM Channel control to select the Power Meter channel to which the intial range parameters will be applied. The maximum input may be 999. Remember that Power Meter channels are numbered from 0 to 999.

Data Type: ViInt32

Input/Output: IN

resettoDefault

Use the Reset to Default control to reset the initial range parameters to the built-in default calculated values.

If you want to reset the inital range parameters to the default values, select "Yes".

If you want to use different values, select "NO".

Data Type: ViBoolean

Input/Output: IN

Values:

Name	Value	Description
VI_TRUE	(1)	Yes
VI_FALSE	(0)	No

initialRange

Use Initial Range control to specify the range for the initial scan.

Data Type: ViReal64

Input/Output: IN

rangeDecrement

Use the Range Decrement control to specify a range decrement for subsequent scans.

Data Type: ViReal64

Input/Output: IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_setScanAttenuation

Syntax:

ViStatus _VI_FUNC hp816x_setScanAttenuation(ViSession *ihandle*, ViReal64 *Scan_Attenuation*);

Use the Set Sweep Speed function (hp816x_setScanAttenuation) to specify the attenuation applied by a backloadable TLS with a built-in optical attenuator during a multi frame lambda scan operation.

Parameter	Description
ihandle	IHandle is the session handle returned from the Initialize Mainframe function (**hp816x_init**). A new session handle is created every time the mainframe is initialized. For LabView Programmers: This input terminal needs to be connected to the session handle, IHandle, that is returned from the Initialize Mainframe function (**hp816x_init**). **Data Type:** ViSession **Input/Output:** IN
*Scan*_Attenuation	Use this control to specify the attenuation applied in decibels (dB). **Data Type:** ViReal64 **Input/Output:** IN

Return Value:

VI_SUCCESS: No error

Non VI_SUCCESS: Indicates error condition. To determine error message, pass the return value to routine "hp816x_error_message"

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hp816x_ATT_displayToOffset
hp816x_PWM_displayToReference
hp816x_PWM_fetchValue
hp816x_PWM_functionStop
hp816x_PWM_ignoreError
hp816x_PWM_readAll
hp816x_PWM_readValue
hp816x_PWM_slaveChannelCheck
hp816x_PWM_zeroing
hp816x_PWM_zeroingAll
hp816x_RLM_fetchReturnLoss
hp816x_RLM_fetchValue
hp816x_RLM_functionStop
hp816x_RLM_readReturnLoss
hp816x_RLM_readValue
hp816x_RLM_zeroing
hp816x_RLM_zeroingAll
hp816x_TLS_configureBNC
hp816x_TLS_sweepControl
hp816x_TLS_sweepNextStep
hp816x_TLS_sweepPreviousStep
hp816x_TLS_sweepWait
hp816x_TLS_zeroing
hp816x_TLS_zeroingAll
hp816x_TLS_zeroing Ex
hp816x_WaitForOPC
hp816x_calculate_RL
hp816x_calibrate_RLM
hp816x_close
hp816x_cls
hp816x_cmd
hp816x_cmdInt
hp816x_cmdInt32_Q
hp816x_cmdReal
hp816x_cmdReal64_Q
hp816x_cmdString_Q
hp816x_convertQuestionableStatus_Q
hp816x_cp_ATT_refFromExtPM

hp816x_dbmToWatt
hp816x_dcl
hp816x_delete_ATT_offsetTblEntries
hp816x_displayToLambdaZero
hp816x_driverLogg
hp816x_enableDisableDisplay
hp816x_enableHighSweepSpeed
hp816x_enable_RLM_sweep
hp816x_errorQueryDetect
hp816x_error_message
hp816x_error_query
hp816x_excludeChannel
hp816x_executeLambdaScan
hp816x_executeMfLambdaScan
hp816x_fetch_ATT_outputPower
hp816x_forceTransaction
hp816x_generateTrigger
hp816x_getChannelLocation
hp816x_getDate
hp816x_getHandle
hp816x_getInstrumentId_Q
hp816x_getLambdaScanParameters_Q
hp816x_getLambdaScanResult
hp816x_getLockState
hp816x_getMFLambdaScanParameters_Q
hp816x_getModuleStatus_Q
hp816x_getNoOfRegPWMChannels_Q
hp816x_getSlotInformation_Q
hp816x_getTime
hp816x_getWlRespTblSize
hp816x_getWlRespTblSizeEx
hp816x_get_ATT_absPowerMode_Q
hp816x_get_ATT_attenuationOffset_Q
hp816x_get_ATT_attenuation_Q
hp816x_get_ATT_attenuatorSpeed_Q
hp816x_get_ATT_avtime_Q
hp816x_get_ATT_controlLoopState_Q
hp816x_get_ATT_offsetFromWavelength_Q
hp816x_get_ATT_offsetTable_Q
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hp816x_get_ATT_powerOffset_Q
hp816x_get_ATT_powerReference_Q
hp816x_get_ATT_powerUnit_Q

hp816x_get_ATT_power_Q
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hp816x_get_ATT_wavelengthOffsetState_Q
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hp816x_get_FLS_laserSource_Q
hp816x_get_FLS_laserState_Q
hp816x_get_FLS_modulationSettings_Q
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hp816x_get_RLM_laserSourceParameters_Q
hp816x_get_RLM_laserState_Q
hp816x_get_RLM_loggingResults_Q
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hp816x_get_RLM_parameters_Q
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hp816x_get_SWT_type
hp816x_get_TLS_BNC_config_Q
hp816x_get_TLS_accClass
hp816x_get_TLS_attenuationSettings_Q
hp816x_get_TLS_autoCalState
hp816x_get_TLS_ccLevel_Q
hp816x_get_TLS_darkState_Q
hp816x_get_TLS_frequencyOffset_Q
hp816x_get_TLS_lambdaLoggingState_Q
hp816x_get_TLS_lambdaLoggingStateEx_Q
hp816x_get_TLS_lambdaZero_Q
hp816x_get_TLS_laserRiseTime
hp816x_get_TLS_laserState_Q
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hp816x_get_TLS_parameters_Q
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hp816x_moduleSelftest
hp816x_nodeInputConfiguration
hp816x_opc_Q
hp816x_prepareLambdaScan
hp816x_prepareMfLambdaScan
hp816x_preset

hp816x_readWlRepTblCSV
hp816x_readWlRepTblCSV_Ex
hp816x_readWlRespTable
hp816x_readWlRespTableEx
hp816x_read_ATT_outputPower
hp816x_registerMainframe
hp816x_reset
hp816x_returnEquidistantData
hp816x_revision_Q
hp816x_revision_query
hp816x_self_test
hp816x_setBaudrate
hp816x_setDate
hp816x_setInitialRangeParams
hp816x_setScanAttenuation
hp816x_setSweepSpeed
hp816x_setTime
hp816x_set_ATT_absPowerMode
hp816x_set_ATT_attenuation
hp816x_set_ATT_attenuationOffset
hp816x_set_ATT_attenuatorSpeed
hp816x_set_ATT_avTime
hp816x_set_ATT_controlLoopState
hp816x_set_ATT_offsByRefPM
hp816x_set_ATT_power
hp816x_set_ATT_powerOffset
hp816x_set_ATT_powerReference
hp816x_set_ATT_powerUnit
hp816x_set_ATT_pwrOffsRefPM
hp816x_set_ATT_shutterAtPowerOn
hp816x_set_ATT_shutterState
hp816x_set_ATT_triggerConfig
hp816x_set_ATT_wavelength
hp816x_set_ATT_wavelengthOffset
hp816x_set_ATT_wavelengthOffsetState
hp816x_set_ATT_wlOffsRefPowermeter
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hp816x_set_FLS_laserSource
hp816x_set_FLS_laserState
hp816x_set_FLS_modulation
hp816x_set_FLS_parameters
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hp816x_set_PWM_calibration
hp816x_set_PWM_internalTrigger
hp816x_set_PWM_logging
hp816x_set_PWM_minMax
hp816x_set_PWM_parameters
hp816x_set_PWM_powerRange
hp816x_set_PWM_powerUnit
hp816x_set_PWM_referenceSource
hp816x_set_PWM_referenceValue
hp816x_set_PWM_stability
hp816x_set_PWM_triggerConfiguration
hp816x_set_PWM_wavelength
hp816x_set_RLM_FPDelta
hp816x_set_RLM_averagingTime
hp816x_set_RLM_internalTrigger
hp816x_set_RLM_laserSourceParameters
hp816x_set_RLM_laserState
hp816x_set_RLM_logging
hp816x_set_RLM_minMax
hp816x_set_RLM_modulationState
hp816x_set_RLM_parameters
hp816x_set_RLM_powerRange
hp816x_set_RLM_rlReference
hp816x_set_RLM_stability
hp816x_set_RLM_triggerConfiguration
hp816x_set_RLM_wavelength
hp816x_set_SWT_route
hp816x_set_TLS_attenuation
hp816x_set_TLS_autoCalibration
hp816x_set_TLS_ccLevel
hp816x_set_TLS_dark
hp816x_set_TLS_frequencyOffset
hp816x_set_TLS_lambdaLoggingState
hp816x_set_TLS_lambdaLoggingStateEx
hp816x_set_TLS_laserRiseTime
hp816x_set_TLS_laserState
hp816x_set_TLS_modulation
hp816x_set_TLS_opticalOutput
hp816x_set_TLS_parameters
hp816x_set_TLS_power
hp816x_set_TLS_SBS_Control
hp816x_set_TLS_SBSLevel
hp816x_set_TLS_sweep
hp816x_set_TLS_triggerConfiguration

hp816x_set_TLS_wavelength
hp816x_spectralCalibration
hp816x_spectralCalibrationEx
hp816x_standardTriggerConfiguration
hp816x_standardTriggerConfiguration_Q
hp816x_start_PWM_internalTrigger
hp816x_start_RLM_internalTrigger
hp816x_timeOut
hp816x_timeOut_Q
hp816x_trigOutConfiguration
hp816x_unregisterMainframe
hp816x_wattToDBm
hp816x_zero_ATT_all
hp816x_zero_ATT_powermeter

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Function Reference (Hierarchical)

The hiarchical function reference lists the hp816x VXI*plug&play* functions by classes as defined by the hp816x function panel (hp816x.fp). The function panel runs under VEE, LabWindows®, and LabVIEW®.

DRIVER: hp816x

Initialize
Get Session Handle

revision query
Self Test

Utility
Timeout
Get Instrument ID
List Visa Addresses
Set Baudrate
Query Timeout
Device Clear
Instrument Error Query
Error Message
Automatic Error Detection
dBm to Watt
WattToDBm
Set Date
Get Date
Set Time
Get Time
Clear Error Queue
Force Transaction
Enable Driver Logging

Pass Through Functions
Send String Command
Send String Query
Send Integer Command
Send Integer Query
Send Real Command
Send Real Query

Mainframe Specific Functions
Information for N77 Series Instruments
Reset

Preset
Mainframe Self-Test
Module Self-Test
Revision Query
Operation Complete
Lock-Unlock Instrument
Get Lock State
Enable-Disable Display
Get Slot Information
Get Module Status
Convert Questionable Status
Standard Trigger Configuration
Get Standard Trigger Configuration
Custom Trigger
Trigger Input Configuration
Trigger Output Configuration
Generate Trigger
Attenuators
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Set Attenuation
Get Attenuation
Set Attenuation Offset
Get Attenuation Offset
Set Attenuation Speed
Get Attenuation Speed
Display to Offset
Set Wavelength
Get Wavelength
Set Power Unit
Get Power Unit
Set Absolute Power Mode
Get Absolute Power Mode
Set Power
Get Power
Copy Ref. from Powermeter
Set Power Reference
Get Power Reference
Set Shutter State
Get Shutter State
Set Shutter State at Power On
Get Shutter State at Power On
Get Operational Status
Wavelength-Offset Table Control
Set Wavel. Offs. Ref. PM

Get Wavel. Offs. Ref. PM
Set Wavelength Offset State
Get Wavelength Offset State
Set Wavelength Offset
Get WL/Offset from Index
Get Offset from Wavelength
Delete Table Entries
Get Offset Table Entries
Get Table Size
Builtin Powermeter Control
Read ATT Output Power
Fetch ATT Output Power
Set Power Offset
Get Power Offset
Set Power Offset Ref. PM
Set Att. Offs. by Ext PM
Set Control Loop State
Get Control Loop State
Set Averaging Time
Get Averaging Time
Set ATT Trigger Configuration
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Zero Builtin Powermeter
Get Last Zero Result
Zero All
Multi Wavelength Calibration
Spectral Calibration
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Set PWM Calibration
Get PWM Calibration
Set PWM Power Range
Get PWM Power Range
Set PWM Power Unit

Get PWM Power Unit
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Get PWM Reference Source
Set PWM Reference Value
Get PWM Reference Value
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Ignore Instrument Error
PWM Data Acquisition
Set PWM Logging
Get PWM Logging Results
Set PWM Stability
Get PWM Stability Results
Set PWM MinMax
Get PWM MinMax Results
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Spectral CalibrationEx
Get Wavelength Resp. Tbl SizeEx
Read Wavelength Response TableEx
Read WI Resp Table CSV_Ex
Fixed Laser Sources
Set FLS Parameters
Get FLS Parameters
Set FLS Modulation
Get FLS Modulation Settings
Set FLS Laser Source
Get FLS Laser Source
Get FLS Wavelength
Set FLS Attenuation
Get FLS Attenuation
Set FLS Trigger State
Get FLS Trigger State
Get FLS Power

Set FLS Laser State
Get FLS Laser State
Switch Modules
Get Switch Type
Set Route
Get Route
Get Route Table
Return Loss Modules
Set RLM Parameters
Get RLM Parameters
Set RLM Internal Trigger
Set RLM Averaging Time
Get RLM Averaging Time
Set RLM Wavelength
Get RLM Wavelength
Set RLM Power Range
Get RLM Power Range
Set RLM Trigger Configuration
Start RLM Trigger
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Fetch RLM Return Loss
Read RLM Power Value
Fetch RLM Power Value
Set RLM Return Loss Reference
Get RLM Return Loss Reference
Set RLM Front Panel Delta
Get RLM Front Panel Delta
Calibrate RLM
RLM Zeroing
RLM Zeroing All
Enable RLM Sweep
Advanced Functions
Get RLM Reflectance Values
Get RLM Termination Values
Get RLM DUT Values
Calculate Return Loss
Optional Laser Source Control
Get RLM Source Wavelength
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Get RLM Laser Source Parameters
Set RLM Modulation State
Get RLM Modulation State
Set RLM Laser State
Get RLM Laser State

RLM Data Acquisition
Set RLM Logging
Get RLM Logging Results
Set RLM Stability
Get RLM Stability Results
Set RLM MinMax
Get RLM MinMax Results
Stop RLM Function
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Set TLS Wavelength
Get TLS Wavelength
Set TLS Power
Get TLS Power
Set TLS Optical Output
Get TLS Optical Output
Get TLS Max Power in Range
Set TLS Laser State
Get TLS Laser State
Set Laser Rise Time
Get Laser Rise Time
Set TLS Trigger Configuration
Get TLS Trigger Configuration
Set TLS Attenuation
Get TLS Attenuation Settings
Set TLS Dark
Get TLS Dark State
Get TLS Temperatures
Get TLS Temperatures Ex
TLS Zeroing
TLS ZeroingAll
TLS Zeroing Ex
Set Auto Calibration
Get Auto Calibration State
Get Accuracy Class
TLS Frequency Offset
TLS Display to Lambda Zero
Get TLS Lambda Zero
Set TLS Frequency Offset
Get TLS Frequency Offset

TLS Sweep Functions
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TLS Sweep Control
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TLS Next Step
TLS Previous Step
Wait Sweep Finished
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Set TLS Modulation
Get TLS Modulation Settings
TLS Configure BNC Output
TLS Get BNC Configuration
Set Coherence Control Lev (DFB)
Get Coherence Control Lev (DFB)
TLS SBS Suppression
Set TLS SBS Control
Get TLS SBS Control
Set TLS SBS Level
Get TLS SBS Level
TLS Maximum Power Curve
Get Number of Power Points
Get Logged Power Data
TLS Logging Functions
Set Lambda Logging State
Set Lambda Logging State Ex
Get Lambda Logging State
Get Lambda Logging State Ex
Get Number of Wavelength Values
Get Number of Wavelength Values Ex
Get Logged Wavelength Data
Get Logged Wavelength Data Ex
Applications
Set Lambda Scan Wavelength
Enable High Sweep Speed
Lambda Scan
Prepare Lambda Scan
Get Lambda Scan Parameters
Execute Lambda Scan
Multi Frame Lambda Scan
Information for N774x Series Instruments
Equally Spaced Datapoints
Register Mainframe
Unregister Mainframe
Set Sweep Speed

Prepare Multi Frame Lambda Scan
Get MF Lambda Scan Parameters
Execute Multi Frame Lambda Scan
Get Lambda Scan Result
Get Number of PWM Channels
Get Channel Location
Exclude Channel
Set Initial Range Parameters
Set Scan Attenuation
Close

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Alphabetical Reference | Hierarchical Reference

Where to find Online and User Manual Information

Online Information

http://www.keysight.com/find/optical

Contact information is available via the Keysight Technologies customer website:

http://www.keysight.com/find/contactus

User Manual Information

Using the Keysight Technologies 816x VXI*plug&play* Instrument Driver requires you to use both the online help and the instrument user's manual. It is assumed that VISA and a GP-IB card if required has been installed and properly configured. The following is a high-level directory of where specific information can be found:

SCPI programming information, SCPI Command Reference, and SCPI example programs	*Keysight 8163A/B Lightwave Multimeter, Keysight 8164A/B Lightwave Measurement System, & Keysight 8166A/B Lightwave Multichannel System Programming Guide* *Keysight N77xx Series Programming Guide*
SCPI example programs	*SCPI 1995 Volume2: Command Reference*
Keysight Technologies 816x VXIplug&play instrument driver programming information	*Online Help*
Keysight Technologies 816x VXIplug&play instrument driver example programs	*Online Help*
Keysight Technologies 816x VXIplug&play instrument driver function reference	*Online Help*
VISA Language Information	*VISA User's Manual*
VEE programming information	*VEE User's Manual*

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In Case of Trouble

A variety of support is available through the Web including troubleshooting hints, technical information and access to the most up-to-date drivers.

For general troubleshooting support and technical information see:

http://www.keysight.com/find/optical

Contact Keysight:

http://www.keysight.com/find/contactus

For access to the most up to date instrument drivers see:

http://www.keysight.com/find/octfirmware

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